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(54) **PREDICTIVE POWER MAP GENERATION
AND CONTROL SYSTEM**

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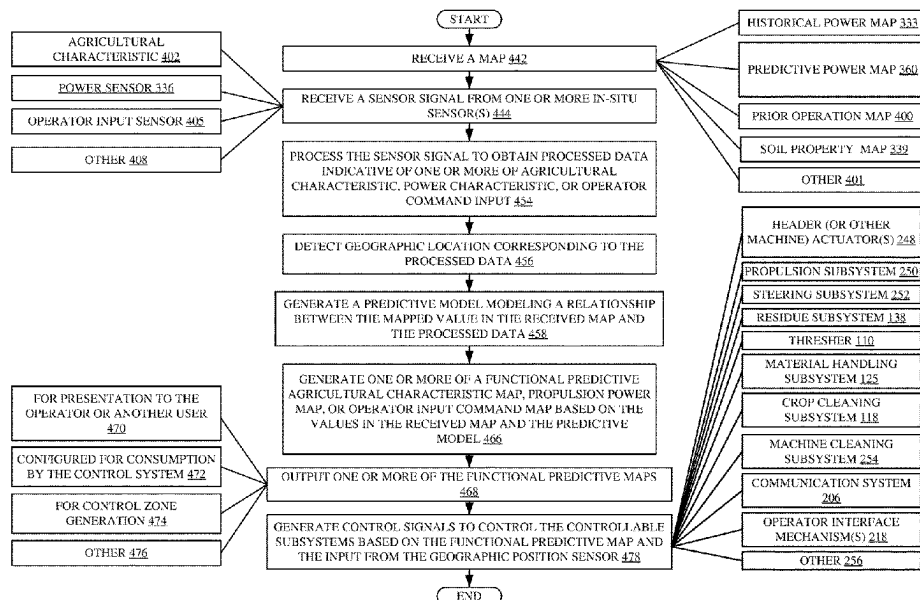
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See application file for complete search history.

(57) **ABSTRACT**

One or more information maps are obtained by an agricul-
tural work machine. The one or more information maps map
one or more agricultural characteristic values at different
geographic locations of a field. An in-situ sensor on the
agricultural work machine senses an agricultural character-
istic as the agricultural work machine moves through the
field. A predictive map generator generates a predictive map
that predicts a predictive agricultural characteristic at dif-
ferent locations in the field based on a relationship between
the values in the one or more information maps and the
agricultural characteristic sensed by the in-situ sensor. The
predictive map can be output and used in automated
machine control.

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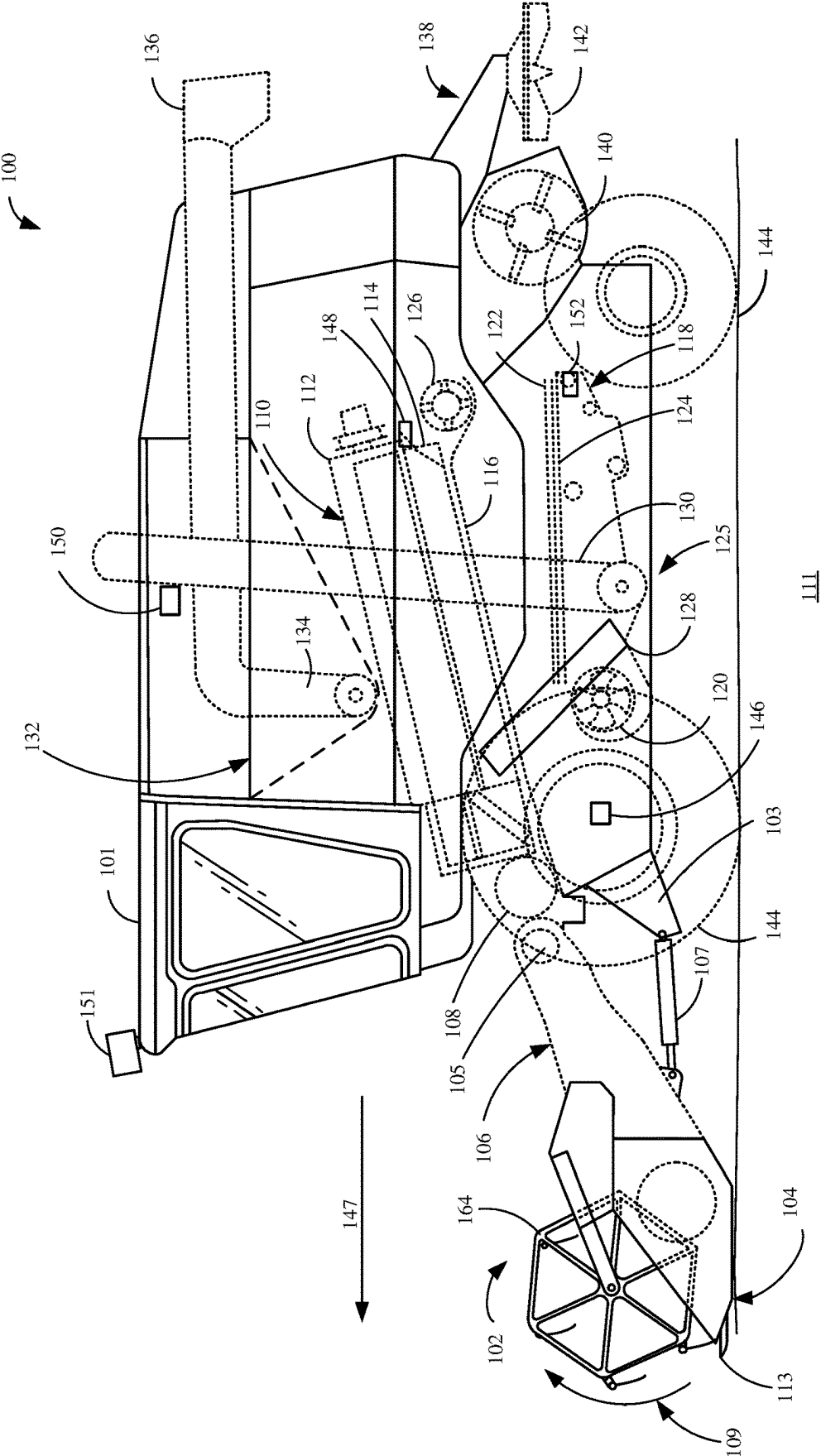


FIG. 1

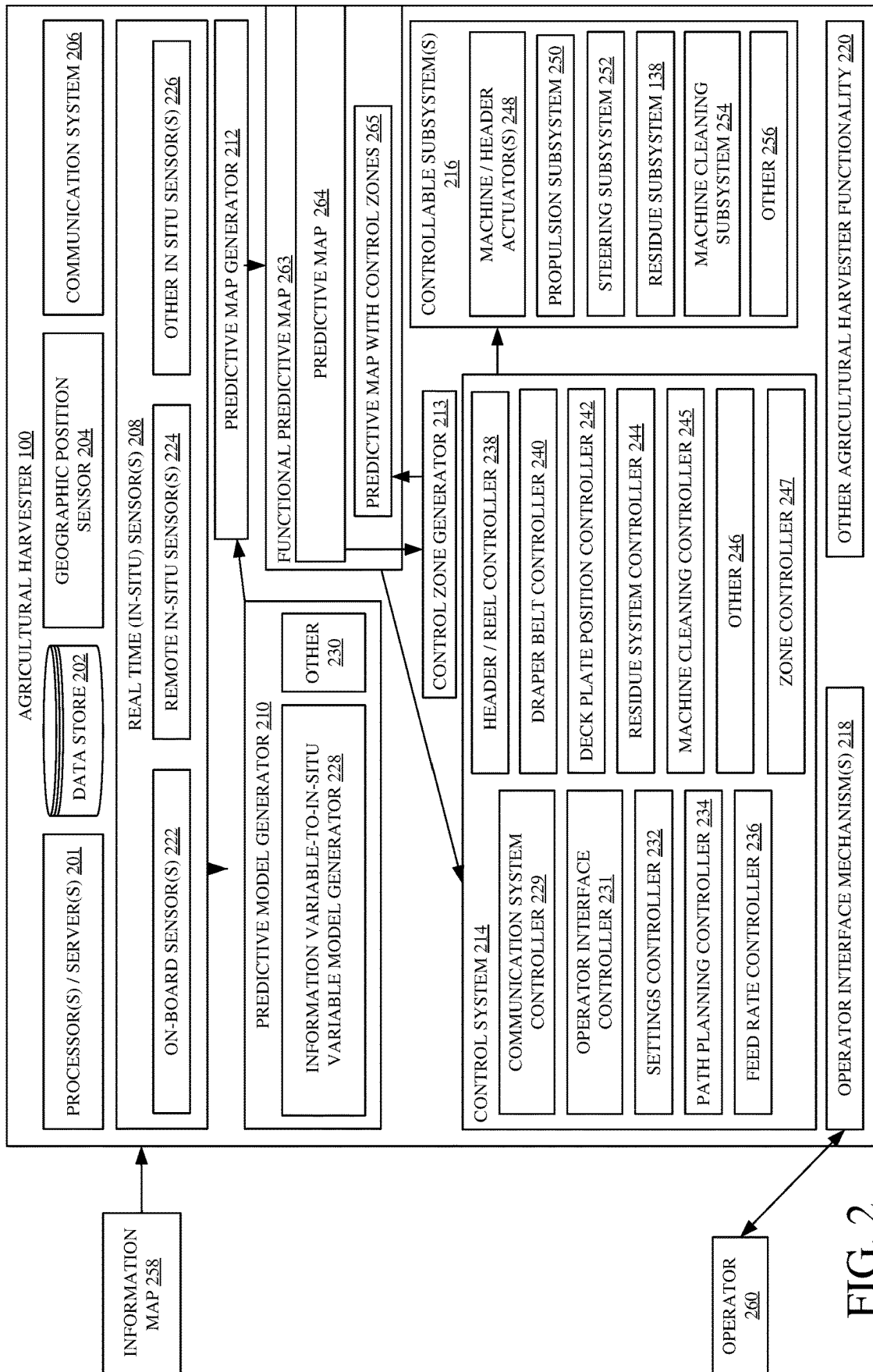


FIG. 2

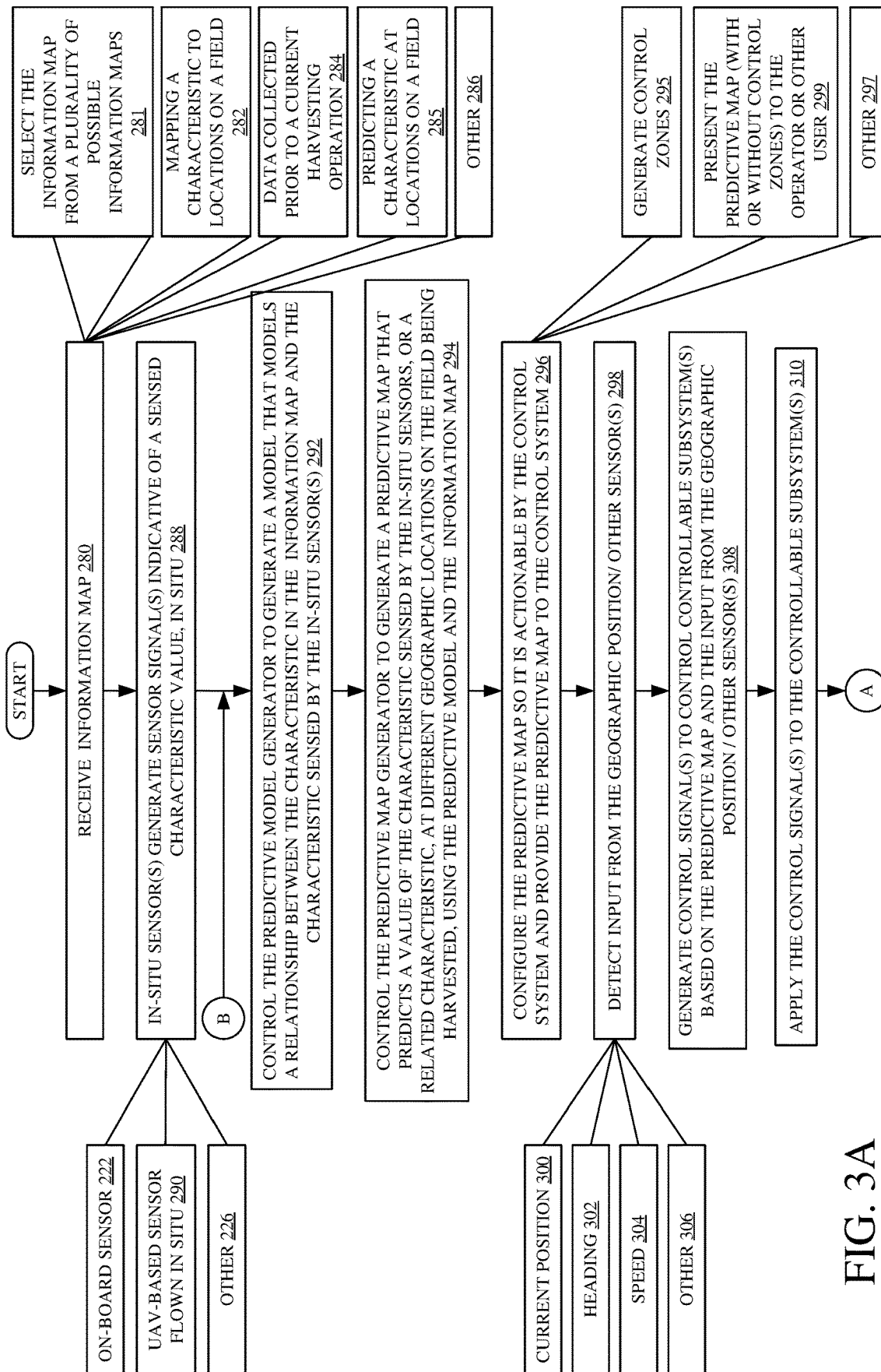


FIG. 3A

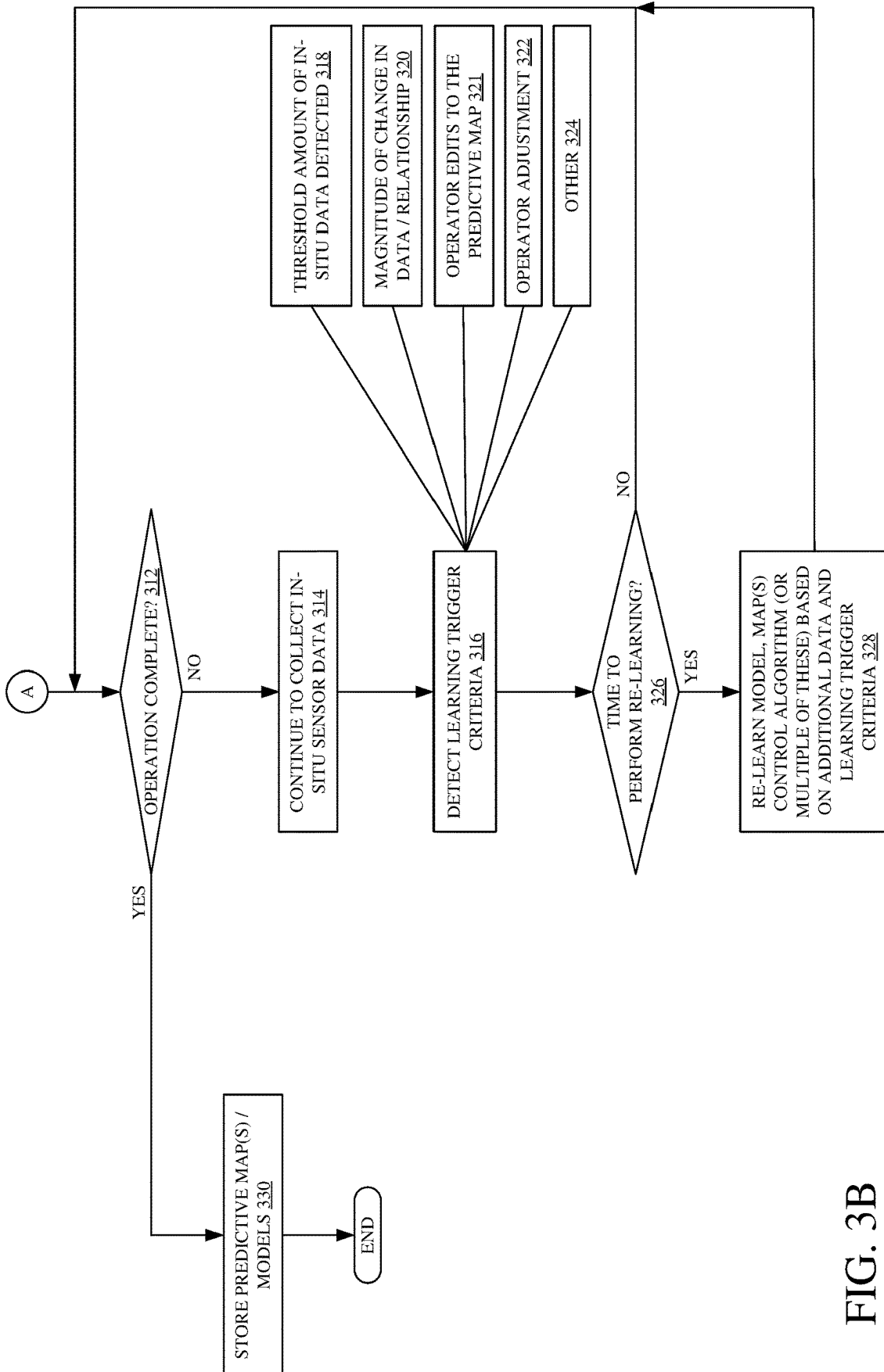


FIG. 3B

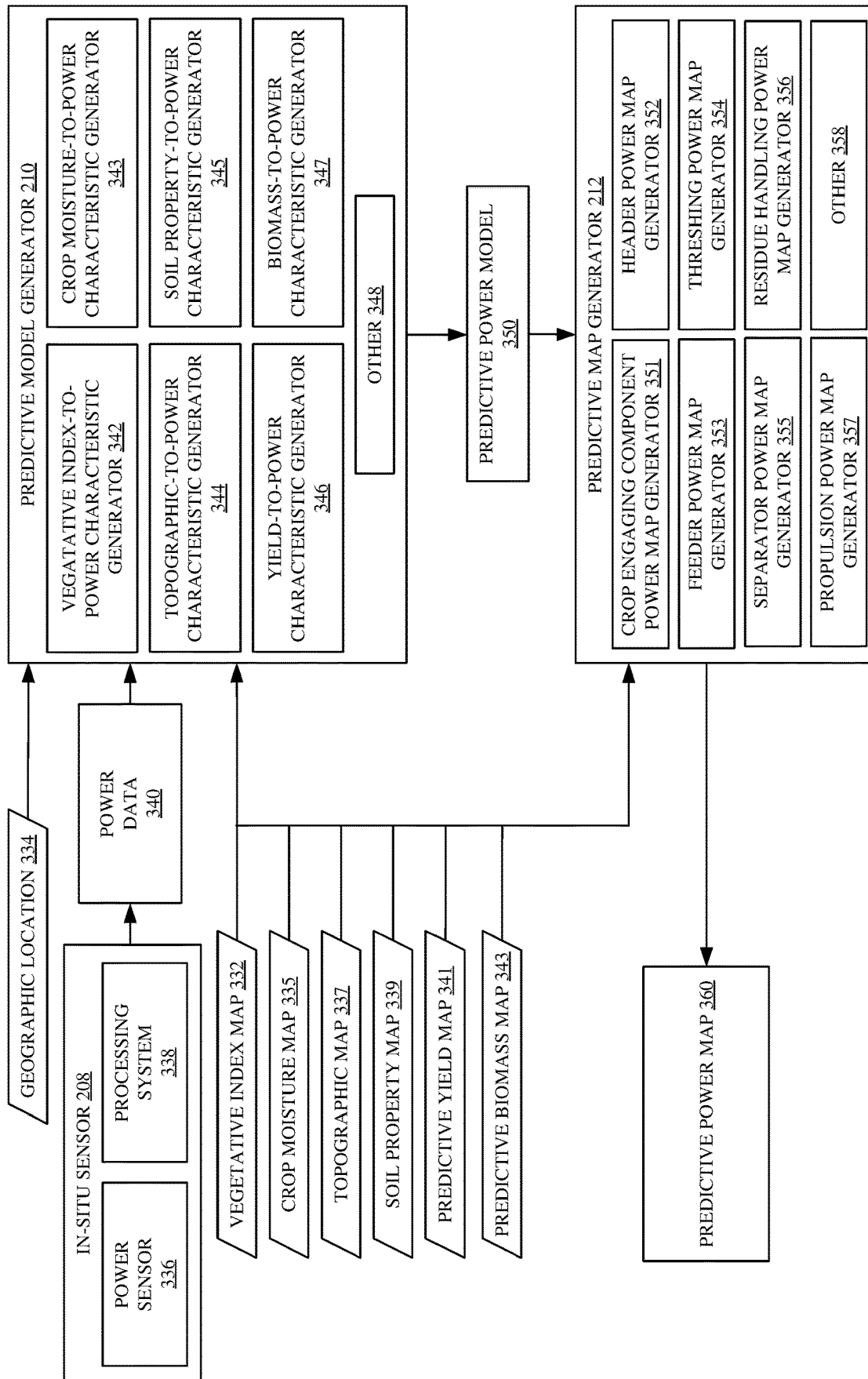
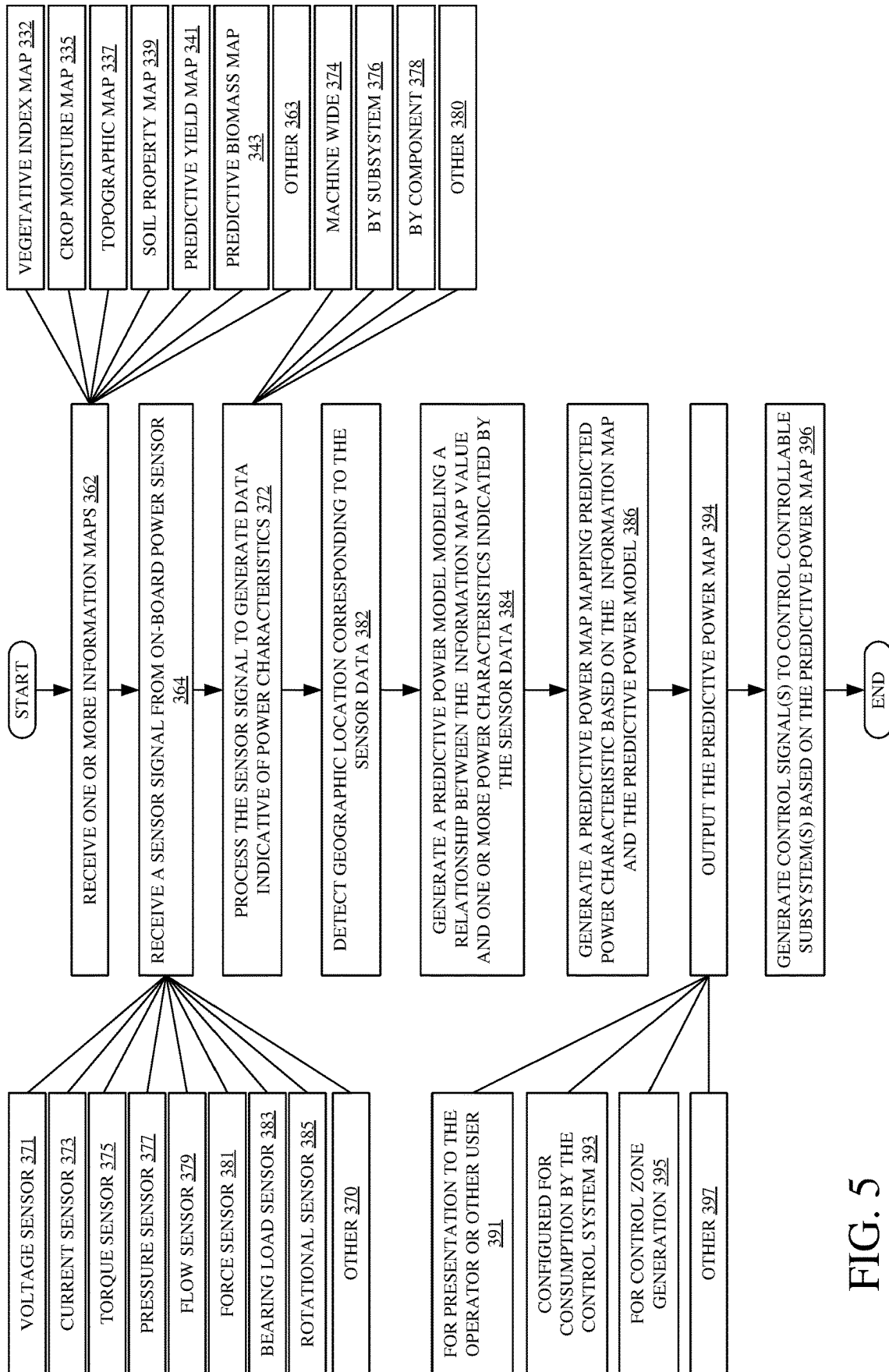


FIG. 4



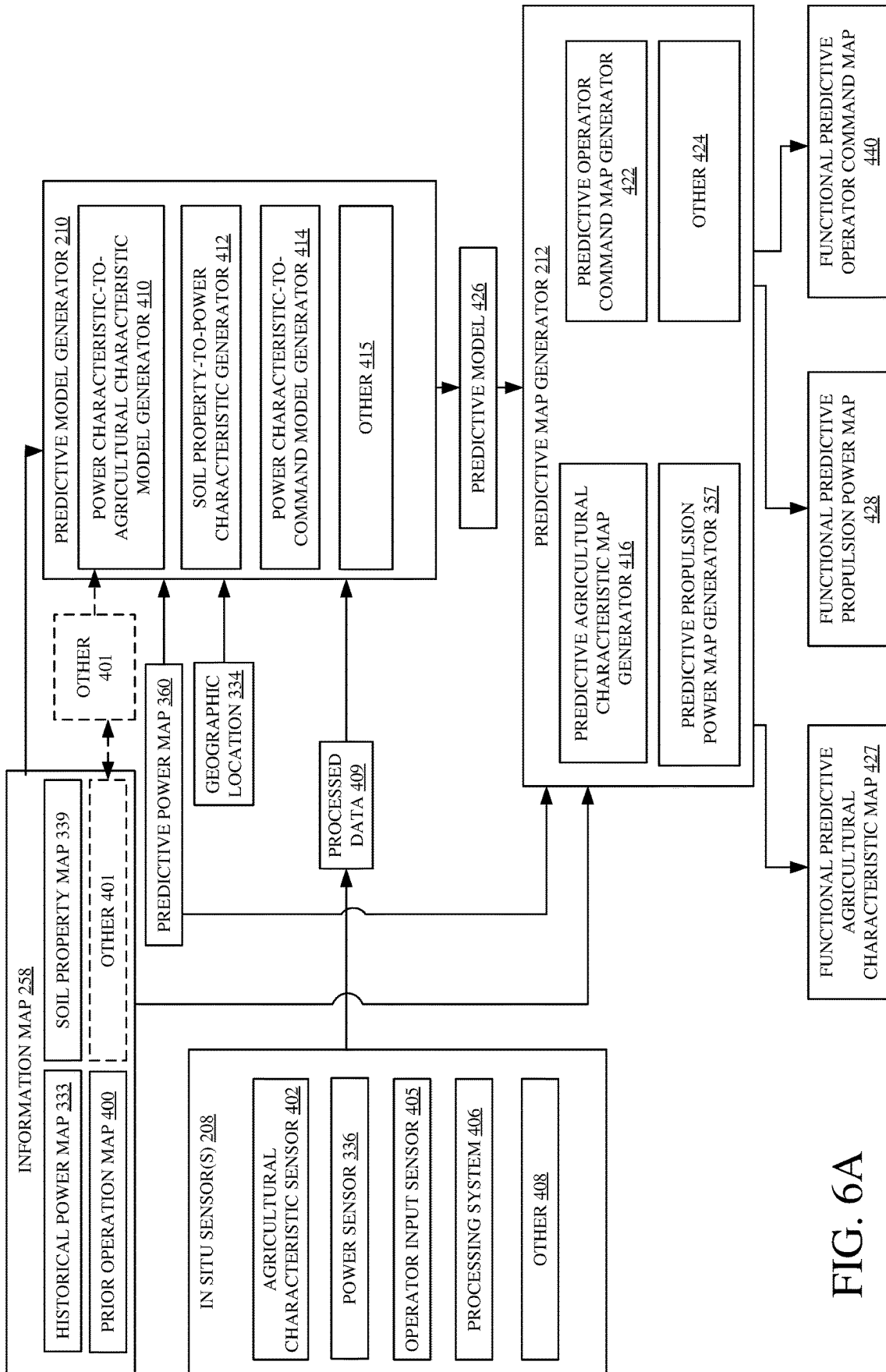


FIG. 6A

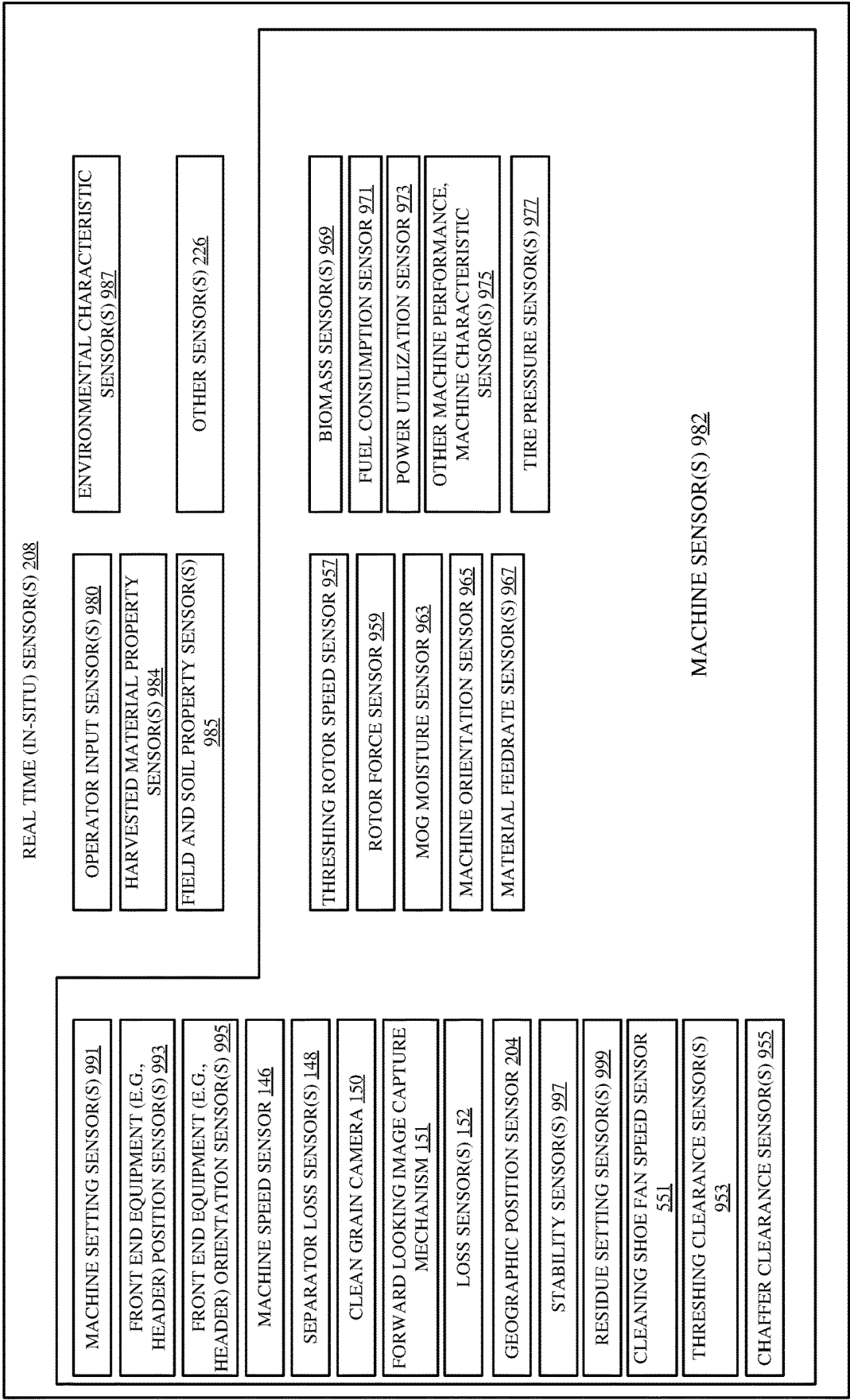
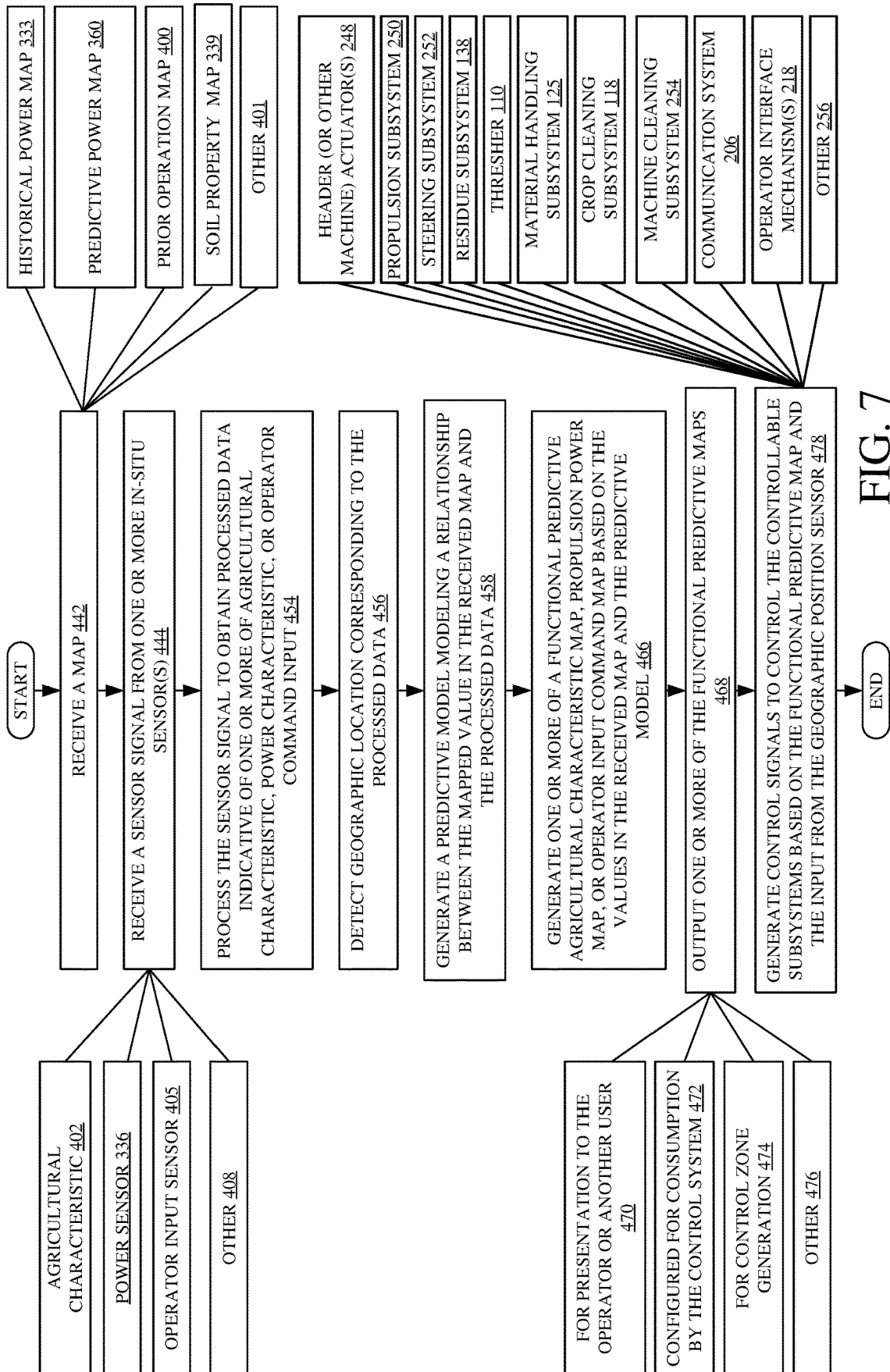


FIG. 6B



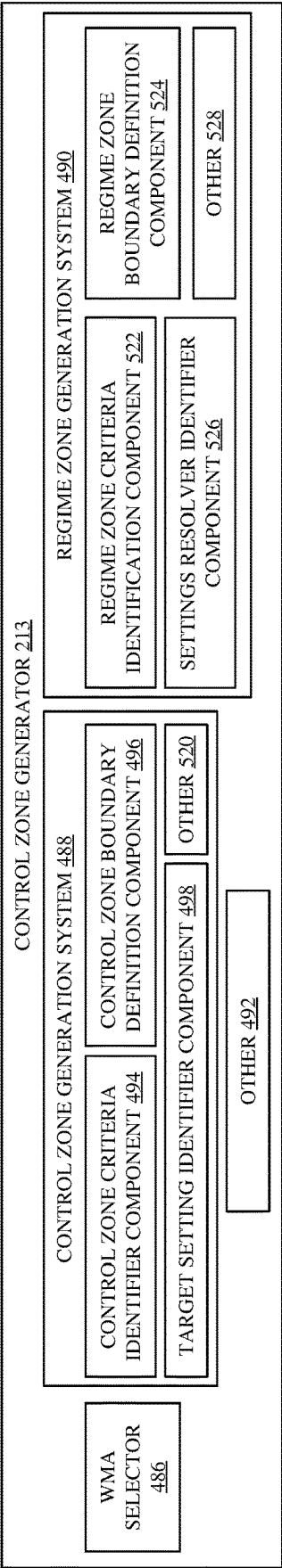


FIG. 8

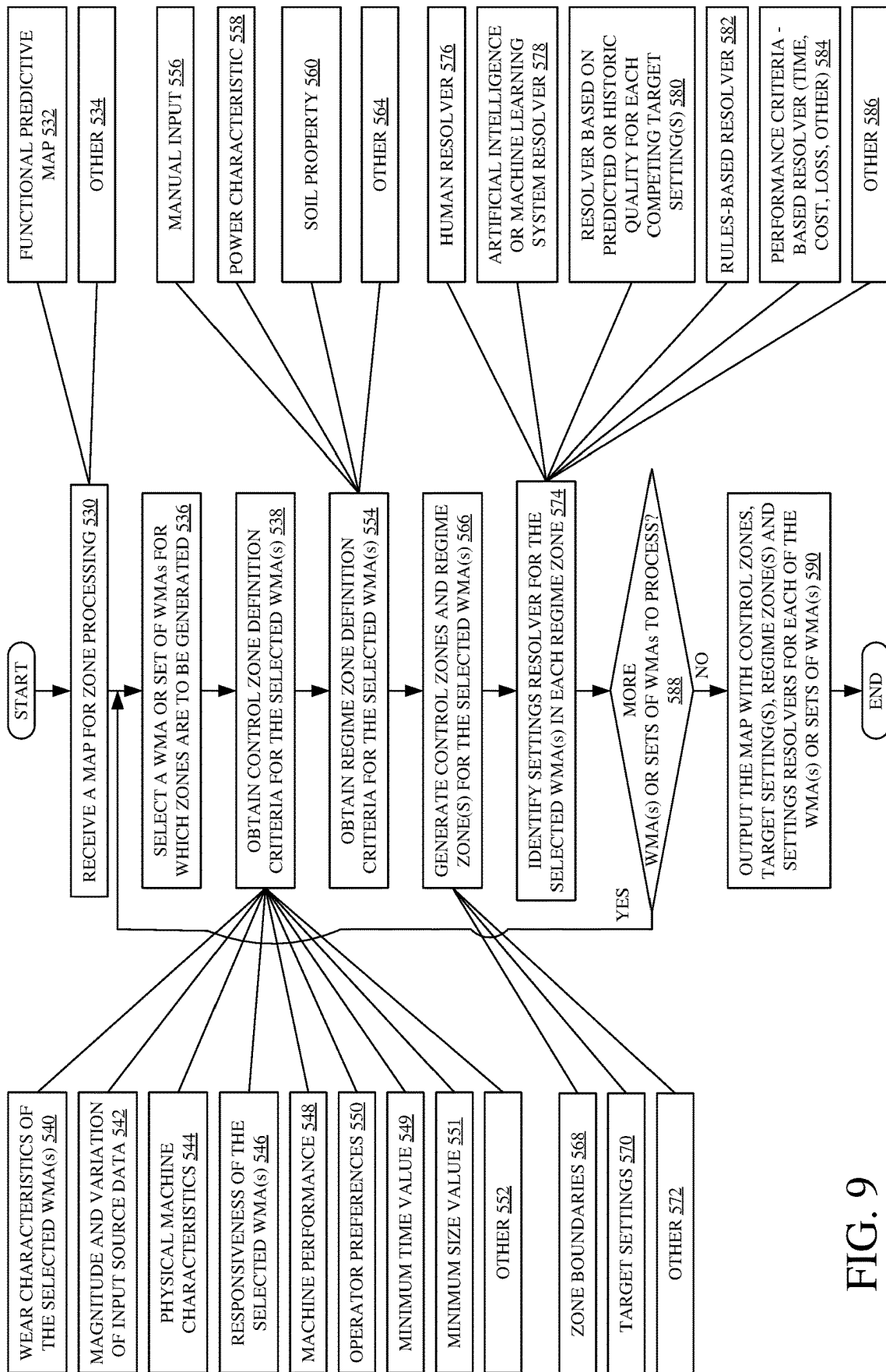
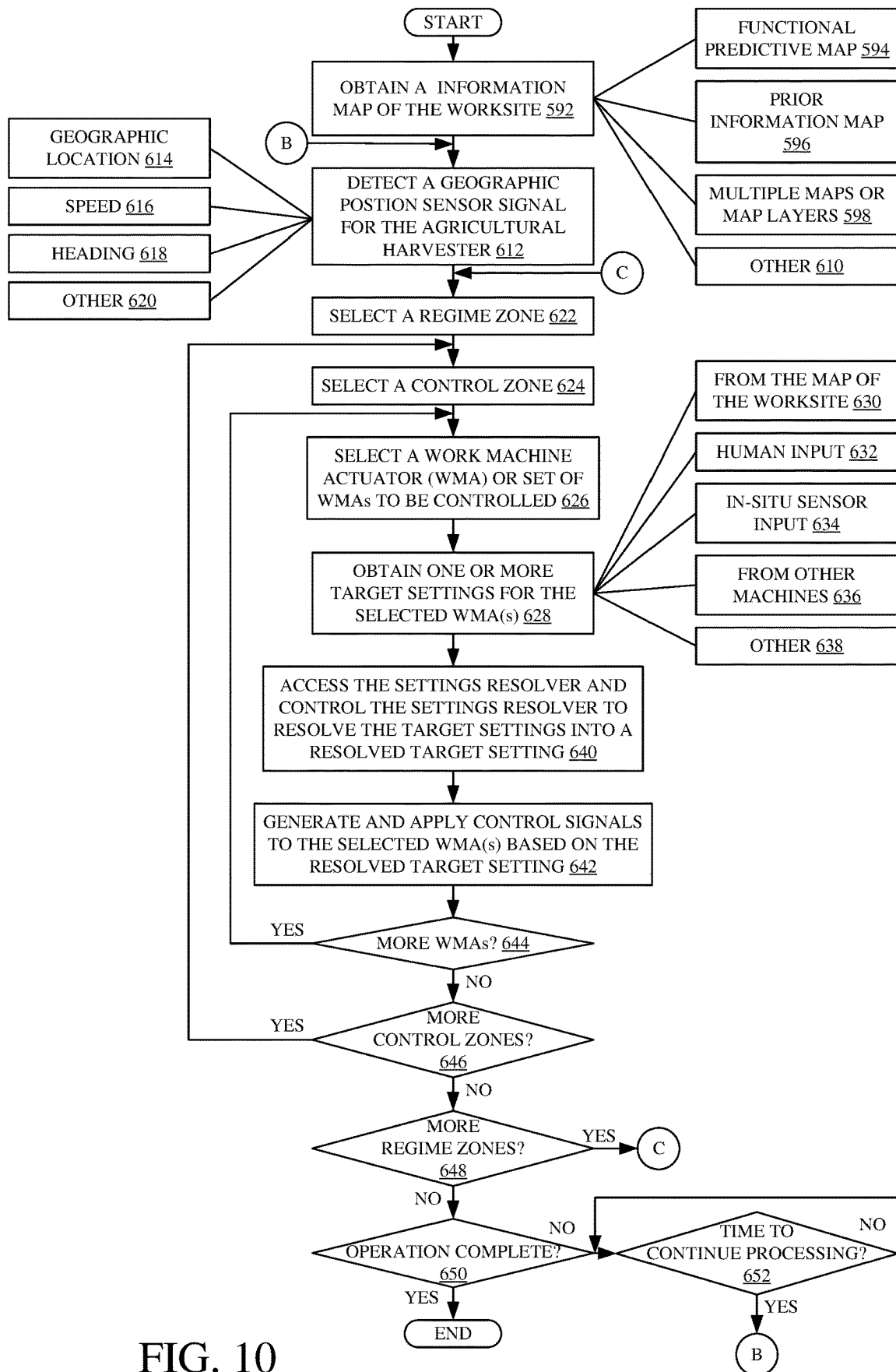


FIG. 9



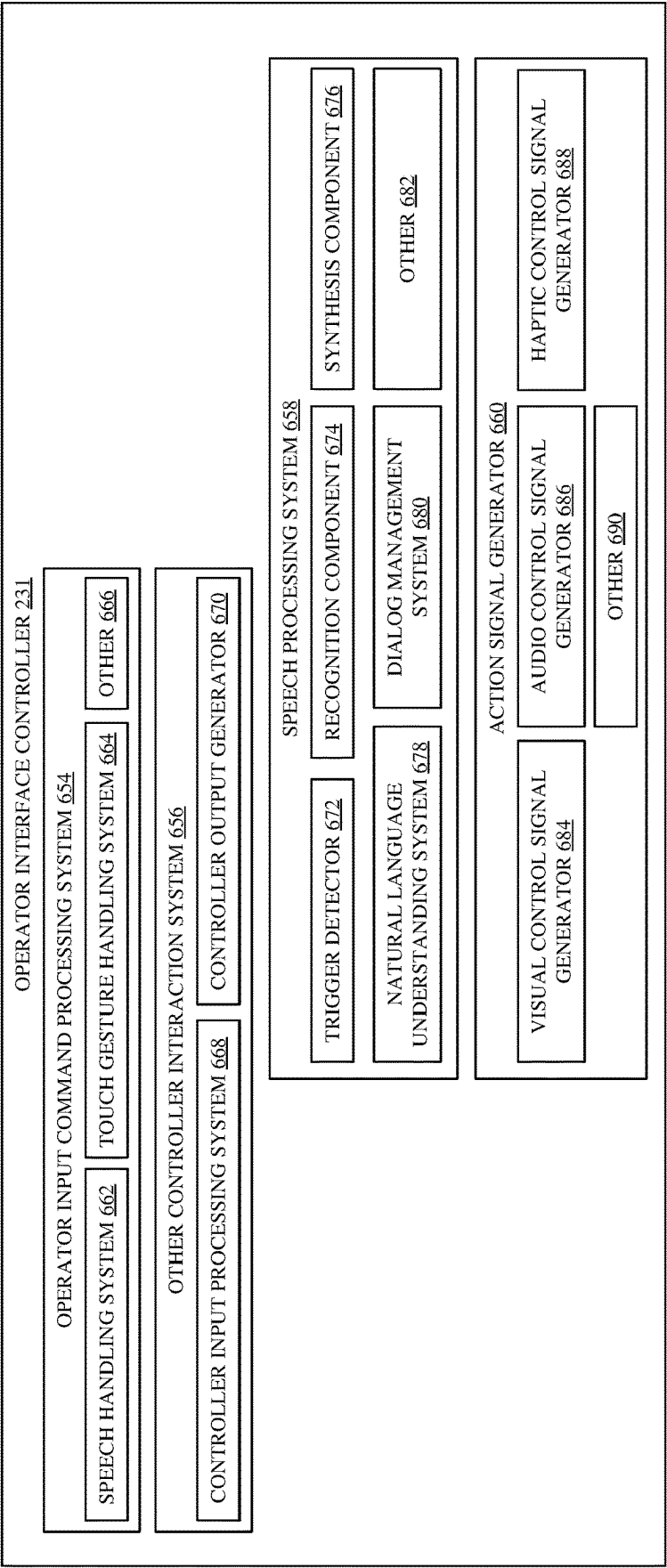
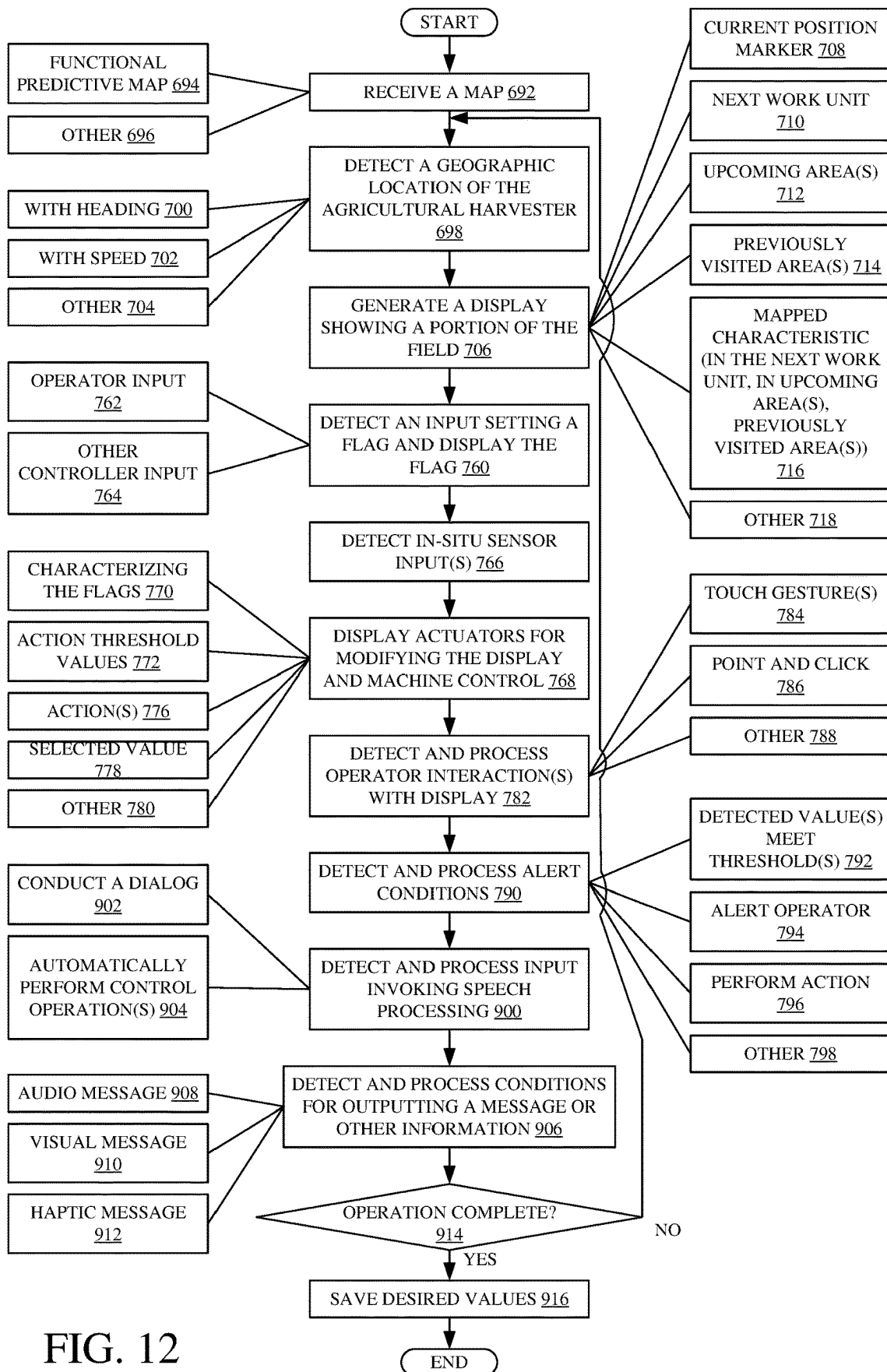


FIG. 11



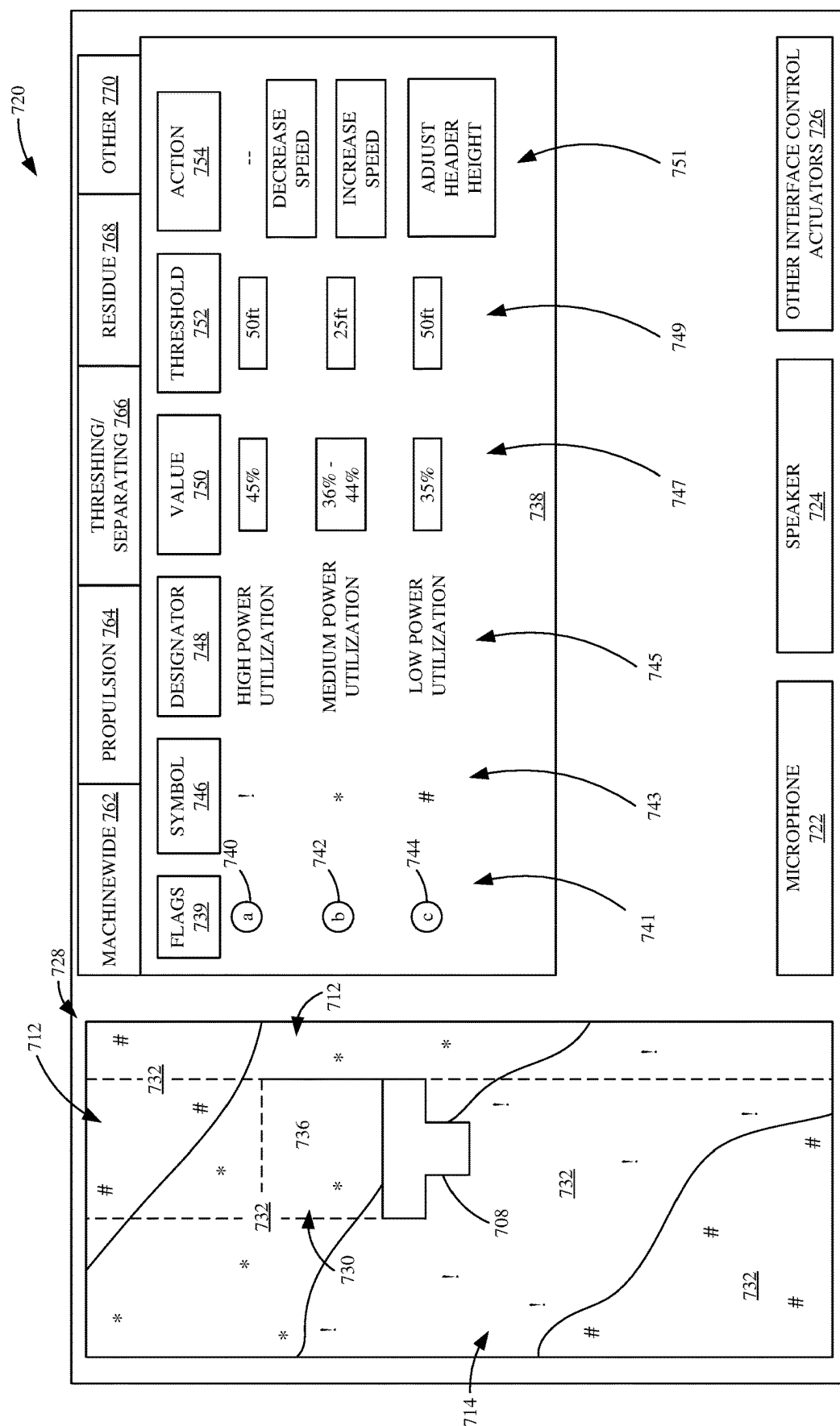


FIG. 13

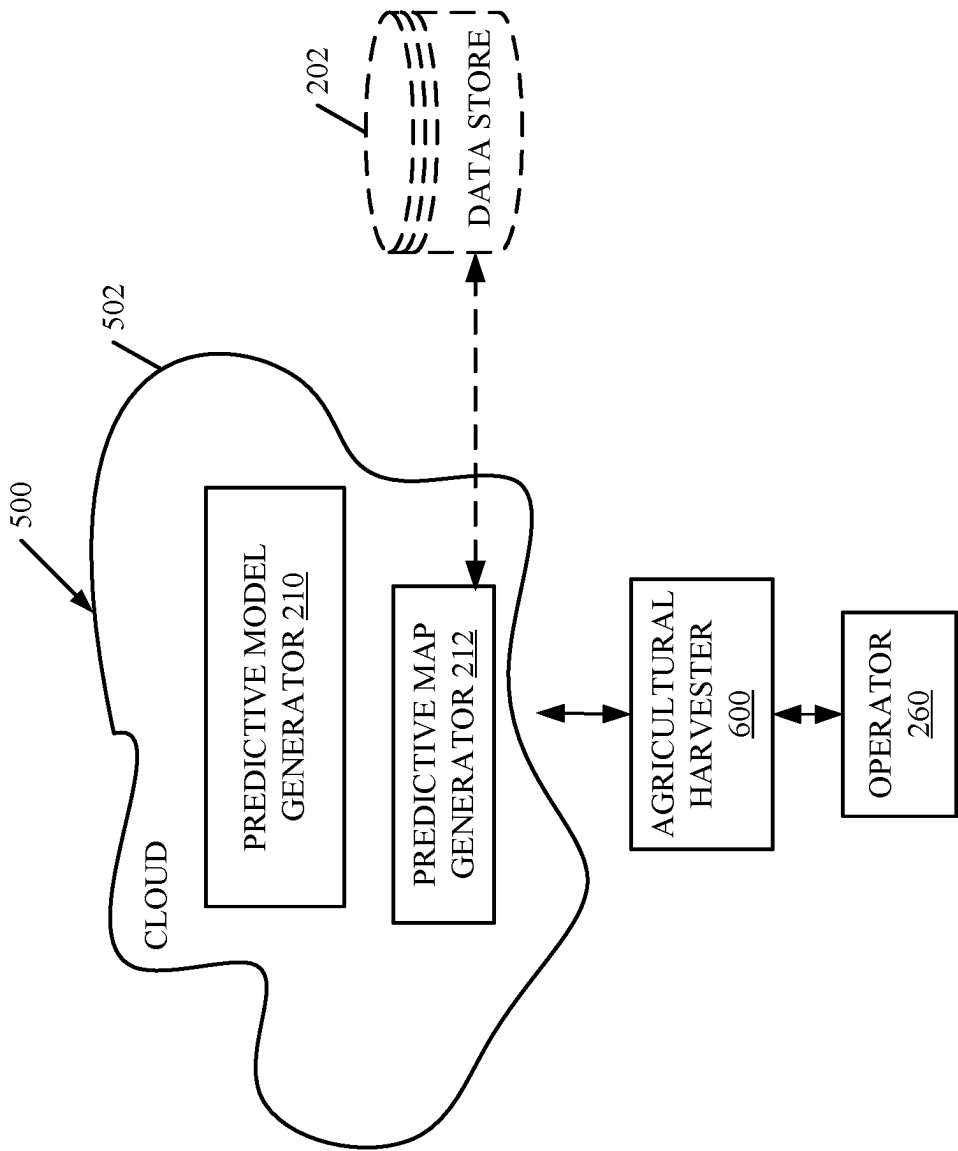


FIG. 14

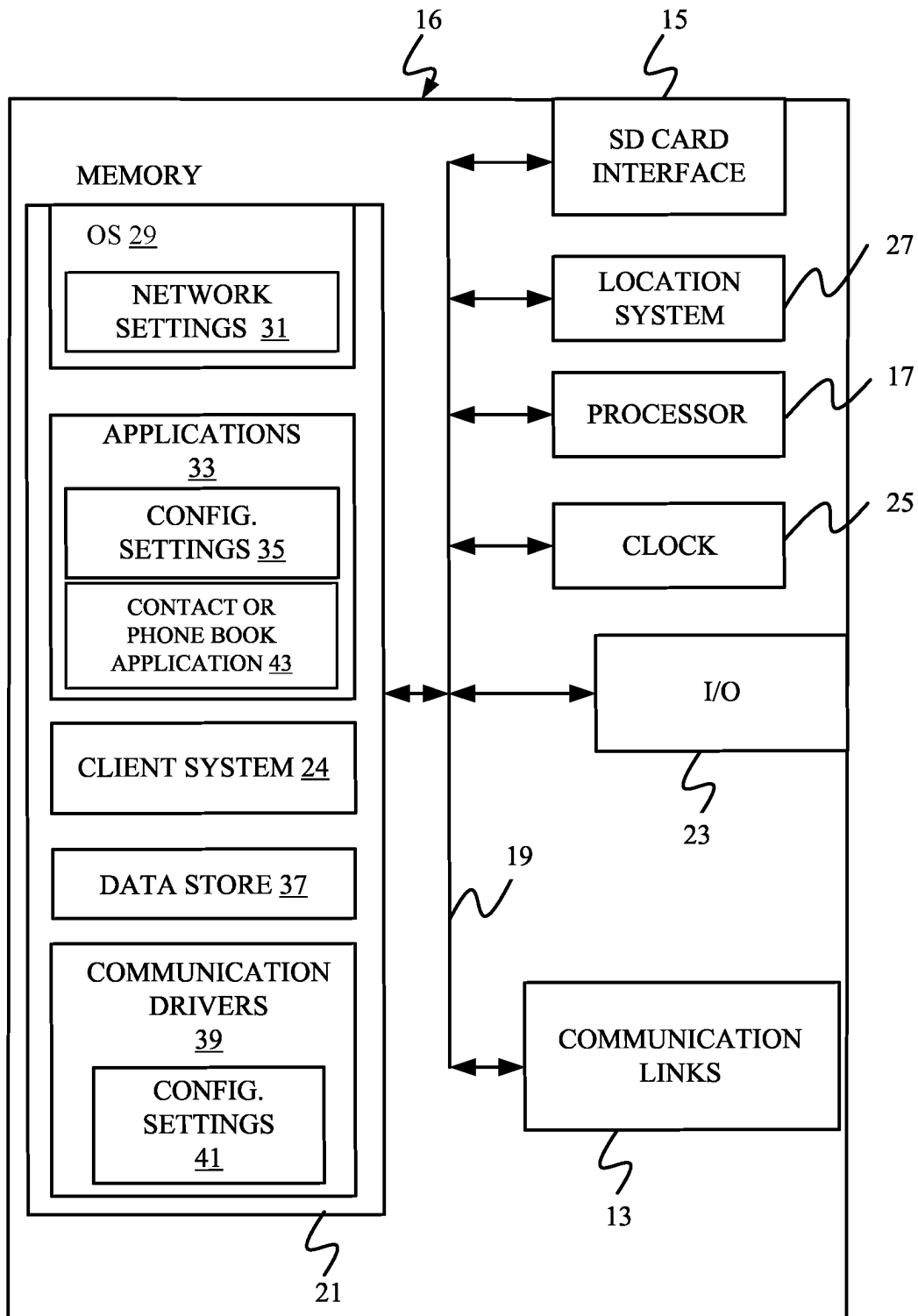


FIG. 15

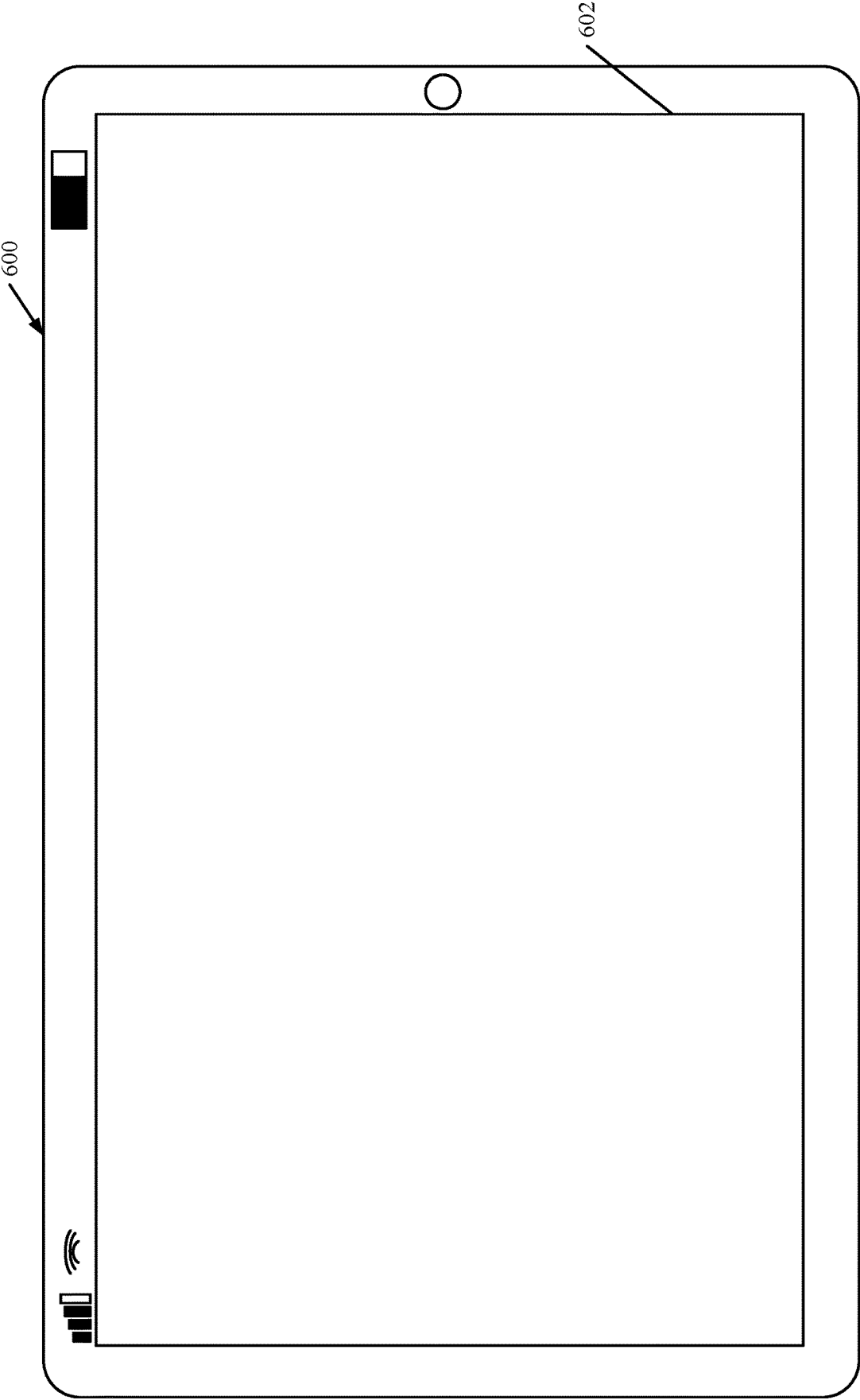


FIG. 16

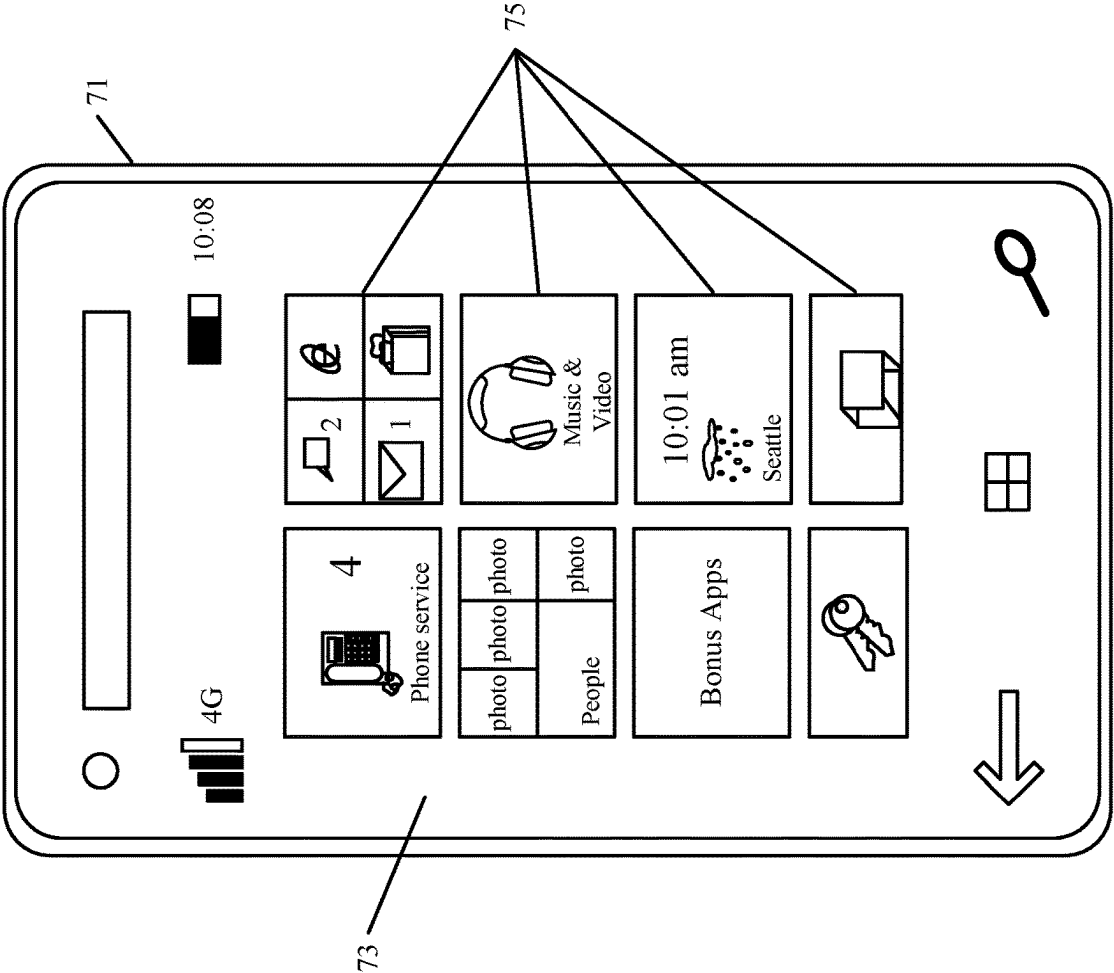


FIG. 17

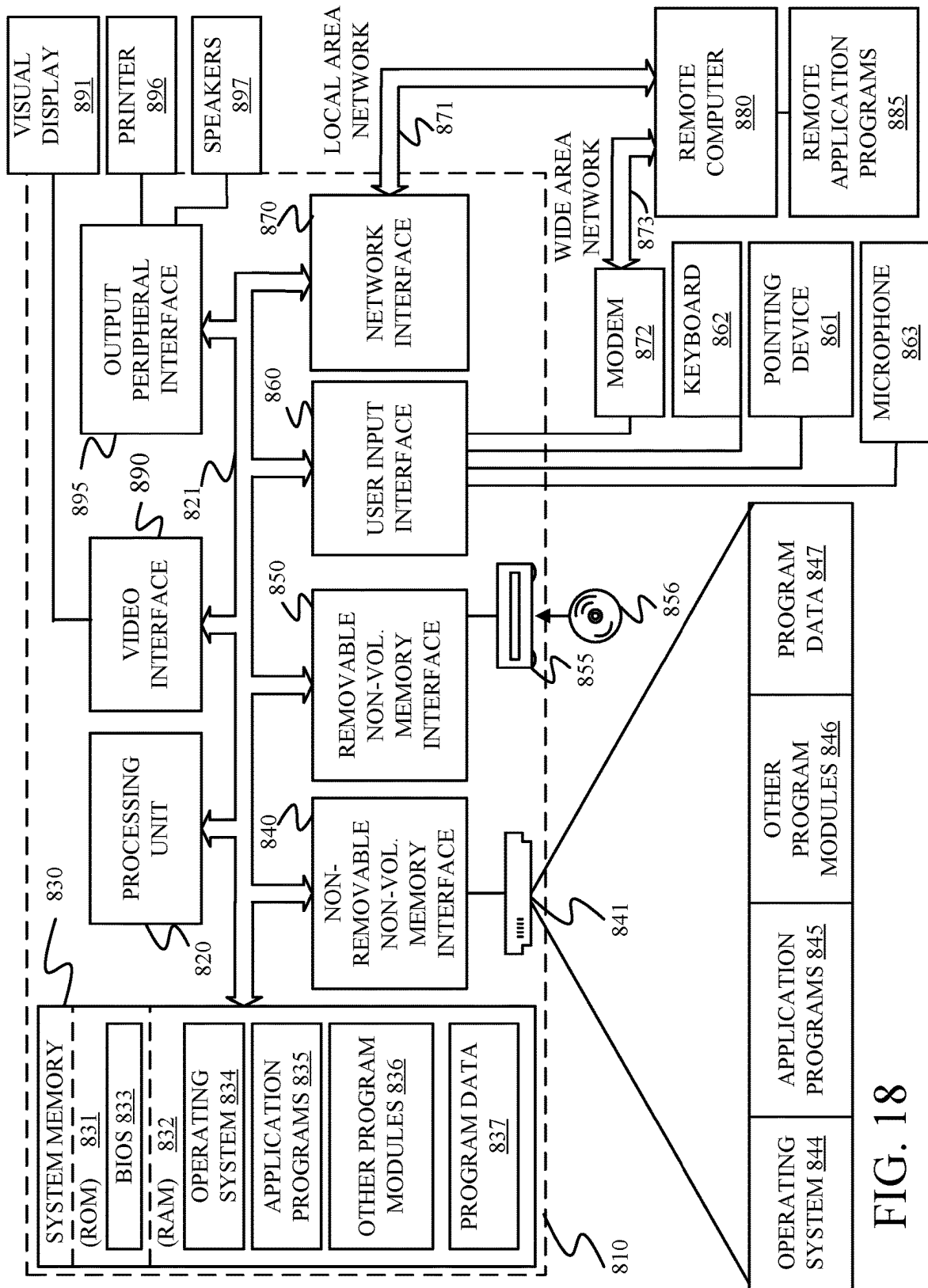


FIG. 18

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PREDICTIVE POWER MAP GENERATION AND CONTROL SYSTEM

FIELD OF THE DESCRIPTION

The present description relates to agricultural machines, forestry machines, construction machines and turf management machines.

BACKGROUND

There are a wide variety of different types of agricultural machines. Some agricultural machines include harvesters, such as combine harvesters, sugar cane harvesters, cotton harvesters, self-propelled forage harvesters, and windrowers. Some harvesters can also be fitted with different types of heads to harvest different types of crops.

Agricultural harvesters typically include an engine or other power source which produces a finite amount of power which is provided to the various subsystems of the agricultural harvester. Maintaining an efficient power distribution to the various subsystems from a finite amount of power can be difficult to attain across changing field conditions.

The discussion above is merely provided for general background information and is not intended to be used as an aid in determining the scope of the claimed subject matter.

SUMMARY

One or more information maps are obtained by an agricultural work machine. The one or more information maps map one or more agricultural characteristic values at different geographic locations of a field. An in-situ sensor on the agricultural work machine senses an agricultural characteristic as the agricultural work machine moves through the field. A predictive map generator generates a predictive map that predicts a predictive agricultural characteristic at different locations in the field based on a relationship between the values in the one or more information maps and the agricultural characteristic sensed by the in-situ sensor. The predictive map can be output and used in automated machine control.

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. The claimed subject matter is not limited to examples that solve any or all disadvantages noted in the background.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a partial pictorial, partial schematic illustration of one example of a combine harvester.

FIG. 2 is a block diagram showing some portions of an agricultural harvester in more detail, according to some examples of the present disclosure.

FIGS. 3A-3B (collectively referred to herein as FIG. 3) show a flow diagram illustrating an example of operation of an agricultural harvester in generating a map.

FIG. 4 is a block diagram showing one example of a predictive model generator and a predictive metric map generator.

FIG. 5 is a flow diagram showing an example of operation of an agricultural harvester in receiving a vegetative index, crop moisture, soil property, topography, predictive yield or

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predictive biomass map, detecting a power characteristic, and generating a functional predictive power map for use in controlling the agricultural harvester during a harvesting operation.

FIG. 6A is a block diagram showing one example of a predictive model generator and a predictive map generator.

FIG. 6B is a block diagram showing some examples of in-situ sensors.

FIG. 7 shows a flow diagram illustrating one example of operation of an agricultural harvester involving generating a functional predictive map using a prior information map and an in-situ sensor input.

FIG. 8 is a block diagram showing one example of a control zone generator.

FIG. 9 is a flow diagram illustrating one example of the operation of the control zone generator shown in FIG. 8.

FIG. 10 illustrates a flow diagram showing an example of operation of a control system in selecting a target settings value to control an agricultural harvester.

FIG. 11 is a block diagram showing one example of an operator interface controller.

FIG. 12 is a flow diagram illustrating one example of an operator interface controller.

FIG. 13 is a pictorial illustration showing one example of an operator interface display.

FIG. 14 is a block diagram showing one example of an agricultural harvester in communication with a remote server environment.

FIGS. 15-17 show examples of mobile devices that can be used in an agricultural harvester.

FIG. 18 is a block diagram showing one example of a computing environment that can be used in an agricultural harvester.

DETAILED DESCRIPTION

For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the examples illustrated in the drawings, and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the disclosure is intended. Any alterations and further modifications to the described devices, systems, methods, and any further application of the principles of the present disclosure are fully contemplated as would normally occur to one skilled in the art to which the disclosure relates. In particular, it is fully contemplated that the features, components, steps, or a combination thereof described with respect to one example may be combined with the features, components, steps, or a combination thereof described with respect to other examples of the present disclosure.

The present description relates to using in-situ data taken concurrently with an agricultural operation, in combination with prior data, to generate a predictive map and, more particularly, a predictive power map. In some examples, the predictive power map can be used to control an agricultural work machine, such as an agricultural harvester. As discussed above, the power generation of a harvester has a finite limit and overall performance may be degraded when one or more subsystems have increased power demands.

Performance of a harvester may be deleteriously affected based on a number of different criteria. For example, areas of dense crop plants, weeds, or combinations thereof, may have deleterious effects on the operation of the harvester because subsystems require more power to process larger amounts of material, which includes crop plants and weeds. Vegetative index may signal where areas of dense crop

plants, weeds, or combinations thereof may exist. Or for example, crop plants or weeds that have higher moisture content also take more power to process. Or for example, soil properties, such as type or moisture, can affect the power usage by the steering and propulsion systems. For instance, wet clay soils can cause additional slippage compared to dry soils which reduces the efficiency of the drive train. Or for example, topography of the field can change the power characteristics of an agricultural harvester. For instance, as the harvester ascends a hill some power needs to be diverted to the propulsion system to maintain a constant speed. Or for example, an area of the field having a higher grain yield may require that more power be diverted to the crop processing subsystems. Or for example, an area of the field containing a large biomass may require that more power be diverted to the crop processing subsystems.

Some current systems provide vegetative index maps. A vegetative index map illustratively maps vegetative index values (which may be indicative of vegetative growth) across different geographic locations in a field of interest. One example of a vegetative index includes a normalized difference vegetation index (NDVI). There are many other vegetative indices that are within the scope of the present disclosure. In some examples, a vegetative index may be derived from sensor readings of one or more bands of electromagnetic radiation reflected by the plants. Without limitations, these bands may be in the microwave, infrared, visible or ultraviolet portions of the electromagnetic spectrum.

A vegetative index map can be used to identify the presence and location of vegetation. In some examples, these maps enable weeds to be identified and georeferenced in the presence of bare soil, crop residue, or other plants, including crop or other weeds. For instance, at the end of a growing season, when a crop is mature, the crop plants may show a relatively low level of live, growing vegetation. However, weeds often persist in a growing state after the maturity of the crop. Therefore, if a vegetative index map is generated relatively late in the growing season, the vegetative index map may be indicative of the location of weeds in the field.

Some current systems provide crop moisture maps. A crop moisture map illustratively maps crop moisture across different geographic locations in a field of interest. In one example, crop moisture can be sensed prior to a harvesting operation by an unmanned aerial vehicle (UAV) equipped with a moisture sensor. As the UAV travels across the field, the crop moisture readings are geolocated to create a crop moisture map. This is an example only and the crop moisture map can be created in other ways as well, for example, the crop moisture across a field can be predicted based on precipitation, soil moisture or combinations thereof.

Some current systems provide topographic maps. A topographic map illustratively maps elevations, or other topographical characteristics of the ground across different geographic locations in a field of interest. Since ground slope is indicative of a change in elevation, having two or more elevation values allows for calculation of slope across the areas having known elevation values. Greater granularity of slope can be accomplished by having more areas with known elevation values. As an agricultural harvester travels across the terrain in known directions, the pitch and roll of the agricultural harvester can be determined based on the slope of the ground (i.e., areas of changing elevation). Topographical characteristics, when referred to below, can include, but are not limited to, the elevation, slope (e.g.,

including the machine orientation relative to the slope), and ground profile (e.g., roughness).

Some current systems provide soil property maps. A soil property map illustratively maps soil property values (which may be indicative of topographic characteristics, soil type, soil moisture, soil cover, soil structure, as well as various other soil properties) across different geographic locations in a field of interest. The soil property maps thus provide geo-referenced soil properties across a field of interest. Topographic characteristics can include, for example, elevation data of the field, including elevation of different locations across the field, for instance, an elevation of a particular location in the field relative to a reference, such as sea level. Topographic characteristics can also include slope data of the field, including slope data of different locations across the field, for instance, slope gradient of a particular location in the field. Topographic characteristics can include various other topographic data. Soil type can refer to taxonomic units in soil science, wherein each soil type includes defined sets of shared properties. Soil types can include, for example, sandy soil, clay soil, silt soil, peat soil, chalk soil, loam soil, and various other soil types. Soil moisture can refer to the amount of water that is held or otherwise contained in the soil. Soil moisture can also be referred to as soil wetness. Soil cover can refer to the amount of items or materials covering the soil, including, vegetation material, such as crop residue or cover crop, debris, as well as various other items or materials. Commonly, in agricultural terms, soil cover includes a measure of remaining crop residue, such as a remaining mass of plant stalks, as well as a measure of cover crop. Soil structure can refer to the arrangement of solid parts of the soil and the pore space located between the solid parts of the soil. Soil structure can include the way in which individual particles, such as individual particles of sand, silt, and clay, are assembled. Soil structure can be described in terms of grade (degree of aggregation), class (average size of aggregates), and form (types of aggregates), as well as a variety of other descriptions. These are merely examples. Various other characteristics and properties of the soil can be mapped as soil property values on a soil property map.

These soil property maps can be generated on the basis of data collected during another operation corresponding to the field of interest, for example, previous agricultural operations in the same season, such as planting operations or spraying operations, as well as previous agricultural operations performed in past seasons, such as a previous harvesting operation. The agricultural machines performing those agricultural operations can have on-board sensors that detect characteristics indicative of soil properties, for example, characteristics indicative of soil type, soil moisture, soil cover, soil structure, as well as various other characteristics indicative of various other soil properties. Additionally, operating characteristics, machine settings, or machine performance characteristics of the agricultural machines during previous operations along with other data can be used to generate a soil property map. For instance, header height data indicative of a height of an agricultural harvester's header across different geographic locations in the field of interest during a previous harvesting operation along with weather data that indicates weather conditions such as precipitation data or wind data during an interim period (such as the period since the time of the previous harvesting operation and the generation of the soil property map) can be used to generate a soil moisture map. For example, by knowing the height of the header, the amount of remaining plant residue, such as crop stalks, can be known or estimated

and, along with precipitation data, a level of soil moisture can be predicted. This is merely an example.

The present discussion also includes predictive maps that predict a characteristic based on a prior information map and a relationship to an in-situ sensor. Two of these maps include a predictive yield map and a predictive biomass map. In one example, the predictive yield map is generated by receiving a prior vegetative index map and sensing a yield during a harvesting operation and determining a relationship between the prior vegetative index map and the yield sensor signal, and using the relationship to generate the predictive yield map based on the relationship and the prior vegetative index map. In one example, the predictive biomass map is generated by receiving a prior vegetative index map and sensing a biomass and determining a relationship between the prior vegetative index map and the biomass sensor signal, and using the relationship to generate the predictive biomass map based on the relationship and the prior vegetative index map. The predictive yield and biomass maps can be created based on other prior information maps or generated in other ways as well. For example, the predictive yield and biomass maps can be generated based on a satellite or growth model.

The present discussion thus proceeds with respect to examples in which a system receives one or more of a vegetative index, crop moisture, soil property, topography, predictive yield or predictive biomass map, and also uses an in-situ sensor to detect a variable indicative of a power characteristic, during a harvesting operation. The system generates a model that models a relationship between the vegetative index values, crop moisture values, soil property values, predictive yield values, or predictive biomass values from the maps and the in-situ data from the in-situ sensor. The model is used to generate a functional predictive power map that predicts an anticipated power characteristic of the agricultural harvester in the field. The functional predictive power characteristic map, generated during the harvesting operation, can be presented to an operator or other user or used in automatically controlling an agricultural harvester during the harvesting operation, or both.

FIG. 1 is a partial pictorial, partial schematic, illustration of a self-propelled agricultural harvester **100**. In the illustrated example, agricultural harvester **100** is a combine harvester. Further, although combine harvesters are provided as examples throughout the present disclosure, it will be appreciated that the present description is also applicable to other types of harvesters, such as cotton harvesters, sugarcane harvesters, self-propelled forage harvesters, windrowers, or other agricultural work machines. Consequently, the present disclosure is intended to encompass the various types of harvesters described and is, thus, not limited to combine harvesters. Moreover, the present disclosure is directed to other types of work machines, such as agricultural seeders and sprayers, construction equipment, forestry equipment, and turf management equipment where generation of a predictive map may be applicable. Consequently, the present disclosure is intended to encompass these various types of harvesters and other work machines and is, thus, not limited to combine harvesters.

As shown in FIG. 1, agricultural harvester **100** illustratively includes an operator compartment **101**, which can have a variety of different operator interface mechanisms, for controlling agricultural harvester **100**. Agricultural harvester **100** includes front-end equipment, such as a header **102**, and a cutter generally indicated at **104**. Agricultural harvester **100** also includes a feeder house **106**, a feed accelerator **108**, and a thresher generally indicated at **110**. The feeder house **106** and the feed accelerator **108** form part

of a material handling subsystem **125**. Header **102** is pivotally coupled to a frame **103** of agricultural harvester **100** along pivot axis **105**. One or more actuators **107** drive movement of header **102** about axis **105** in the direction generally indicated by arrow **109**. Thus, a vertical position of header **102** (the header height) above ground **111** over which the header **102** travels is controllable by actuating actuator **107**. While not shown in FIG. 1, agricultural harvester **100** may also include one or more actuators that operate to apply a tilt angle, a roll angle, or both to the header **102** or portions of header **102**. Tilt refers to an angle at which the cutter **104** engages the crop. The tilt angle is increased, for example, by controlling header **102** to point a distal edge **113** of cutter **104** more toward the ground. The tilt angle is decreased by controlling header **102** to point the distal edge **113** of cutter **104** more away from the ground. The roll angle refers to the orientation of header **102** about the front-to-back longitudinal axis of agricultural harvester **100**.

Thresher **110** illustratively includes a threshing rotor **112** and a set of concaves **114**. Further, agricultural harvester **100** also includes a separator **116**. Agricultural harvester **100** also includes a cleaning subsystem or cleaning shoe (collectively referred to as cleaning subsystem **118**) that includes a cleaning fan **120**, chaffer **122**, and sieve **124**. The material handling subsystem **125** also includes discharge beater **126**, tailings elevator **128**, clean grain elevator **130**, as well as unloading auger **134** and spout **136**. The clean grain elevator moves clean grain into clean grain tank **132**. Agricultural harvester **100** also includes a residue subsystem **138** that can include chopper **140** and spreader **142**. Agricultural harvester **100** also includes a propulsion subsystem that includes an engine that drives ground engaging components **144**, such as wheels or tracks. In some examples, a combine harvester within the scope of the present disclosure may have more than one of any of the subsystems mentioned above. In some examples, agricultural harvester **100** may have left and right cleaning subsystems, separators, etc., which are not shown in FIG. 1.

In operation, and by way of overview, agricultural harvester **100** illustratively moves through a field in the direction indicated by arrow **147**. As agricultural harvester **100** moves, header **102** (and the associated reel **164**) engages the crop to be harvested and gathers the crop toward cutter **104**. An operator of agricultural harvester **100** can be a local human operator, a remote human operator, or an automated system. An operator command is a command from an operator. The operator of agricultural harvester **100** may determine one or more of a height setting, a tilt angle setting, or a roll angle setting for header **102**. For example, the operator inputs a setting or settings to a control system, described in more detail below, that controls actuator **107**. The control system may also receive a setting from the operator for establishing the tilt angle and roll angle of the header **102** and implement the inputted settings by controlling associated actuators, not shown, that operate to change the tilt angle and roll angle of the header **102**. The actuator **107** maintains header **102** at a height above ground **111** based on a height setting and, where applicable, at desired tilt and roll angles. Each of the height, roll, and tilt settings may be implemented independently of the others. The control system responds to header error (e.g., the difference between the height setting and measured height of header **104** above ground **111** and, in some examples, tilt angle and roll angle errors) with a responsiveness that is determined based on a selected sensitivity level. If the sensitivity level is set at a greater level of sensitivity, the control system

responds to smaller header position errors, and attempts to reduce the detected errors more quickly than when the sensitivity is at a lower level of sensitivity.

Returning to the description of the operation of agricultural harvester 100, after crops are cut by cutter 104, the severed crop material is moved through a conveyor in feeder house 106 toward feed accelerator 108, which accelerates the crop material into thresher 110. The crop material is threshed by rotor 112 rotating the crop against concaves 114. The threshed crop material is moved by a separator rotor in separator 116 where a portion of the residue is moved by discharge beater 126 toward the residue subsystem 138. The portion of residue transferred to the residue subsystem 138 is chopped by residue chopper 140 and spread on the field by spreader 142. In other configurations, the residue is released from the agricultural harvester 100 in a windrow. In other examples, the residue subsystem 138 can include weed seed eliminators (not shown) such as seed baggers or other seed collectors, or seed crushers or other seed destroyers.

Grain falls to cleaning subsystem 118. Chaffer 122 separates some larger pieces of material from the grain, and sieve 124 separates some of finer pieces of material from the clean grain. Clean grain falls to an auger that moves the grain to an inlet end of clean grain elevator 130, and the clean grain elevator 130 moves the clean grain upwards, depositing the clean grain in clean grain tank 132. Residue is removed from the cleaning subsystem 118 by airflow generated by cleaning fan 120. Cleaning fan 120 directs air along an airflow path upwardly through the sieves and chaffers. The airflow carries residue rearwardly in agricultural harvester 100 toward the residue handling subsystem 138.

Tailings elevator 128 returns tailings to thresher 110 where the tailings are re-threshed. Alternatively, the tailings also may be passed to a separate re-threshing mechanism by a tailings elevator or another transport device where the tailings are re-threshed as well.

FIG. 1 also shows that, in one example, agricultural harvester 100 includes ground speed sensor 146, one or more separator loss sensors 148, a clean grain camera 150, a forward looking image capture mechanism 151, which may be in the form of a stereo or mono camera, and one or more loss sensors 152 provided in the cleaning subsystem 118.

Ground speed sensor 146 senses the travel speed of agricultural harvester 100 over the ground. Ground speed sensor 146 may sense the travel speed of the agricultural harvester 100 by sensing the speed of rotation of the ground engaging components (such as wheels or tracks), a drive shaft, an axle, or other components. In some instances, the travel speed may be sensed using a positioning system, such as a global positioning system (GPS), a dead reckoning system, a long range navigation (LORAN) system, or a wide variety of other systems or sensors that provide an indication of travel speed.

Loss sensors 152 illustratively provide an output signal indicative of the quantity of grain loss occurring in both the right and left sides of the cleaning subsystem 118. In some examples, sensors 152 are strike sensors which count grain strikes per unit of time or per unit of distance traveled to provide an indication of the grain loss occurring at the cleaning subsystem 118. The strike sensors for the right and left sides of the cleaning subsystem 118 may provide individual signals or a combined or aggregated signal. In some examples, sensors 152 may include a single sensor as opposed to separate sensors provided for each cleaning subsystem 118.

Separator loss sensor 148 provides a signal indicative of grain loss in the left and right separators, not separately shown in FIG. 1. The separator loss sensors 148 may be associated with the left and right separators and may provide separate grain loss signals or a combined or aggregate signal. In some instances, sensing grain loss in the separators may also be performed using a wide variety of different types of sensors as well.

Agricultural harvester 100 may also include other sensors and measurement mechanisms. For instance, agricultural harvester 100 may include one or more of the following sensors: a header height sensor that senses a height of header 102 above ground 111; stability sensors that sense oscillation or bouncing motion (and amplitude) of agricultural harvester 100; a residue setting sensor that is configured to sense whether agricultural harvester 100 is configured to chop the residue, produce a windrow, etc.; a cleaning shoe fan speed sensor to sense the speed of fan 120; a concave clearance sensor that senses clearance between the rotor 112 and concaves 114; a threshing rotor speed sensor that senses a rotor speed of rotor 112; a chaffer clearance sensor that senses the size of openings in chaffer 122; a sieve clearance sensor that senses the size of openings in sieve 124; a material other than grain (MOG) moisture sensor that senses a moisture level of the MOG passing through agricultural harvester 100; one or more machine setting sensors configured to sense various configurable settings of agricultural harvester 100; a machine orientation sensor that senses the orientation of agricultural harvester 100; and crop property sensors that sense a variety of different types of crop properties, such as crop type, crop moisture, and other crop properties. Crop property sensors may also be configured to sense characteristics of the severed crop material as the crop material is being processed by agricultural harvester 100. For example, in some instances, the crop property sensors may sense grain quality such as broken grain, MOG levels; grain constituents such as starches and protein; and grain feed rate as the grain travels through the feeder house 106, clean grain elevator 130, or elsewhere in the agricultural harvester 100. The crop property sensors may also sense the feed rate of biomass through feeder house 106, through the separator 116 or elsewhere in agricultural harvester 100. The crop property sensors may also sense the feed rate as a mass flow rate of grain through elevator 130 or through other portions of the agricultural harvester 100 or provide other output signals indicative of other sensed variables.

Examples of sensors used to detect or sense the power characteristics include, but are not limited to, a voltage sensor, a current sensor, a torque sensor, a hydraulic pressure sensor, a hydraulic flow sensor, a force sensor, a bearing load sensor and a rotational sensor. Power characteristics can be measured at varying levels of granularity. For instance, power usage can be sensed machine-wide, subsystem-wide or by individual components of the subsystems.

Prior to describing how agricultural harvester 100 generates a functional predictive power map, and uses the functional predictive power map for control, a brief description of some of the items on agricultural harvester 100, and their operation, will first be described. The description of FIGS. 2 and 3 describe receiving a general type of prior information map and combining information from the prior information map with a georeferenced sensor signal generated by an in-situ sensor, where the sensor signal is indicative of a characteristic in the field, such as power characteristics of the agricultural harvester. Characteristics of the field may include, but are not limited to, characteristics of a field such as slope, weed intensity, weed type, soil moisture, surface

quality; characteristics of crop properties such as crop height, crop moisture, crop density, crop state; characteristics of grain properties such as grain moisture, grain size, grain test weight; and characteristics of machine performance such as loss levels, job quality, fuel consumption, and power utilization. A relationship between the characteristic values obtained from in-situ sensor signals and the prior information map values is identified, and that relationship is used to generate a new functional predictive map. A functional predictive map predicts values at different geographic locations in a field, and one or more of those values may be used for controlling a machine, such as one or more subsystems of an agricultural harvester. In some instances, a functional predictive map can be presented to a user, such as an operator of an agricultural work machine, which may be an agricultural harvester. A functional predictive map may be presented to a user visually, such as via a display, haptically, or audibly. The user may interact with the functional predictive map to perform editing operations and other user interface operations. In some instances, a functional predictive map can be used for one or more of controlling an agricultural work machine, such as an agricultural harvester, presentation to an operator or other user, and presentation to an operator or user for interaction by the operator or user.

After the general approach is described with respect to FIGS. 2 and 3, a more specific approach for generating a functional predictive power characteristic map that can be presented to an operator or user, or used to control agricultural harvester 100, or both is described with respect to FIGS. 4 and 5. Again, while the present discussion proceeds with respect to the agricultural harvester and, particularly, a combine harvester, the scope of the present disclosure encompasses other types of agricultural harvesters or other agricultural work machines.

FIG. 2 is a block diagram showing some portions of an example agricultural harvester 100. FIG. 2 shows that agricultural harvester 100 illustratively includes one or more processors or servers 201, data store 202, geographic position sensor 204, communication system 206, and one or more in-situ sensors 208 that sense one or more agricultural characteristics of a field concurrent with a harvesting operation. An agricultural characteristic can include any characteristic that can have an effect of the harvesting operation. Some examples of agricultural characteristics include characteristics of the harvesting machine, the field, the plants on the field, and the weather. Other types of agricultural characteristics are also included. An agricultural characteristic can include any characteristic that can have an effect of the harvesting operation. Some examples of agricultural characteristics include characteristics of the harvesting machine, the field, the plants on the field, the weather among others. The in-situ sensors 208 generate values corresponding to the sensed characteristics. The agricultural harvester 100 also includes a predictive model or relationship generator (collectively referred to hereinafter as “predictive model generator 210”), predictive map generator 212, control zone generator 213, control system 214, one or more controllable subsystems 216, and an operator interface mechanism 218. The agricultural harvester 100 can also include a wide variety of other agricultural harvester functionality 220. The in-situ sensors 208 include, for example, on-board sensors 222, remote sensors 224, and other sensors 226 that sense characteristics of a field during the course of an agricultural operation. Predictive model generator 210 illustratively includes a prior information variable-to-in-situ variable model generator 228, and predictive model generator 210

can include other items 230. Control system 214 includes communication system controller 229, operator interface controller 231, a settings controller 232, path planning controller 234, feed rate controller 236, header and reel controller 238, draper belt controller 240, deck plate position controller 242, residue system controller 244, machine cleaning controller 245, zone controller 247, and system 214 can include other items 246. Controllable subsystems 216 include machine and header actuators 248, propulsion subsystem 250, steering subsystem 252, residue subsystem 138, machine cleaning subsystem 254, and subsystems 216 can include a wide variety of other subsystems 256.

FIG. 2 also shows that agricultural harvester 100 can receive prior information map 258. As described below, the prior information map 258 includes, for example, a vegetative index map or a vegetation map from a prior operation. However, prior information map 258 may also encompass other types of data that were obtained prior to a harvesting operation or a map from a prior operation. FIG. 2 also shows that an operator 260 may operate the agricultural harvester 100. The operator 260 interacts with operator interface mechanisms 218. In some examples, operator interface mechanisms 218 may include joysticks, levers, a steering wheel, linkages, pedals, buttons, dials, keypads, user actuable elements (such as icons, buttons, etc.) on a user interface display device, a microphone and speaker (where speech recognition and speech synthesis are provided), among a wide variety of other types of control devices. Where a touch sensitive display system is provided, operator 260 may interact with operator interface mechanisms 218 using touch gestures. These examples described above are provided as illustrative examples and are not intended to limit the scope of the present disclosure. Consequently, other types of operator interface mechanisms 218 may be used and are within the scope of the present disclosure.

Prior information map 258 may be downloaded onto agricultural harvester 100 and stored in data store 202, using communication system 206 or in other ways. In some examples, communication system 206 may be a cellular communication system, a system for communicating over a wide area network or a local area network, a system for communicating over a near field communication network, or a communication system configured to communicate over any of a variety of other networks or combinations of networks. Communication system 206 may also include a system that facilitates downloads or transfers of information to and from a secure digital (SD) card or a universal serial bus (USB) card or both.

Geographic position sensor 204 illustratively senses or detects the geographic position or location of agricultural harvester 100. Geographic position sensor 204 can include, but is not limited to, a global navigation satellite system (GNSS) receiver that receives signals from a GNSS satellite transmitter. Geographic position sensor 204 can also include a real-time kinematic (RTK) component that is configured to enhance the precision of position data derived from the GNSS signal. Geographic position sensor 204 can include a dead reckoning system, a cellular triangulation system, or any of a variety of other geographic position sensors.

In-situ sensors 208 may be any of the sensors described above with respect to FIG. 1. In-situ sensors 208 include on-board sensors 222 that are mounted on-board agricultural harvester 100. Such sensors may include, for instance, a perception sensor (e.g., a forward looking mono or stereo camera system and image processing system), image sensors that are internal to agricultural harvester 100 (such as the clean grain camera or cameras mounted to identify weed

seeds that are exiting agricultural harvester 100 through the residue subsystem or from the cleaning subsystem). The in-situ sensors 208 also include remote in-situ sensors 224 that capture in-situ information. In-situ data include data taken from a sensor on-board the harvester or taken by any sensor where the data are detected during the harvesting operation.

Predictive model generator 210 generates a model that is indicative of a relationship between the values sensed by the in-situ sensor 208 and a metric mapped to the field by the prior information map 258. For example, if the prior information map 258 maps a vegetative index value to different locations in the field, and the in-situ sensor 208 is sensing a value indicative of header power usage, then prior information variable-to-in-situ variable model generator 228 generates a predictive power model that models the relationship between the vegetative index value and the header power usage value. The predictive power model can also be generated based on vegetative index values from the prior information map 258 and multiple in-situ data values generated by in-situ sensors 208. Then, predictive map generator 212 uses the predictive power model generated by predictive model generator 210 to generate a functional predictive power map that predicts the value of a power characteristic, such as power usage by a subsystem, sensed by the in-situ sensors 208 at different locations in the field based upon the prior information map 258.

In some examples, the type of values in the functional predictive map 263 may be the same as the in-situ data type sensed by the in-situ sensors 208. In some instances, the type of values in the functional predictive map 263 may have different units from the data sensed by the in-situ sensors 208. In some examples, the type of values in the functional predictive map 263 may be different from the data type sensed by the in-situ sensors 208 but have a relationship to the type of data type sensed by the in-situ sensors 208. For example, in some examples, the data type sensed by the in-situ sensors 208 may be indicative of the type of values in the functional predictive map 263. In some examples, the type of data in the functional predictive map 263 may be different than the data type in the prior information map 258. In some instances, the type of data in the functional predictive map 263 may have different units from the data in the prior information map 258. In some examples, the type of data in the functional predictive map 263 may be different from the data type in the prior information map 258 but has a relationship to the data type in the prior information map 258. For example, in some examples, the data type in the prior information map 258 may be indicative of the type of data in the functional predictive map 263. In some examples, the type of data in the functional predictive map 263 is different than one of, or both of the in-situ data type sensed by the in-situ sensors 208 and the data type in the prior information map 258. In some examples, the type of data in the functional predictive map 263 is the same as one of, or both of, of the in-situ data type sensed by the in-situ sensors 208 and the data type in prior information map 258. In some examples, the type of data in the functional predictive map 263 is the same as one of the in-situ data type sensed by the in-situ sensors 208 or the data type in the prior information map 258, and different than the other.

Continuing with the preceding example, in which prior information map 258 is a vegetative index map and in-situ sensor 208 senses a value indicative of header power usage, predictive map generator 212 can use the vegetative index values in prior information map 258, and the model generated by predictive model generator 210, to generate a

functional predictive map 263 that predicts the header power usage at different locations in the field. Predictive map generator 212 thus outputs predictive map 264.

As shown in FIG. 2, predictive map 264 predicts the value of a sensed characteristic (sensed by in-situ sensors 208), or a characteristic related to the sensed characteristic, at various locations across the field based upon a prior information value in prior information map 258 at those locations and the predictive model. For example, if predictive model generator 210 has generated a predictive model indicative of a relationship between a vegetative index value and header power usage, then, given the vegetative index value at different locations across the field, predictive map generator 212 generates a predictive map 264 that predicts the value of the header power usage at different locations across the field. The vegetative index value, obtained from the vegetative index map, at those locations and the relationship between vegetative index value and header power usage, obtained from the predictive model, are used to generate the predictive map 264.

Some variations in the data types that are mapped in the prior information map 258, the data types sensed by in-situ sensors 208, and the data types predicted on the predictive map 264 will now be described.

In some examples, the data type in the prior information map 258 is different from the data type sensed by in-situ sensors 208, yet the data type in the predictive map 264 is the same as the data type sensed by the in-situ sensors 208. For instance, the prior information map 258 may be a vegetative index map, and the variable sensed by the in-situ sensors 208 may be a power characteristic. The predictive map 264 may then be a predictive power characteristic map that maps predicted power characteristic values to different geographic locations in the field. predictive map 264

Also, in some examples, the data type in the prior information map 258 is different from the data type sensed by in-situ sensors 208, and the data type in the predictive map 264 is different from both the data type in the prior information map 258 and the data type sensed by the in-situ sensors 208. For instance, the prior information map 258 may be a topographic map, and the variable sensed by the in-situ sensors 208 may be a thresher power characteristic. The predictive map 264 may then be a predictive biomass map that maps predicted biomass values to different geographic locations in the field. predictive map 264

In some examples, the prior information map 258 is from a prior pass through the field during a prior operation and the data type is different from the data type sensed by in-situ sensors 208, yet the data type in the predictive map 264 is the same as the data type sensed by the in-situ sensors 208. For instance, the prior information map 258 may be a topographic map generated during planting or spraying, and the variable sensed by the in-situ sensors 208 may be a power characteristic. The predictive map 264 may then be a predictive power characteristic map that maps predicted power characteristic values to different geographic locations in the field. predictive map 264

In some examples, the prior information map 258 is from a prior pass through the field during a prior operation and the data type is the same as the data type sensed by in-situ sensors 208, and the data type in the predictive map 264 is also the same as the data type sensed by the in-situ sensors 208. For instance, the prior information map 258 may be a power characteristic map generated during a previous year, and the variable sensed by the in-situ sensors 208 may be a power characteristic. The predictive map 264 may then be a predictive power characteristic map that maps predicted

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power characteristic values to different geographic locations in the field. In such an example, the relative power characteristic differences in the georeferenced prior information map **258** from the prior year can be used by predictive model generator **210** to generate a predictive model that models a relationship between the relative power characteristic differences on the prior information map **258** and the power characteristic values sensed by in-situ sensors **208** during the current harvesting operation. The predictive model is then used by predictive map generator **210** to generate a predictive power characteristic map.

In another example, the prior information map **258** may be a threshing/separating subsystem power usage map generated during a prior operation, and the variable sensed by the in-situ sensors **208** may be threshing/separating subsystems power usage. The predictive map **264** may then be a predictive threshing/separating subsystems power usage map that maps predicted threshing/separating subsystems power usage values to different geographic locations in the field.

In some examples, predictive map **264** can be provided to the control zone generator **213**. Control zone generator **213** groups adjacent portions of an area into one or more control zones based on data values of predictive map **264** that are associated with those adjacent portions. A control zone may include two or more contiguous portions of an area, such as a field, for which a control parameter corresponding to the control zone for controlling a controllable subsystem is constant. For example, a response time to alter a setting of controllable subsystems **216** may be inadequate to satisfactorily respond to changes in values contained in a map, such as predictive map **264**. In that case, control zone generator **213** parses the map and identifies control zones that are of a defined size to accommodate the response time of the controllable subsystems **216**. In another example, control zones may be sized to reduce wear from excessive actuator movement resulting from continuous adjustment. In some examples, there may be a different set of control zones for each controllable subsystem **216** or for groups of controllable subsystems **216**. The control zones may be added to the predictive map **264** to obtain predictive control zone map **265**. Predictive control zone map **265** can thus be similar to predictive map **264** except that predictive control zone map **265** includes control zone information defining the control zones. Thus, a functional predictive map **263**, as described herein, may or may not include control zones. Both predictive map **264** and predictive control zone map **265** are functional predictive maps **263**. In one example, a functional predictive map **263** does not include control zones, such as predictive map **264**. In another example, a functional predictive map **263** does include control zones, such as predictive control zone map **265**. In some examples, multiple crops may be simultaneously present in a field if an intercrop production system is implemented. In that case, predictive map generator **212** and control zone generator **213** are able to identify the location and characteristics of the two or more crops and then generate predictive map **264** and predictive control zone map **265** accordingly.

It will also be appreciated that control zone generator **213** can cluster values to generate control zones and the control zones can be added to predictive control zone map **265**, or a separate map, showing only the control zones that are generated. In some examples, the control zones may be used for controlling or calibrating agricultural harvester **100** or both. In other examples, the control zones may be presented to the operator **260** and used to control or calibrate agricul-

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tural harvester **100**, and, in other examples, the control zones may be presented to the operator **260** or another user or stored for later use.

Predictive map **264** or predictive control zone map **265** or both are provided to control system **214**, which generates control signals based upon the predictive map **264** or predictive control zone map **265** or both. In some examples, communication system controller **229** controls communication system **206** to communicate the predictive map **264** or predictive control zone map **265** or control signals based on the predictive map **264** or predictive control zone map **265** to other agricultural harvesters that are harvesting in the same field. In some examples, communication system controller **229** controls the communication system **206** to send the predictive map **264**, predictive control zone map **265**, or both to other remote systems.

Operator interface controller **231** is operable to generate control signals to control operator interface mechanisms **218**. The operator interface controller **231** is also operable to present the predictive map **264** or predictive control zone map **265** or other information derived from or based on the predictive map **264**, predictive control zone map **265**, or both to operator **260**. Operator **260** may be a local operator or a remote operator. As an example, controller **231** generates control signals to control a display mechanism to display one or both of predictive map **264** and predictive control zone map **265** for the operator **260**. Controller **231** may generate operator actuatable mechanisms that are displayed and can be actuated by the operator to interact with the displayed map. The operator can edit the map by, for example, correcting a power characteristic displayed on the map, based on the operator's observation. Settings controller **232** can generate control signals to control various settings on the agricultural harvester **100** based upon predictive map **264**, the predictive control zone map **265**, or both. For instance, settings controller **232** can generate control signals to control machine and header actuators **248**. In response to the generated control signals, the machine and header actuators **248** operate to control, for example, one or more of the sieve and chaffer settings, concave clearance, rotor settings, cleaning fan speed settings, header height, header functionality, reel speed, reel position, draper functionality (where agricultural harvester **100** is coupled to a draper header), corn header functionality, internal distribution control and other actuators **248** that affect the other functions of the agricultural harvester **100**. Path planning controller **234** illustratively generates control signals to control steering subsystem **252** to steer agricultural harvester **100** according to a desired path. Path planning controller **234** can control a path planning system to generate a route for agricultural harvester **100** and can control propulsion subsystem **250** and steering subsystem **252** to steer agricultural harvester **100** along that route. Feed rate controller **236** can control various subsystems, such as propulsion subsystem **250** and machine actuators **248**, to control a feed rate based upon the predictive map **264** or predictive control zone map **265** or both. For instance, as agricultural harvester **100** approaches an area having a predicted subsystem power usage value above a selected threshold, feed rate controller **236** may reduce the speed of agricultural harvester **100** to maintain power allocation to the predicted power usage requirements of the one or more subsystems. Header and reel controller **238** can generate control signals to control a header or a reel or other header functionality. Draper belt controller **240** can generate control signals to control a draper belt or other draper functionality based upon the predictive map **264**, predictive control zone map **265**, or both. Deck plate position control-

ler 242 can generate control signals to control a position of a deck plate included on a header based on predictive map 264 or predictive control zone map 265 or both, and residue system controller 244 can generate control signals to control a residue subsystem 138 based upon predictive map 264 or predictive control zone map 265, or both. Machine cleaning controller 245 can generate control signals to control machine cleaning subsystem 254. For instance, based upon the different predicted power usage of agricultural harvester 100, a particular type of machine cleaning operation or a frequency with which a cleaning operation is performed may be controlled. Other controllers included on the agricultural harvester 100 can control other subsystems based on the predictive map 264 or predictive control zone map 265 or both as well.

FIGS. 3A and 3B (collectively referred to herein as FIG. 3) show a flow diagram illustrating one example of the operation of agricultural harvester 100 in generating a predictive map 264 and predictive control zone map 265 based upon prior information map 258.

At 280, agricultural harvester 100 receives prior information map 258. Examples of prior information map 258 or receiving prior information map 258 are discussed with respect to blocks 281, 282, 284 and 286. As discussed above, prior information map 258 maps values of a variable, corresponding to a first characteristic, to different locations in the field, as indicated at block 282. As indicated at block 281, receiving the prior information map 258 may involve selecting one or more of a plurality of possible prior information maps that are available. For instance, one prior information map may be a vegetative index map generated from aerial imagery. Another prior information map may be a map generated during a prior pass through the field which may have been performed by a different machine performing a previous operation in the field, such as a sprayer or other machine. The process by which one or more prior information maps are selected can be manual, semi-automated, or automated. The prior information map 258 is based on data collected prior to a current harvesting operation. This is indicated by block 284. For instance, the data may be collected based on aerial images taken during a previous year, or earlier in the current growing season, or at other times. As indicated by block 285, the prior information map can be a predictive map that predicts a characteristic based on a prior information map and a relationship to an in-situ sensor. A process of generating a predictive map is presented in FIG. 5. This process could also be performed with other sensors and other prior maps to generate, for example, predictive yield maps or predictive biomass maps. These predictive maps can be used as prior maps in other predictive processes, as indicated by block 285. The data may be based on data detected in ways other than using aerial images. For instance, the data for the prior information map 258 can be transmitted to agricultural harvester 100 using communication system 206 and stored in data store 202. The data for the prior information map 258 can be provided to agricultural harvester 100 using communication system 206 in other ways as well, and this is indicated by block 286 in the flow diagram of FIG. 3. In some examples, the prior information map 258 can be received by communication system 206.

Upon commencement of a harvesting operation, in-situ sensors 208 generate sensor signals indicative of one or more in-situ data values indicative of a characteristic, for example, a power characteristic, such as a power usage by one or more subsystems, as indicated by block 288. Examples of in-situ sensors 288 are discussed with respect to blocks 222, 290, and 226. As explained above, the in-situ

sensors 208 include on-board sensors 222; remote in-situ sensors 224, such as UAV-based sensors flown at a time to gather in-situ data, shown in block 290; or other types of in-situ sensors, designated by in-situ sensors 226. In some examples, data from on-board sensors is georeferenced using position, heading, or speed data from geographic position sensor 204.

Predictive model generator 210 controls the prior information variable-to-in-situ variable model generator 228 to generate a model that models a relationship between the mapped values contained in the prior information map 258 and the in-situ values sensed by the in-situ sensors 208 as indicated by block 292. The characteristics or data types represented by the mapped values in the prior information map 258 and the in-situ values sensed by the in-situ sensors 208 may be the same characteristics or data type or different characteristics or data types.

The relationship or model generated by predictive model generator 210 is provided to predictive map generator 212. Predictive map generator 212 generates a predictive map 264 that predicts a value of the characteristic sensed by the in-situ sensors 208 at different geographic locations in a field being harvested, or a different characteristic that is related to the characteristic sensed by the in-situ sensors 208, using the predictive model and the prior information map 258, as indicated by block 294.

It should be noted that, in some examples, the prior information map 258 may include two or more different maps or two or more different map layers of a single map. Each map layer may represent a different data type from the data type of another map layer or the map layers may have the same data type that were obtained at different times. Each map in the two or more different maps or each layer in the two or more different map layers of a map maps a different type of variable to the geographic locations in the field. In such an example, predictive model generator 210 generates a predictive model that models the relationship between the in-situ data and each of the different variables mapped by the two or more different maps or the two or more different map layers. Similarly, the in-situ sensors 208 can include two or more sensors each sensing a different type of variable. Thus, the predictive model generator 210 generates a predictive model that models the relationships between each type of variable mapped by the prior information map 258 and each type of variable sensed by the in-situ sensors 208. Predictive map generator 212 can generate a functional predictive map 263 that predicts a value for each sensed characteristic sensed by the in-situ sensors 208 (or a characteristic related to the sensed characteristic) at different locations in the field being harvested using the predictive model and each of the maps or map layers in the prior information map 258.

Predictive map generator 212 configures the predictive map 264 so that the predictive map 264 is actionable (or consumable) by control system 214. Predictive map generator 212 can provide the predictive map 264 to the control system 214 or to control zone generator 213 or both. Some examples of different ways in which the predictive map 264 can be configured or output are described with respect to blocks 296, 295, 299 and 297. For instance, predictive map generator 212 configures predictive map 264 so that predictive map 264 includes values that can be read by control system 214 and used as the basis for generating control signals for one or more of the different controllable subsystems of the agricultural harvester 100, as indicated by block 296.

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Control zone generator **213** can divide the predictive map **264** into control zones based on the values on the predictive map **264**. Contiguously-geolocated values that are within a threshold value of one another can be grouped into a control zone. The threshold value can be a default threshold value, or the threshold value can be set based on an operator input, based on an input from an automated system, or based on other criteria. A size of the zones may be based on a responsiveness of the control system **214**, the controllable subsystems **216**, based on wear considerations, or on other criteria as indicated by block **295**. Predictive map generator **212** configures predictive map **264** for presentation to an operator or other user. Control zone generator **213** can configure predictive control zone map **265** for presentation to an operator or other user. This is indicated by block **299**. When presented to an operator or other user, the presentation of the predictive map **264** or predictive control zone map **265** or both may contain one or more of the predictive values on the predictive map **264** correlated to geographic location, the control zones on predictive control zone map **265** correlated to geographic location, and settings values or control parameters that are used based on the predicted values on map **264** or zones on predictive control zone map **265**. The presentation can, in another example, include more abstracted information or more detailed information. The presentation can also include a confidence level that indicates an accuracy with which the predictive values on predictive map **264** or the zones on predictive control zone map **265** conform to measured values that may be measured by sensors on agricultural harvester **100** as agricultural harvester **100** moves through the field. Further where information is presented to more than one location, an authentication and authorization system can be provided to implement authentication and authorization processes. For instance, there may be a hierarchy of individuals that are authorized to view and change maps and other presented information. By way of example, an on-board display device may show the maps in near real time locally on the machine, or the maps may also be generated at one or more remote locations, or both. In some examples, each physical display device at each location may be associated with a person or a user permission level. The user permission level may be used to determine which display markers are visible on the physical display device and which values the corresponding person may change. As an example, a local operator of agricultural harvester **100** may be unable to see the information corresponding to the predictive map **264** or make any changes to machine operation. A supervisor, such as a supervisor at a remote location, however, may be able to see the predictive map **264** on the display but be prevented from making any changes. A manager, who may be at a separate remote location, may be able to see all of the elements on predictive map **264** and also be able to change the predictive map **264**. In some instances, the predictive map **264** accessible and changeable by a manager located remotely may be used in machine control. This is one example of an authorization hierarchy that may be implemented. The predictive map **264** or predictive control zone map **265** or both can be configured in other ways as well, as indicated by block **297**.

At block **298**, input from geographic position sensor **204** and other in-situ sensors **208** are received by the control system. Particularly, at block **300**, control system **214** detects an input from the geographic position sensor **204** identifying a geographic location of agricultural harvester **100**. Block **302** represents receipt by the control system **214** of sensor inputs indicative of trajectory or heading of agricultural harvester **100**, and block **304** represents receipt

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by the control system **214** of a speed of agricultural harvester **100**. Block **306** represents receipt by the control system **214** of other information from various in-situ sensors **208**.

At block **308**, control system **214** generates control signals to control the controllable subsystems **216** based on the predictive map **264** or predictive control zone map **265** or both and the input from the geographic position sensor **204** and any other in-situ sensors **208**. At block **310**, control system **214** applies the control signals to the controllable subsystems. It will be appreciated that the particular control signals that are generated, and the particular controllable subsystems **216** that are controlled, may vary based upon one or more different things. For example, the control signals that are generated and the controllable subsystems **216** that are controlled may be based on the type of predictive map **264** or predictive control zone map **265** or both that is being used. Similarly, the control signals that are generated and the controllable subsystems **216** that are controlled and the timing of the control signals can be based on various latencies of crop flow through the agricultural harvester **100** and the responsiveness of the controllable subsystems **216**.

By way of example, a generated predictive map **264** in the form of a predictive power map can be used to control one or more subsystems **216**. For instance, the predictive power map can include power usage requirement values georeferenced to locations within the field being harvested. The power usage requirement values from the predictive power map can be extracted and used to control the steering and propulsion subsystems **252** and **250**. By controlling the steering and propulsion subsystems **252** and **250**, a feed rate of material moving through the agricultural harvester **100** can be controlled. Similarly, the header height can be controlled to take in more or less material, and, thus, the header height can also be controlled to control feed rate of material through the agricultural harvester **100**. In other examples, if the predictive map **264** maps predicted header power usage to positions in the field, power allocation to the header can be implemented. For example, if the values present in the predictive power map indicate one or more areas having higher power usage requirements for the header subsystems, then header and reel controller **238** can allocate more power from the engine to the header subsystems, this may require allocating less power to other subsystems, such as by reducing speed and reducing power to the propulsion subsystem. The preceding example involving header control using a predictive power map is provided merely as an example. Consequently, a wide variety of other control signals can be generated using values obtained from a predictive power map or other type of predictive map to control one or more of the controllable subsystems **216**.

At block **312**, a determination is made as to whether the harvesting operation has been completed. If harvesting is not completed, the processing advances to block **314** where in-situ sensor data from geographic position sensor **204** and in-situ sensors **208** (and perhaps other sensors) continue to be read.

In some examples, at block **316**, agricultural harvester **100** can also detect learning trigger criteria to perform machine learning on one or more of the predictive map **264**, predictive control zone map **265**, the model generated by predictive model generator **210**, the zones generated by control zone generator **213**, one or more control algorithms implemented by the controllers in the control system **214**, and other triggered learning.

The learning trigger criteria can include any of a wide variety of different criteria. Some examples of detecting trigger criteria are discussed with respect to blocks **318**, **320**,

321, 322 and 324. For instance, in some examples, triggered learning can involve recreation of a relationship used to generate a predictive model when a threshold amount of in-situ sensor data are obtained from in-situ sensors **208**. In such examples, receipt of an amount of in-situ sensor data from the in-situ sensors **208** that exceeds a threshold triggers or causes the predictive model generator **210** to generate a new predictive model that is used by predictive map generator **212**. Thus, as agricultural harvester **100** continues a harvesting operation, receipt of the threshold amount of in-situ sensor data from the in-situ sensors **208** triggers the creation of a new relationship represented by a predictive model generated by predictive model generator **210**. Further, new predictive map **264**, predictive control zone map **265**, or both can be regenerated using the new predictive model. Block **318** represents detecting a threshold amount of in-situ sensor data used to trigger creation of a new predictive model.

In other examples, the learning trigger criteria may be based on how much the in-situ sensor data from the in-situ sensors **208** are changing, such as over time or compared to previous values. For example, if variations within the in-situ sensor data (or the relationship between the in-situ sensor data and the information in prior information map **258**) are within a selected range or is less than a defined amount, or below a threshold value, then a new predictive model is not generated by the predictive model generator **210**. As a result, the predictive map generator **212** does not generate a new predictive map **264**, predictive control zone map **265**, or both. However, if variations within the in-situ sensor data are outside of the selected range, are greater than the defined amount, or are above the threshold value, for example, then the predictive model generator **210** generates a new predictive model using all or a portion of the newly received in-situ sensor data that the predictive map generator **212** uses to generate a new predictive map **264**. At block **320**, variations in the in-situ sensor data, such as a magnitude of an amount by which the data exceeds the selected range or a magnitude of the variation of the relationship between the in-situ sensor data and the information in the prior information map **258**, can be used as a trigger to cause generation of a new predictive model and predictive map. Keeping with the examples described above, the threshold, the range, and the defined amount can be set to default values; set by an operator or user interaction through a user interface; set by an automated system; or set in other ways.

Other learning trigger criteria can also be used. For instance, if predictive model generator **210** switches to a different prior information map (different from the originally selected prior information map **258**), then switching to the different prior information map may trigger re-learning by predictive model generator **210**, predictive map generator **212**, control zone generator **213**, control system **214**, or other items. In another example, transitioning of agricultural harvester **100** to a different topography or to a different control zone may be used as learning trigger criteria as well.

In some instances, operator **260** can also edit the predictive map **264** or predictive control zone map **265** or both. The edits can change a value on the predictive map **264**, change a size, shape, position, or existence of a control zone on predictive control zone map **265**, or both. Block **321** shows that edited information can be used as learning trigger criteria.

In some instances, it may also be that operator **260** observes that automated control of a controllable subsystem, is not what the operator desires. In such instances, the operator **260** may provide a manual adjustment to the

controllable subsystem reflecting that the operator **260** desires the controllable subsystem to operate in a different way than is being commanded by control system **214**. Thus, manual alteration of a setting by the operator **260** can cause one or more of predictive model generator **210** to relearn a model, predictive map generator **212** to regenerate map **264**, control zone generator **213** to regenerate one or more control zones on predictive control zone map **265**, and control system **214** to relearn a control algorithm or to perform machine learning on one or more of the controller components **232** through **246** in control system **214** based upon the adjustment by the operator **260**, as shown in block **322**. Block **324** represents the use of other triggered learning criteria.

In other examples, relearning may be performed periodically or intermittently based, for example, upon a selected time interval such as a discrete time interval or a variable time interval, as indicated by block **326**.

If relearning is triggered, whether based upon learning trigger criteria or based upon passage of a time interval, as indicated by block **326**, then one or more of the predictive model generator **210**, predictive map generator **212**, control zone generator **213**, and control system **214** performs machine learning to generate a new predictive model, a new predictive map, a new control zone, and a new control algorithm, respectively, based upon the learning trigger criteria. The new predictive model, the new predictive map, and the new control algorithm are generated using any additional data that has been collected since the last learning operation was performed. Performing relearning is indicated by block **328**.

If the harvesting operation has been completed, operation moves from block **312** to block **330** where one or more of the predictive map **264**, predictive control zone map **265**, and predictive model generated by predictive model generator **210** are stored. The predictive map **264**, predictive control zone map **265**, and predictive model may be stored locally on data store **202** or sent to a remote system using communication system **206** for later use.

It will be noted that while some examples herein describe predictive model generator **210** and predictive map generator **212** receiving a prior information map in generating a predictive model and a functional predictive map, respectively, in other examples, the predictive model generator **210** and predictive map generator **212** can receive, in generating a predictive model and a functional predictive map, respectively, other types of maps, including predictive maps, such as a functional predictive map generated during the harvesting operation. The other types of maps, including predictive maps, such as a functional predictive map generated during the harvesting operation, can be used by the predictive model generator **210** and the predictive map generator **212** similarly as the prior information map, in that a predictive model and a predictive map can be generated using the other types of maps, including predictive map, such as a functional predictive map generated during the harvesting operation, as an input.

FIG. 4 is a block diagram of a portion of the agricultural harvester **100** shown in FIG. 1. Particularly, FIG. 4 shows, among other things, examples of the predictive model generator **210** and the predictive map generator **212** in more detail. FIG. 4 also illustrates information flow among the various components shown. The predictive model generator **210** receives one or more of a vegetative index map **332**, a crop moisture map **335**, a topographic map **337**, a soil property map **339**, a predictive yield map **341**, or a predictive biomass map **343** as a prior information map. Vegetative

index map 332 includes georeferenced vegetative index values. Crop moisture map 335 includes georeferenced crop moisture values. Topographic map 337 includes georeferenced topographic characteristic values. Soil property map 339 includes georeferenced soil property values.

Predictive yield map 341 includes georeferenced predictive yield values. Predictive yield map 341 can be generated using a process described in FIGS. 2 and 3, where the prior information map includes a vegetative index map or a historical yield map and the in-situ sensor includes a yield sensor. Predictive yield map 341 can be generated in other ways as well.

Predictive biomass map 343 includes georeferenced predictive biomass values. Predictive biomass map 343 can be generated using a process described in FIGS. 2 and 3, where the prior information map includes a vegetative index map the in-situ sensor includes a rotor drive pressure or optical sensor, that generate sensor signals indicative of biomass. Predictive biomass map 343 can be generated in other ways as well.

Predictive model generator 210 also receives a geographic location 334, or an indication of a geographic location, from geographic position sensor 204. In-situ sensors 208 illustratively include a power characteristic sensor, such as power sensor 336, as well as a processing system 338. Power sensor 336 senses one or more power characteristics of one or more components of agricultural harvester 100. In some instances, power sensor 336 may be located on board the agricultural harvester 100. The processing system 338 processes sensor data generated from power sensor 336 to generate processed data, some examples of which are described below. Power sensor 336 can include, but is not limited to, one or more of a voltage sensor, a current sensor, a torque sensor, a fluid pressure sensor, a fluid flow sensor, a force sensor, a bearing load sensor and a rotational sensor. The outputs of one or more of these or other sensors can be combined to determine one or more power characteristics.

The present discussion proceeds with respect to an example in which power sensor 336 is one or more of the above listed. It will be appreciated that these are just examples, and other examples of power sensor 336, are contemplated herein as well. As shown in FIG. 4, the example predictive model generator 210 includes one or more of a vegetative index-power characteristic model generator 342, crop moisture-power characteristic model generator 343, topographic-power characteristic model generator 344, soil property characteristic-power characteristic model generator 345, yield-power characteristic model generator 346, biomass-power characteristic model generator 347. In other examples, the predictive model generator 210 may include additional, fewer, or different components than those shown in the example of FIG. 4. Consequently, in some examples, the predictive model generator 210 may include other items 348 as well, which may include other types of predictive model generators to generate other types of power models.

Model generator 342 identifies a relationship between a power characteristic, at a geographic location corresponding to where power sensor 336 sensed the characteristic, and vegetative index values from the vegetative index map 332 corresponding to the same location in the field where the power characteristic was sensed. Based on this relationship established by model generator 342, model generator 342 generates a predictive power model 350. The predictive power model 350 is used by predictive map generator 212 to predict power characteristics at different locations in the

field based upon the georeferenced vegetative index values contained in the vegetative index map 332 at the same locations in the field.

Model generator 343 identifies a relationship between a power characteristic, at a geographic location corresponding to where power sensor 336 sensed the characteristic, and crop moisture values from the crop moisture map 335 corresponding to the same location in the field where the power characteristic was sensed. Based on this relationship established by model generator 343, model generator 343 generates a predictive power model 350. The predictive power model 350 is used by predictive map generator 212 to predict power characteristics at different locations in the field based upon the georeferenced crop moisture values contained in the crop moisture map 335 at the same locations in the field.

Model generator 344 identifies a relationship between a power characteristic, at a geographic location corresponding to where power sensor 336 sensed the characteristic, and a topographic feature value from the topographic map 337 corresponding to the same location in the field where the power characteristic was sensed. Based on this relationship established by model generator 344, model generator 344 generates a predictive power model 350. The predictive power model 350 is used by predictive map generator 212 to predict power characteristics at different locations in the field based upon the georeferenced topographic feature values contained in the topographic map 337 at the same locations in the field.

Model generator 345 identifies a relationship between a power characteristic, at a geographic location corresponding to where power sensor 336 sensed the characteristic, and a soil property value from the soil property map 339 corresponding to the same location in the field where the power characteristic was sensed. Based on this relationship established by model generator 345, model generator 345 generates a predictive power model 350. The predictive power model 350 is used by predictive map generator 212 to predict power characteristics at different locations in the field based upon the soil property values contained in the soil property map 339 at the same locations in the field.

Model generator 346 identifies a relationship between a power characteristic, at a geographic location corresponding to where power sensor 336 sensed the characteristic, and a yield value from the yield map 341 corresponding to the same location in the field where the power characteristic was sensed. Based on this relationship established by model generator 346, model generator 346 generates a predictive power model 350. The predictive power model 350 is used by predictive map generator 212 to predict power characteristics at different locations in the field based upon the yield value contained in the predictive yield map 341 at the same locations in the field.

Model generator 347 identifies a relationship between a power characteristic, at a geographic location corresponding to where power sensor 336 sensed the characteristic, and a biomass value from the biomass map 343 corresponding to the same location in the field where the power characteristic was sensed. Based on this relationship established by model generator 347, model generator 347 generates a predictive power model 350. The predictive power model 350 is used by predictive map generator 212 to predict power characteristics at different locations in the field based upon the biomass value contained in the predictive biomass map 343 at the same locations in the field.

In light of the above, the predictive model generator 210 is operable to produce a plurality of predictive power

models, such as one or more of the predictive power models generated by model generators 342, 343, 344, 345, 346, and 347. In another example, two or more of the predictive power models described above may be combined into a single predictive power model that predicts two or more power characteristics based upon the different values at different locations in the field. Any of these power models, or combinations thereof, are represented collectively by power model 350 in FIG. 4.

The predictive power model 350 is provided to predictive map generator 212. In the example of FIG. 4, predictive map generator 212 includes a crop engaging component map generator 351, a header power map generator 352, a feeder power map generator 353, a threshing power map generator 354, a separator power map generator 355, a residue handling power map generator 356, and a propulsion power map generator 357. In other examples, the predictive map generator 212 may include additional, fewer, or different map generators. Thus, in some examples, the predictive map generator 212 may include other items 358 which may include other types of map generators to generate power maps for other types of power characteristics.

Crop engaging component map generator 351 receives the predictive power model 350, which predicts power characteristics based upon values in one or more of the vegetative index map 332, crop moisture map 335, topographic map 337, soil property map 339, predictive yield map 341, or predictive biomass map 343, and generates a predictive map that predicts the power characteristics of a crop engaging component at different locations in the field. For example, the crop engaging components could include a cutter and a reel and the crop engaging component map generator 351 generates a map of estimated power usage by the reel and cutter based on a predictive power model 350 that defines a relationship between crop moisture and power usage by the reel and cutter.

Header power map generator 352 receives the predictive power model 350, which predicts power characteristics based upon values in one or more of the vegetative index map 332, crop moisture map 335, topographic map 337, soil property map 339, predictive yield map 341, or predictive biomass map 343, and generates a predictive map that predicts the power characteristics of the header at different locations in the field. For example, the crop engaging components could include one or more of: a cutter, a reel, draper belts, augers, and header positioning actuators and the header power map generator 352 generates a map of estimated power usage by one or more of: the reel, the cutter, draper belts, augers, and header positioning actuators based on a predictive power model 350 that defines a relationship between crop moisture and topography, and power usage by one or more of the reel, the cutter, draper belts, auger, and header positioning actuators.

Feeder power map generator 353 receives the predictive power model 350, which predicts power characteristics based upon values in one or more of the vegetative index map 332, crop moisture map 335, topographic map 337, soil property map 339, predictive yield map 341, or predictive biomass map 343, and generates a predictive map that predicts the power characteristics of the feeder at different locations in the field.

Threshing power map generator 354 receives the predictive power model 350, which predicts power characteristics based upon values in one or more of the vegetative index map 332, crop moisture map 335, topographic map 337, soil property map 339, predictive yield map 341, or predictive biomass map 343, and generates a predictive map that

predicts the power characteristics of the threshing subsystems at different locations in the field. For example, the threshing subsystems could include one or more threshing drums, concave adjustment actuators, and beaters, and the threshing power map generator 352 generates a map of estimated power usage by the one or more threshing drums, concave adjustment actuators, and beaters based on a predictive power model 350 that defines a relationship between vegetative index, and power usage by the one or more threshing drums, concave adjustment actuators, and beaters. Or for example, the threshing subsystems could include a threshing drum and a set of concaves at a given clearance, and the threshing power map generator 352 generates a map of estimated power usage by the threshing drum with the set of concaves at the given clearance based on a predictive power model 350 that defines a relationship between predictive biomass, and power usage by the threshing drum with the set of concaves at the given clearance. Or for example, the threshing subsystems could include one or more beaters at a given configuration, and the threshing power map generator 352 generates a map of estimated power usage by the one or more beaters at the given configuration based on a predictive power model 350 that defines a relationship between predictive biomass, and power usage by the one or more beaters at the given configuration. Separator power map generator 355 receives the predictive power model 350, which predicts power characteristics based upon values in one or more of the vegetative index map 332, crop moisture map 335, topographic map 337, soil property map 339, predictive yield map 341, or predictive biomass map 343, and generates a predictive map that predicts the power characteristics of the separator subsystems at different locations in the field. For example, the separator subsystems could include one or more fans, sieves, chaffers, and straw walkers, and the separator power map generator 355 generates a map of estimated power usage by the one or more fans, sieves, chaffers, and straw walkers based on a predictive power model 350 that defines a relationship between predictive yield value, and power usage by the one or more fans, sieves, chaffers, and straw walkers. For example, the separator subsystems could include one or more fans running at a given speed and sieves, chaffers, and straw walkers in a given configuration, and the separator power map generator 355 generates a map of estimated power usage by the one or more fans running at the given speed and sieves, chaffers, and straw walkers in the given configuration based on a predictive power model 350 that defines a relationship between predictive yield value, and power usage by the one or more fans running at the given speed and sieves, chaffers, and straw walkers in the given configuration.

Residue handling power map generator 356 receives the predictive power model 350, which predicts power characteristics based upon values in one or more of the vegetative index map 332, crop moisture map 335, topographic map 337, soil property map 339, predictive yield map 341, or predictive biomass map 343, and generates a predictive map that predicts the power characteristics of the residue handling subsystems at different locations in the field. For example, the residue handling power map generator 356 generates a map of estimated power usage by a residue spreader based on a predictive power model 350 that defines a relationship between predictive biomass value, and power usage by the residue spreader. For example, the residue handling power map generator 356 generates a map of estimated power usage by a residue chopper based on a

predictive power model **350** that defines a relationship between predictive yield value, and power usage by the residue chopper.

Propulsion power map generator **357** receives the predictive power model **350**, which predicts power characteristics based upon values in one or more of the vegetative index map **332**, crop moisture map **335**, topographic map **337**, soil property map **339**, predictive yield map **341**, or predictive biomass map **343**, and generates a predictive map that predicts the power characteristics of the propulsion subsystems at different locations in the field. For example, the propulsion power map generator **357** generates a map of estimated power usage by the propulsion subsystem based on a predictive power model **350** that defines a relationship between topographic map value, and power usage by the propulsion system.

Predictive map generator **212** outputs one or more predictive power maps **360** that are predictive of one or more power characteristics. Each of the predictive power maps **360** predicts the respective power characteristic at different locations in a field. Each of the generated predictive power maps **360** may be provided to control zone generator **213**, control system **214**, or both. Control zone generator **213** generates control zones and incorporates those control zones into the functional predictive map, i.e., predictive map **360**, to provide a predictive map **360** with control zones. The functional predictive map **360** (with or without control zones) may be provided to control system **214**, which generates control signals to control one or more of the controllable subsystems **216** based upon the functional predictive map **360** (with or without control zones).

FIG. 5 is a flow diagram of an example of operation of predictive model generator **210** and predictive map generator **212** in generating the predictive power model **350** and the predictive power map **360**. At block **362**, predictive model generator **210** and predictive map generator **212** receives one or more of a prior vegetative index map **332**, a crop moisture map **335**, a topographic map **337**, a soil property map **339**, a predictive yield map **341**, a predictive biomass map **343**, or some other map **363**. At block **364**, processing system **338** receives one or more sensor signals from power sensor **336**. As discussed above, the power sensor **336** may include one or more of a voltage sensor **371**, a current sensor **373**, a torque sensor **375**, a hydraulic pressure sensor **377**, a hydraulic flow sensor **379**, a force sensor **381**, a bearing load sensor **383**, a rotational sensor **385**, or another type of power sensor **370**. The outputs of one or more of these or other sensors can be combined to determine one or more power characteristics

At block **372**, processing system **338** processes the one or more received sensor signals to generate data indicative of a power characteristic. As indicated by block **374**, the power characteristic may be identified at a machine wide level. For example, the entire power usage by the entire agricultural harvester. A power usage at this level can be used to calculate fuel consumption, efficiency, etc. As indicated by block **376**, the power characteristic may be identified at a subsystem level. A characteristic at this level, for instance, can be used to allocate power across subsystems. As indicated by block **378**, the power characteristic may be identified at a component level. The sensor data can include other data at other levels as well as indicated by block **380**.

At block **382**, predictive model generator **210** also obtains the geographic location corresponding to the sensor data. For instance, the predictive model generator **210** can obtain the geographic position from geographic position sensor **204**

and determine, based upon machine delays, machine speed, etc., a precise geographic location where the sensor data **340** was captured or derived.

At block **384**, predictive model generator **210** generates one or more predictive power models, such as power model **350**, that model a relationship between a vegetative index value, a crop moisture value, a soil property value, a predictive yield value, or a predictive biomass value obtained from a prior information map, such as prior information map **258**, and a power characteristic being sensed by the in-situ sensor **208** or a related characteristic. For instance, predictive model generator **210** may generate a predictive power model that models the relationship between a vegetative index value and a sensed characteristic including power usage indicated by the sensor data obtained from in-situ sensor **208**.

At block **386**, the predictive power model, such as predictive power model **350**, is provided to predictive map generator **212** which generates a predictive power map **360** that maps a predicted power characteristic based on a vegetative index map, a crop moisture map, a soil property map, a predictive yield map, or a predictive biomass map, and the predictive power model **350**. For instance, in some examples, the predictive power map **360** predicts power usage/requirements of various subsystems. Further, the predictive power map **360** can be generated during the course of an agricultural operation. Thus, as an agricultural harvester is moving through a field performing an agricultural operation, the predictive power map **360** is generated as the agricultural operation is being performed.

At block **394**, predictive map generator **212** outputs the predictive power map **360**. At block **391** predictive power map generator **212** outputs the predictive power map for presentation to and possible interaction by operator **260**. At block **393**, predictive map generator **212** may configure the map for consumption by control system **214**. At block **395**, predictive map generator **212** can also provide the map **360** to control zone generator **213** for generation and incorporation of control zones. At block **397**, predictive map generator **212** configures the map **360** in other ways as well. The predictive power map **360** (with or without the control zones) is provided to control system **214**. At block **396**, control system **214** generates control signals to control the controllable subsystems **216** based upon the predictive power map **360** (with or without control zones).

It can thus be seen that the present system takes a prior information map that maps a characteristic such as a vegetative index value, a crop moisture value, a soil property value, a predictive yield value, or a predictive biomass value or information from a prior operation pass to different locations in a field. The present system also uses one or more in-situ sensors that sense in-situ sensor data that is indicative of a power characteristic, such as power usage, power requirement, or power loss, and generates a model that models a relationship between the characteristic sensed using the in-situ sensor, or a related characteristic, and the characteristic mapped in the prior information map. Thus, the present system generates a functional predictive map using a model, in-situ data, and a prior information map and may configure the generated functional predictive map for consumption by a control system, for presentation to a local or remote operator or other user, or both. For example, the control system may use the map to control one or more systems of a combine harvester.

FIG. 6A is a block diagram of an example portion of the agricultural harvester **100** shown in FIG. 1. Particularly, FIG. 6A shows, among other things, examples of predictive

model generator **210** and predictive map generator **212**. In the illustrated example, the prior information map **258** is one or more of a historical power map **333**, a soil property map **339**, or a prior operation map **400**. Prior operation map **400** may include power characteristic values at various locations in the field. FIG. **6A** also shows that predictive model generator **210** and predictive map generator can receive, in addition to prior information map **258**, functional predictive power map **360**. Functional predictive power map **360** can be used similarly as prior information map **258** in that model generator **210** models a relationship between information provided by functional predictive power map **360** and characteristics sensed by in-situ sensors **208**, and map generator **212** can thus use the model to generate a functional predictive map that predicts the characteristics sensed by the in-situ sensors **208**, or a characteristic indicative of the sensed characteristic, at different locations in the field based upon one or more of the values in the functional predictive power map **360** at those locations in the field and based on the predictive model. As illustrated in FIG. **6A**, predictive model generator **210** and predictive map generator **212** can also receive other maps **401**, for instance other prior information maps or other predictive maps, such as other predictive power maps generated in ways other than functional predictive power map **360**.

Also, in the example shown in FIG. **6A**, in-situ sensor **208** can include one or more of an agricultural characteristic sensor **402**, power sensor **336**, an operator input sensor **405**, and a processing system **406**. In-situ sensors **208** can include other sensors **408** as well.

Agricultural characteristic sensor **402** senses values indicative of agricultural characteristics. As discussed above, power sensor **336** senses power characteristics of one or more components of agricultural harvester **100**. For example, power sensor **336** senses power characteristics of propulsion subsystem **250** and generates sensor signals indicative of those power characteristics. For example, power sensor **336** can sense a power consumption or power utilization of propulsion subsystem **250** as the agricultural harvester **100** travels across the field and generate a sensor signal indicative thereof.

Operator input sensor **405** illustratively senses various operator inputs. The inputs can be setting inputs for controlling the settings on agricultural harvester **100** or other control inputs, such as steering inputs and other inputs. Thus, when operator **260** changes a setting or provides a commanded input through an operator interface mechanism **218**, such an input is detected by operator input sensor **405**, which provides a sensor signal indicative of that sensed operator input.

Processing system **406** may receive the sensor signals from one or more of agricultural characteristic sensor **402**, power sensor **336**, and operator input sensor **405** and generate an output indicative of the sensed variable. For instance, processing system **406** may receive a sensor input from agricultural characteristic sensor **402** and generate an output indicative of an agricultural characteristic. Processing system **406** may also receive an input from operator input sensor **405** and generate an output indicative of the sensed operator input.

Predictive model generator **210** may include power characteristic-to-agricultural characteristic model generator **410**, soil property-to-power characteristic model generator **412**, and power characteristic-to-command model generator **414**. In other examples, predictive model generator **210** can include additional, fewer, or other model generators **415**. Predictive model generator **210** may receive a geographic

location indicator **334** from geographic position sensor **204** and generate a predictive model **426** that models a relationship between the information in one or more of the prior information maps **258** or the information in functional predictive power map **360** and one or more of: the agricultural characteristic sensed by agricultural characteristic sensor **402**; the power characteristic sensed by power sensor **336**; and operator input commands sensed by operator input sensor **405**.

Power characteristic-to-agricultural characteristic model generator **410** generates a relationship between power characteristic values (which may be on historical power map **333**, on predictive power map **360**, or on prior operation map **400**) and the agricultural characteristic sensed by agricultural characteristic sensor **402**. Power characteristic-to-agricultural characteristic model generator **410** generates a predictive model **426** that corresponds to this relationship.

Soil property-to-power characteristic model generator **412** generates a relationship between soil property values (which may be on soil property map **339**) and the power characteristic sensed by power sensor **336**. Soil property-to-power characteristic model generator **412** generates a predictive model **426** that corresponds to this relationship. In some examples, soil property-to-power characteristic model generator **415** may generate a relationship between specific soil properties, such as soil moisture or soil type, and power characteristics, or specific power characteristics, such as power characteristics of the propulsion subsystem **250**, and, thus, soil property-to-power characteristic model generator **412** generates a corresponding predictive model **426**.

Power characteristic-to-operator command model generator **414** generates a model that models the relationship between a power characteristic as reflected on historical power map **333**, on predictive power map **360**, or on prior operation map **400**, or any combination thereof and operator input commands that are sensed by operator input sensor **405**. Power-to-operator command model generator **414** generates a predictive model **426** that corresponds to this relationship.

Other model generators **415** may include, for example, specific power characteristic model generators, such as propulsion subsystem power characteristics, for instance, a propulsion subsystem power characteristic-to-agricultural characteristic model generator, a propulsion subsystem power characteristic-to-command model generator, or a soil property-to-propulsion subsystem power characteristic. In some examples, other model generators **415** may include specific soil property model generators, such as a soil type-to-power characteristic model generator, a soil moisture-to-power characteristic model generator. In yet other examples, other model generators **415** may include one or more specific soil property-to-specific power characteristic model generators, such as a soil type-to-propulsion subsystem power characteristic model generator or a soil moisture-to-propulsion subsystem power characteristic model generator.

Predictive model **426** generated by the predictive model generator **210** can include one or more of the predictive models that may be generated by power characteristic-to-agricultural characteristic model generator **410**, soil property-to-power characteristic model generator **412**, and power characteristic-to-operator command model generator **414**, and other model generators that may be included as part of other items **415**.

In the example of FIG. **6A**, predictive map generator **212** includes predictive agricultural characteristic map generator **416**, predictive propulsion power map generator **357**, and a

predictive operator command map generator **422**. In other examples, predictive map generator **212** can include additional, fewer, or other map generators **424**.

Predictive agricultural characteristic map generator **416** receives a predictive model **426** that models the relationship between a power characteristic and an agricultural characteristic sensed by agricultural characteristic **402** (such as a predictive model generated by power characteristic-to-agricultural characteristic model generator **410**), and one or more of the prior information maps **258** or functional predictive power map **360**. Predictive agricultural characteristic map generator **416** generates a functional predictive agricultural characteristic map **427** that predicts agricultural characteristic values (or the agricultural characteristics the values are indicative of) at different locations in the field based upon one or more of the power characteristics in one or more of the prior information maps **258** (such as historical power map **333**) or the functional predictive power map **360**, or other maps **401**, at those locations in the field and based on predictive model **426**.

Predictive propulsion power map generator **357** receives a predictive model **426** that models the relationship between a soil property and a power characteristic sensed by power sensor **336** (such as a predictive model generated by soil property-to-power characteristic model generator **412**), and one or more of the prior information maps **258**. Predictive propulsion power map generator **357** generates a functional predictive propulsion power map **428** that predicts a likely propulsion power characteristic at different locations in the field based upon one or more of the soil properties in one or more of the prior information maps **258** (such as soil property map **339**) at those locations in the field and based on predictive model **426**.

Predictive operator command map generator **422** receives a predictive model **426** that models the relationship between the power characteristic and operator command inputs detected by operator input sensor **405** (such as a predictive model generated by power characteristic-to-command model generator **414**), and one or more of the prior information maps **258** or the functional predictive power map **360**, or other map **401**. Predictive operator command map generator **422** generates a functional predictive operator command map **440** that predicts operator command inputs at different locations in the field based upon the power characteristic values from the prior information map **258** (such as historical power map **333**) or the functional predictive power map **360**, or other maps **401**, at those locations in the field and based on the predictive model **426**.

Predictive map generator **212** outputs one or more of the functional predictive maps **427**, **428**, and **440**. Each of the functional predictive maps **427**, **428**, and **440** may be provided to control zone generator **213**, control system **214**, or both. Control zone generator **213** generates and incorporates control zones to provide one or more of a functional predictive map **427** with control zones, a functional predictive map **428** with control zones, or a functional predictive map **440** with control zones. Any or all of functional predictive maps **427**, **428**, and **440**, with or without control zones, may be provided to control system **214**, which generates control signals to control one or more of the controllable subsystems **216** based upon one or all of the functional predictive maps **427**, **428**, and **440**, with or without control zones. Any or all of the maps **427**, **428**, and **440**, with or without control zones, may be presented to operator **260** or another user.

FIG. 6B is a block diagram showing some examples of in-situ sensors **208**. Some of the sensors shown in FIG. 6B,

or different combinations of them, may have both a sensor **402** and a processing system **406**, while others may act as sensor **402** described with respect to FIGS. 6A and 7 where the processing system **406** is separate. Some of the possible in-situ sensors **208** shown in FIG. 6B are shown and described above with respect to previous FIGS. and are similarly numbered. FIG. 6B shows that in-situ sensors **208** can include operator input sensors **480**, machine sensors **482**, harvested material property sensors **484**, field and soil property sensors **485**, environmental characteristic sensors **487**, and they may include a wide variety of other sensors **226**. Operator input sensors **480** may be sensors that sense operator inputs through operator interface mechanisms **218**. Therefore, operator input sensors **480** may sense user movement of linkages, joysticks, a steering wheel, buttons, dials, or pedals. Operator input sensors **480** can also sense user interactions with other operator input mechanisms, such as with a touch sensitive screen, with a microphone where speech recognition is utilized, or any of a wide variety of other operator input mechanisms.

Machine sensors **482** may sense different characteristics of agricultural harvester **100**. For instance, as discussed above, machine sensors **482** may include machine speed sensors **146**, separator loss sensor **148**, clean grain camera **150**, forward looking image capture mechanism **151**, loss sensors **152** or geographic position sensor **204**, examples of which are described above. Machine sensors **482** can also include machine setting sensors **491** that sense machine settings. Some examples of machine settings were described above with respect to FIG. 1. Front-end equipment (e.g., header) position sensor **493** can sense the position of the header **102**, reel **164**, cutter **104**, or other front-end equipment relative to the frame of agricultural harvester **100**. For instance, sensors **493** may sense the height of header **102** above the ground. Machine sensors **482** can also include front-end equipment (e.g., header) orientation sensors **495**. Sensors **495** may sense the orientation of header **102** relative to agricultural harvester **100**, or relative to the ground. Machine sensors **482** may include stability sensors **497**. Stability sensors **497** sense oscillation or bouncing motion (and amplitude) of agricultural harvester **100**. Machine sensors **482** may also include residue setting sensors **499** that are configured to sense whether agricultural harvester **100** is configured to chop the residue, produce a windrow, or deal with the residue in another way. Machine sensors **482** may include cleaning shoe fan speed sensor **551** that senses the speed of cleaning fan **120**. Machine sensors **482** may include concave clearance sensors **553** that sense the clearance between the rotor **112** and concaves **114** on agricultural harvester **100**. Machine sensors **482** may include chaffer clearance sensors **555** that sense the size of openings in chaffer **122**. The machine sensors **482** may include threshing rotor speed sensor **557** that senses a rotor speed of rotor **112**. Machine sensors **482** may include rotor pressure sensor **559** that senses the pressure used to drive rotor **112**. Machine sensors **482** may include sieve clearance sensor **561** that senses the size of openings in sieve **124**. The machine sensors **482** may include MOG moisture sensor **563** that senses a moisture level of the MOG passing through agricultural harvester **100**. Machine sensors **482** may include machine orientation sensor **565** that senses the orientation of agricultural harvester **100**. Machine sensors **482** may include material feed rate sensors **567** that sense the feed rate of material as the material travels through feeder house **106**, clean grain elevator **130**, or elsewhere in agricultural harvester **100**. Machine sensors **482** can include biomass sensors **569** that sense the biomass traveling through feeder

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house 106, through separator 116, or elsewhere in agricultural harvester 100. The machine sensors 482 may include fuel consumption sensor 571 that senses a rate of fuel consumption over time of agricultural harvester 100. Machine sensors 482 may include power utilization sensor 573 that senses power utilization in agricultural harvester 100, such as which subsystems are utilizing power, or the rate at which subsystems are utilizing power, or the distribution of power among the subsystems in agricultural harvester 100. Machine sensors 482 may include tire pressure sensors 577 that sense the inflation pressure in tires 144 of agricultural harvester 100. Machine sensor 482 may include a wide variety of other machine performance sensors, or machine characteristic sensors, indicated by block 575. The machine performance sensors and machine characteristic sensors 575 may sense machine performance or characteristics of agricultural harvester 100.

Harvested material property sensors 484 may sense characteristics of the severed crop material as the crop material is being processed by agricultural harvester 100. The crop properties may include such things as crop type, crop moisture, grain quality (such as broken grain), MOG levels, grain constituents such as starches and protein, MOG moisture, and other crop material properties. Other sensors could sense straw "toughness", adhesion of corn to ears, and other characteristics that might be beneficially used to control processing for better grain capture, reduced grain damage, reduced power consumption, reduced grain loss, etc.

Field and soil property sensors 485 may sense characteristics of the field and soil. The field and soil properties may include soil moisture, soil compactness, the presence and location of standing water, soil type, and other soil and field characteristics.

Environmental characteristic sensors 487 may sense one or more environmental characteristics. The environmental characteristics may include such things as wind direction and wind speed, precipitation, fog, dust level or other obscuring, or other environmental characteristics.

FIG. 7 shows a flow diagram illustrating one example of the operation of predictive model generator 210 and predictive map generator 212 in generating one or more predictive models 426 and one or more functional predictive maps 427, 428, and 440. At block 442, predictive model generator 210 and predictive map generator 212 receive a map. The map received by predictive model generator 210 or predictive map generator 212 in generating one or more predictive models 426 and one or more functional predictive maps 427, 428, and 440 may be prior information map 258, such as historical power map 333, soil property map 339, or a prior operation map 400 created using data obtained during a prior operation in a field. The map received by predictive model generator 210 or predictive map generator 212 in generating one or more predictive models 426 and one or more functional predictive maps 427, 428, and 440 may be functional predictive power map 360. Other maps can be received as well as indicated by block 401, such as other prior information maps or other predictive maps, for instance, other predictive power maps.

At block 444, predictive model generator 210 receives a sensor signal containing sensor data from an in-situ sensor 208. The in-situ sensor can be one or more of an agricultural characteristic sensor 402, a power sensor 336, and an operator input sensor 406. Agricultural characteristic sensor 402 senses an agricultural characteristic. Power characteristic sensor 336 senses a power characteristic, such as a power characteristic of propulsion subsystem 250. Operator input sensor 406 senses an operator input command. Pre-

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dictive model generator 210 can receive other in-situ sensor inputs as well, as indicated by block 408. Some other examples of in-situ sensors 208 are shown in FIG. 6B and discussed elsewhere herein.

At block 454, processing system 406 processes the data contained in the sensor signal or signals received from the in-situ sensor or sensors 208 to obtain processed data 409, shown in FIG. 6A. The data contained in the sensor signal or signals can be in a raw format that is processed to receive processed data 409. For example, a temperature sensor signal includes electrical resistance data, this electrical resistance data can be processed into temperature data. In other examples, processing may comprise digitizing, encoding, formatting, scaling, filtering, or classifying data. The processed data 409 may be indicative of one or more of an agricultural characteristic, a power characteristic, or an operator input command. The processed data 409 is provided to predictive model generator 210.

Returning to FIG. 7, at block 456, predictive model generator 210 also receives a geographic location 334 from geographic position sensor 204, as shown in FIG. 6A. The geographic location 334 may be correlated to the geographic location from which the sensed variable or variables, sensed by in-situ sensors 208, were taken. For instance, the predictive model generator 210 can obtain the geographic location 334 from geographic position sensor 204 and determine, based upon machine delays, machine speed, etc., a precise geographic location from which the processed data 409 was derived.

At block 458, predictive model generator 210 generates one or more predictive models 426 that model a relationship between a mapped value in a received map and a characteristic represented in the processed data 409. For example, in some instances, the mapped value in a received map may be a power characteristic and the predictive model generator 210 generates a predictive model using the mapped value of a received map and a characteristic sensed by in-situ sensors 208, as represented in the processed data 409, or a related characteristic, such as a characteristic that correlates to the characteristic sensed by in-situ sensors 208.

The one or more predictive models 426 are provided to predictive map generator 212. At block 466, predictive map generator 212 generates one or more functional predictive maps. The functional predictive maps may be functional predictive agricultural characteristic map 427, functional predictive propulsion power map 428, and a functional predictive operator command map 440, or any combination of these maps. Functional predictive agricultural characteristic map 427 predicts agricultural characteristic values (or agricultural characteristics indicated by the values) at different locations in the field. Functional predictive propulsion power map 428 predicts likely propulsion power at different locations in the field. Functional predictive operator command map 440 predicts desired or likely operator command inputs at different locations in the field. Further, one or more of the functional predictive maps 427, 428, and 440 can be generated during the course of an agricultural operation. Thus, as agricultural harvester 100 is moving through a field performing an agricultural operation, the one or more predictive maps 427, 428, and 440 are generated as the agricultural operation is being performed.

At block 468, predictive map generator 212 outputs the one or more functional predictive maps 427, 428, and 440. At block 470, predictive map generator 212 may configure the map for presentation to and possible interaction by an operator 260 or another user. At block 472, predictive map generator 212 may configure the map for consumption by

control system 214. At block 474, predictive map generator 212 can provide the one or more predictive maps 427, 428, and 440 to control zone generator 213 for generation and incorporation of control zones. At block 476, predictive map generator 212 configures the one or predictive maps 427, 428, and 440 in other ways. One or more functional predictive maps 427, 428, and 440, with or without control zones, may be presented to operator 260 or another user or provided to control system 214 as well.

At block 478, control system 214 then generates control signals to control the controllable subsystems based upon the one or more functional predictive maps 427, 428 and 440 (or the functional predictive maps 427, 428, and 440 having control zones) as well as an input from the geographic position sensor 204.

In an example in which control system 214 receives a functional predictive map or a functional predictive map with control zones added, the path planning controller 234 controls steering subsystem 252 to steer agricultural harvester 100. In another example in which control system 214 receives a functional predictive map or a functional predictive map with control zones added, the residue system controller 244 controls residue subsystem 138. In another example in which control system 214 receives a functional predictive map or a functional predictive map with control zones added, the settings controller 232 controls thresher settings of thresher 110. In another example in which control system 214 receives a functional predictive map or a functional predictive map with control zones added, the settings controller 232 or another controller 246 controls material handling subsystem 125. In another example in which control system 214 receives a functional predictive map or a functional predictive map with control zones added, the settings controller 232 controls crop cleaning subsystem 118. In another example in which control system 214 receives a functional predictive map or a functional predictive map with control zones added, the machine cleaning controller 245 controls machine cleaning subsystem 254 on agricultural harvester 100. In another example in which control system 214 receives a functional predictive map or a functional predictive map with control zones added, the communication system controller 229 controls communication system 206. In another example in which control system 214 receives a functional predictive map or a functional predictive map with control zones added, the operator interface controller 231 controls operator interface mechanisms 218 on agricultural harvester 100. In another example in which control system 214 receives the functional predictive map or the functional predictive map with control zones added, the deck plate position controller 242 controls machine/header actuators to control a deck plate on agricultural harvester 100. In another example in which control system 214 receives the functional predictive map or the functional predictive map with control zones added, the draper belt controller 240 controls machine/header actuators to control a draper belt on agricultural harvester 100. In another example in which control system 214 receives the functional predictive map or the functional predictive map with control zones added, the other controllers 246 control other controllable subsystems 256 on agricultural harvester 100.

In one example, control system 214 may receive a functional predictive map or a functional predictive map with control zones added, feed rate controller 236 can control various subsystems, such as propulsion subsystem 250 and header or other machine actuators 248, to control a feed rate based upon the functional predictive map (with or without

control zones. For instance, feed rate controller 236 can control propulsion subsystem 250 to adjust a speed of agricultural harvester 100 to control a feed rate. In another example, feed rate controller 236 can control header or other machine actuators 248 to control a height of header 102 above the field to control a feed rate.

FIG. 8 shows a block diagram illustrating one example of control zone generator 213. Control zone generator 213 includes work machine actuator (WMA) selector 486, control zone generation system 488, and regime zone generation system 490. Control zone generator 213 may also include other items 492. Control zone generation system 488 includes control zone criteria identifier component 494, control zone boundary definition component 496, target setting identifier component 498, and other items 520. Regime zone generation system 490 includes regime zone criteria identification component 522, regime zone boundary definition component 524, settings resolver identifier component 526, and other items 528. Before describing the overall operation of control zone generator 213 in more detail, a brief description of some of the items in control zone generator 213 and the respective operations thereof will first be provided.

Agricultural harvester 100, or other work machines, may have a wide variety of different types of controllable actuators that perform different functions. The controllable actuators on agricultural harvester 100 or other work machines are collectively referred to as work machine actuators (WMAs). Each WMA may be independently controllable based upon values on a functional predictive map, or the WMAs may be controlled as sets based upon one or more values on a functional predictive map. Therefore, control zone generator 213 may generate control zones corresponding to each individually controllable WMA or corresponding to the sets of WMAs that are controlled in coordination with one another.

WMA selector 486 selects a WMA or a set of WMAs for which corresponding control zones are to be generated. Control zone generation system 488 then generates the control zones for the selected WMA or set of WMAs. For each WMA or set of WMAs, different criteria may be used in identifying control zones. For example, for one WMA, the WMA response time may be used as the criteria for defining the boundaries of the control zones. In another example, wear characteristics (e.g., how much a particular actuator or mechanism wears as a result of movement thereof) may be used as the criteria for identifying the boundaries of control zones. Control zone criteria identifier component 494 identifies particular criteria that are to be used in defining control zones for the selected WMA or set of WMAs. Control zone boundary definition component 496 processes the values on a functional predictive map under analysis to define the boundaries of the control zones on that functional predictive map based upon the values in the functional predictive map under analysis and based upon the control zone criteria for the selected WMA or set of WMAs.

Target setting identifier component 498 sets a value of the target setting that will be used to control the WMA or set of WMAs in different control zones. For instance, if the selected WMA is propulsion system 250 or header or other machine actuators 248 and the functional predictive map under analysis is a functional predictive power map 360 (with control zones) or functional predictive propulsion power map 428 (with control zones), then the target setting in each control zone may be a target speed setting or a target header height setting based on power values contained in the

functional predictive power map **360** or functional predictive propulsion power map **428**, within the identified control zone.

In some examples, where agricultural harvester **100** is to be controlled based on a current or future location of the agricultural harvester **100**, multiple target settings may be possible for a WMA at a given position. In that case, the target settings may have different values and may be competing. Thus, the target settings need to be resolved so that only a single target setting is used to control the WMA. For example, where the WMA is an actuator in propulsion system **250** that is being controlled in order to control the speed of agricultural harvester **100**, multiple different competing sets of criteria may exist that are considered by control zone generation system **488** in identifying the control zones and the target settings for the selected WMA in the control zones. For instance, different target settings for controlling machine speed may be generated based upon, for example, a detected or predicted power characteristic value (such as a power characteristic value of the propulsion subsystem), a detected or predicted agricultural characteristic value, a detected or predicted soil property value (such as a soil type value or a soil moisture value), a detected or predicted feed rate value, a detected or predictive fuel efficiency value, a detected or predicted grain loss value, or a combination of these. It will be noted that these are merely example and target settings for various WMAs can be based on various other values or combinations of values. However, at any given time, the agricultural harvester **100** cannot travel over the ground at multiple speeds simultaneously or have multiple header heights simultaneously. Rather, at any given time, the agricultural harvester **100** travels at a single speed and has a single header height. Thus, one of the competing target settings is selected to control the speed or header height of agricultural harvester **100**.

Therefore, in some examples, regime zone generation system **490** generates regime zones to resolve multiple different competing target settings. Regime zone criteria identification component **522** identifies the criteria that are used to establish regime zones for the selected WMA or set of WMAs on the functional predictive map under analysis. Some criteria that can be used to identify or define regime zones include, for example, power characteristics, such as power characteristics (e.g., power consumption or power utilization) machine wide (e.g., power characteristics of all power consuming or power utilizing subsystems of agricultural harvester **100**) or power characteristics of one or more particular subsystems of agricultural harvester **100** (such as power characteristics of the propulsion subsystem), agricultural characteristics, soil properties (such as soil type or soil moisture), operator command inputs, crop type or crop variety (for example based on an as-planted map or another source of the crop type or crop variety), weed type, weed intensity, or crop state (such as whether the crop is down, partially down or standing). These are merely some examples of the criteria that can be used to identify or define regime zones. Just as each WMA or set of WMAs may have a corresponding control zone, different WMAs or sets of WMAs may have a corresponding regime zone. Regime zone boundary definition component **524** identifies the boundaries of regime zones on the functional predictive map under analysis based on the regime zone criteria identified by regime zone criteria identification component **522**.

In some examples, regime zones may overlap with one another. For instance, a propulsion power regime zone may overlap with a portion of or an entirety of a crop state regime zone. In such an example, the different regime zones may be

assigned to a precedence hierarchy so that, where two or more regime zones overlap, the regime zone assigned with a greater hierarchical position or importance in the precedence hierarchy has precedence over the regime zones that have lesser hierarchical positions or importance in the precedence hierarchy. The precedence hierarchy of the regime zones may be manually set or may be automatically set using a rules-based system, a model-based system, or another system. As one example, where a downed crop regime zone overlaps with a propulsion power regime zone, the downed crop regime zone may be assigned a greater importance in the precedence hierarchy than the propulsion power regime zone so that the downed crop regime zone takes precedence.

In addition, each regime zone may have a unique settings resolver for a given WMA or set of WMAs. Settings resolver identifier component **526** identifies a particular settings resolver for each regime zone identified on the functional predictive map under analysis and a particular settings resolver for the selected WMA or set of WMAs.

Once the settings resolver for a particular regime zone is identified, that settings resolver may be used to resolve competing target settings, where more than one target setting is identified based upon the control zones. The different types of settings resolvers can have different forms. For instance, the settings resolvers that are identified for each regime zone may include a human choice resolver in which the competing target settings are presented to an operator or other user for resolution. In another example, the settings resolver may include a neural network or other artificial intelligence or machine learning system. In such instances, the settings resolvers may resolve the competing target settings based upon a predicted or historic quality metric corresponding to each of the different target settings. As an example, an increased vehicle speed setting may reduce the time to harvest a field and reduce corresponding time-based labor and equipment costs but may increase grain losses. A reduced vehicle speed setting may increase the time to harvest a field and increase corresponding time-based labor and equipment costs but may decrease grain losses. When grain loss or time to harvest is selected as a quality metric, the predicted or historic value for the selected quality metric, given the two competing vehicle speed settings values, may be used to resolve the speed setting. In some instances, the settings resolvers may be a set of threshold rules that may be used instead of, or in addition to, the regime zones. An example of a threshold rule may be expressed as follows:

If predicted biomass values within 20 feet of the header of the agricultural harvester **100** are greater than x kilograms (where x is a selected or predetermined value), then use the target setting value that is chosen based on feed rate over other competing target settings, otherwise use the target setting value based on grain loss over other competing target setting values.

The settings resolvers may be logical components that execute logical rules in identifying a target setting. For instance, the settings resolver may resolve target settings while attempting to minimize harvest time or minimize the total harvest cost or maximize harvested grain or based on other variables that are computed as a function of the different candidate target settings. A harvest time may be minimized when an amount to complete a harvest is reduced to at or below a selected threshold. A total harvest cost may be minimized where the total harvest cost is reduced to at or below a selected threshold. Harvested grain may be maximized where the amount of harvested grain is increased to at or above a selected threshold.

FIG. 9 is a flow diagram illustrating one example of the operation of control zone generator **213** in generating control zones and regime zones for a map that the control zone generator **213** receives for zone processing (e.g., for a map under analysis).

At block **530**, control zone generator **213** receives a map under analysis for processing. In one example, as shown at block **532**, the map under analysis is a functional predictive map. For example, the map under analysis may be one of the functional predictive maps **360**, **427**, **428**, or **440**. Block **534** indicates that the map under analysis can be other maps as well.

At block **536**, WMA selector **486** selects a WMA or a set of WMAs for which control zones are to be generated on the map under analysis. At block **538**, control zone criteria identification component **494** obtains control zone definition criteria for the selected WMAs or set of WMAs. Block **540** indicates an example in which the control zone criteria are or include wear characteristics of the selected WMA or set of WMAs. Block **542** indicates an example in which the control zone definition criteria are or include a magnitude and variation of input source data, such as the magnitude and variation of the values on the map under analysis or the magnitude and variation of inputs from various in-situ sensors **208**. Block **544** indicates an example in which the control zone definition criteria are or include physical machine characteristics, such as the physical dimensions of the machine, a speed at which different subsystems operate, or other physical machine characteristics. Block **546** indicates an example in which the control zone definition criteria are or include a responsiveness of the selected WMA or set of WMAs in reaching newly commanded setting values. Block **548** indicates an example in which the control zone definition criteria are or include machine performance metrics. Block **550** indicates an example in which the control zone definition criteria are or include operator preferences. Block **552** indicates an example in which the control zone definition criteria are or include other items as well. Block **549** indicates an example in which the control zone definition criteria are time based, meaning that agricultural harvester **100** will not cross the boundary of a control zone until a selected amount of time has elapsed since agricultural harvester **100** entered a particular control zone. In some instances, the selected amount of time may be a minimum amount of time. Thus, in some instances, the control zone definition criteria may prevent the agricultural harvester **100** from crossing a boundary of a control zone until at least the selected amount of time has elapsed. Block **551** indicates an example in which the control zone definition criteria are based on a selected size value. For example, control zone definition criteria that are based on a selected size value may preclude definition of a control zone that is smaller than the selected size. In some instances, the selected size may be a minimum size.

At block **554**, regime zone criteria identification component **522** obtains regime zone definition criteria for the selected WMA or set of WMAs. Block **556** indicates an example in which the regime zone definition criteria are based on a manual input from operator **260** or another user. Block **558** illustrates an example in which the regime zone definition criteria are based on power characteristics such as power characteristics (e.g., power consumption or power utilization) machine wide (e.g., power characteristics across all power consuming or power utilizing subsystems of agricultural harvester **100**) or power characteristics of one or more particular subsystems of agricultural harvester **100** (such as power characteristics of propulsion subsystem **250**).

Block **560** illustrates an example in which the regime zone definition criteria are based on soil properties (such as soil type or soil moisture, or both). Block **564** indicates an example in which the regime zone definition criteria are or include other criteria as well.

At block **566**, control zone boundary definition component **496** generates the boundaries of control zones on the map under analysis based upon the control zone criteria. Regime zone boundary definition component **524** generates the boundaries of regime zones on the map under analysis based upon the regime zone criteria. Block **568** indicates an example in which the zone boundaries are identified for the control zones and the regime zones. Block **570** shows that target setting identifier component **498** identifies the target settings for each of the control zones. The control zones and regime zones can be generated in other ways as well, and this is indicated by block **572**.

At block **574**, settings resolver identifier component **526** identifies the settings resolver for the selected WMAs in each regime zone defined by regimes zone boundary definition component **524**. As discussed above, the regime zone resolver can be a human resolver **576**, an artificial intelligence or machine learning system resolver **578**, a resolver **580** based on predicted or historic quality for each competing target setting, a rules-based resolver **582**, a performance criteria-based resolver **584**, or other resolvers **586**.

At block **588**, WMA selector **486** determines whether there are more WMAs or sets of WMAs to process. If additional WMAs or sets of WMAs are remaining to be processed, processing reverts to block **436** where the next WMA or set of WMAs for which control zones and regime zones are to be defined is selected. When no additional WMAs or sets of WMAs for which control zones or regime zones are to be generated are remaining, processing moves to block **590** where control zone generator **213** outputs a map with control zones, target settings, regime zones, and settings resolvers for each of the WMAs or sets of WMAs. As discussed above, the outputted map can be presented to operator **260** or another user; the outputted map can be provided to control system **214**; or the outputted map can be output in other ways.

FIG. **10** illustrates one example of the operation of control system **214** in controlling agricultural harvester **100** based upon a map that is output by control zone generator **213**. Thus, at block **592**, control system **214** receives a map of the worksite. In some instances, the map can be a functional predictive map that may include control zones and regime zones, as represented by block **594**. In some instances, the received map may be a functional predictive map that excludes control zones and regime zones. Block **596** indicates an example in which the received map of the worksite can be a prior information map having control zones and regime zones identified on it. Block **598** indicates an example in which the received map can include multiple different maps or multiple different map layers. Block **610** indicates an example in which the received map can take other forms as well.

At block **612**, control system **214** receives a sensor signal from geographic position sensor **204**. The sensor signal from geographic position sensor **204** can include data that indicates the geographic location **614** of agricultural harvester **100**, the speed **616** of agricultural harvester **100**, the heading **618** or agricultural harvester **100**, or other information **620**. At block **622**, zone controller **247** selects a regime zone, and, at block **624**, zone controller **247** selects a control zone on the map based on the geographic position sensor signal. At block **626**, zone controller **247** selects a WMA or a set of

WMAs to be controlled. At block 628, zone controller 247 obtains one or more target settings for the selected WMA or set of WMAs. The target settings that are obtained for the selected WMA or set of WMAs may come from a variety of different sources. For instance, block 630 shows an example in which one or more of the target settings for the selected WMA or set of WMAs is based on an input from the control zones on the map of the worksite. Block 632 shows an example in which one or more of the target settings is obtained from human inputs from operator 260 or another user. Block 634 shows an example in which the target settings are obtained from an in-situ sensor 208. Block 636 shows an example in which the one or more target settings is obtained from one or more sensors on other machines working in the same field either concurrently with agricultural harvester 100 or from one or more sensors on machines that worked in the same field in the past. Block 638 shows an example in which the target settings are obtained from other sources as well.

At block 640, zone controller 247 accesses the settings resolver for the selected regime zone and controls the settings resolver to resolve competing target settings into a resolved target setting. As discussed above, in some instances, the settings resolver may be a human resolver in which case zone controller 247 controls operator interface mechanisms 218 to present the competing target settings to operator 260 or another user for resolution. In some instances, the settings resolver may be a neural network or other artificial intelligence or machine learning system, and zone controller 247 submits the competing target settings to the neural network, artificial intelligence, or machine learning system for selection. In some instances, the settings resolver may be based on a predicted or historic quality metric, on threshold rules, or on logical components. In any of these latter examples, zone controller 247 executes the settings resolver to obtain a resolved target setting based on the predicted or historic quality metric, based on the threshold rules, or with the use of the logical components.

At block 642, with zone controller 247 having identified the resolved target setting, zone controller 247 provides the resolved target setting to other controllers in control system 214, which generate and apply control signals to the selected WMA or set of WMAs based upon the resolved target setting. For instance, where the selected WMA is a machine or header actuator 248, zone controller 247 provides the resolved target setting to settings controller 232 or header/real controller 238 or both to generate control signals based upon the resolved target setting, and those generated control signals are applied to the machine or header actuators 248. At block 644, if additional WMAs or additional sets of WMAs are to be controlled at the current geographic location of the agricultural harvester 100 (as detected at block 612), then processing reverts to block 626 where the next WMA or set of WMAs is selected. The processes represented by blocks 626 through 644 continue until all of the WMAs or sets of WMAs to be controlled at the current geographical location of the agricultural harvester 100 have been addressed. If no additional WMAs or sets of WMAs are to be controlled at the current geographic location of the agricultural harvester 100 remain, processing proceeds to block 646 where zone controller 247 determines whether additional control zones to be considered exist in the selected regime zone. If additional control zones to be considered exist, processing reverts to block 624 where a next control zone is selected. If no additional control zones are remaining to be considered, processing proceeds to block 648 where a determination as to whether additional

regime zones are remaining to be considered. Zone controller 247 determines whether additional regime zones are remaining to be considered. If additional regimes zone are remaining to be considered, processing reverts to block 622 where a next regime zone is selected.

At block 650, zone controller 247 determines whether the operation that agricultural harvester 100 is performing is complete. If not, the zone controller 247 determines whether a control zone criterion has been satisfied to continue processing, as indicated by block 652. For instance, as mentioned above, control zone definition criteria may include criteria defining when a control zone boundary may be crossed by the agricultural harvester 100. For example, whether a control zone boundary may be crossed by the agricultural harvester 100 may be defined by a selected time period, meaning that agricultural harvester 100 is prevented from crossing a zone boundary until a selected amount of time has transpired. In that case, at block 652, zone controller 247 determines whether the selected time period has elapsed. Additionally, zone controller 247 can perform processing continually. Thus, zone controller 247 does not wait for any particular time period before continuing to determine whether an operation of the agricultural harvester 100 is completed. At block 652, zone controller 247 determines that it is time to continue processing, then processing continues at block 612 where zone controller 247 again receives an input from geographic position sensor 204. It will also be appreciated that zone controller 247 can control the WMAs and sets of WMAs simultaneously using a multiple-input, multiple-output controller instead of controlling the WMAs and sets of WMAs sequentially.

FIG. 11 is a block diagram showing one example of an operator interface controller 231. In an illustrated example, operator interface controller 231 includes operator input command processing system 654, other controller interaction system 656, speech processing system 658, and action signal generator 660. Operator input command processing system 654 includes speech handling system 662, touch gesture handling system 664, and other items 666. Other controller interaction system 656 includes controller input processing system 668 and controller output generator 670. Speech processing system 658 includes trigger detector 672, recognition component 674, synthesis component 676, natural language understanding system 678, dialog management system 680, and other items 682. Action signal generator 660 includes visual control signal generator 684, audio control signal generator 686, haptic control signal generator 688, and other items 690. Before describing operation of the example operator interface controller 231 shown in FIG. 11 in handling various operator interface actions, a brief description of some of the items in operator interface controller 231 and the associated operation thereof is first provided.

Operator input command processing system 654 detects operator inputs on operator interface mechanisms 218 and processes those inputs for commands. Speech handling system 662 detects speech inputs and handles the interactions with speech processing system 658 to process the speech inputs for commands. Touch gesture handling system 664 detects touch gestures on touch sensitive elements in operator interface mechanisms 218 and processes those inputs for commands.

Other controller interaction system 656 handles interactions with other controllers in control system 214. Controller input processing system 668 detects and processes inputs from other controllers in control system 214, and controller output generator 670 generates outputs and provides those

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outputs to other controllers in control system **214**. Speech processing system **658** recognizes speech inputs, determines the meaning of those inputs, and provides an output indicative of the meaning of the spoken inputs. For instance, speech processing system **658** may recognize a speech input from operator **260** as a settings change command in which operator **260** is commanding control system **214** to change a setting for a controllable subsystem **216**. In such an example, speech processing system **658** recognizes the content of the spoken command, identifies the meaning of that command as a settings change command, and provides the meaning of that input back to speech handling system **662**. Speech handling system **662**, in turn, interacts with controller output generator **670** to provide the commanded output to the appropriate controller in control system **214** to accomplish the spoken settings change command.

Speech processing system **658** may be invoked in a variety of different ways. For instance, in one example, speech handling system **662** continuously provides an input from a microphone (being one of the operator interface mechanisms **218**) to speech processing system **658**. The microphone detects speech from operator **260**, and the speech handling system **662** provides the detected speech to speech processing system **658**. Trigger detector **672** detects a trigger indicating that speech processing system **658** is invoked. In some instances, when speech processing system **658** is receiving continuous speech inputs from speech handling system **662**, speech recognition component **674** performs continuous speech recognition on all speech spoken by operator **260**. In some instances, speech processing system **658** is configured for invocation using a wakeup word. That is, in some instances, operation of speech processing system **658** may be initiated based on recognition of a selected spoken word, referred to as the wakeup word. In such an example, where recognition component **674** recognizes the wakeup word, the recognition component **674** provides an indication that the wakeup word has been recognized to trigger detector **672**. Trigger detector **672** detects that speech processing system **658** has been invoked or triggered by the wakeup word. In another example, speech processing system **658** may be invoked by an operator **260** actuating an actuator on a user interface mechanism, such as by touching an actuator on a touch sensitive display screen, by pressing a button, or by providing another triggering input. In such an example, trigger detector **672** can detect that speech processing system **658** has been invoked when a triggering input via a user interface mechanism is detected. Trigger detector **672** can detect that speech processing system **658** has been invoked in other ways as well.

Once speech processing system **658** is invoked, the speech input from operator **260** is provided to speech recognition component **674**. Speech recognition component **674** recognizes linguistic elements in the speech input, such as words, phrases, or other linguistic units. Natural language understanding system **678** identifies a meaning of the recognized speech. The meaning may be a natural language output, a command output identifying a command reflected in the recognized speech, a value output identifying a value in the recognized speech, or any of a wide variety of other outputs that reflect the understanding of the recognized speech. For example, the natural language understanding system **678** and speech processing system **658**, more generally, may understand of the meaning of the recognized speech in the context of agricultural harvester **100**.

In some examples, speech processing system **658** can also generate outputs that navigate operator **260** through a user experience based on the speech input. For instance, dialog

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management system **680** may generate and manage a dialog with the user in order to identify what the user wishes to do. The dialog may disambiguate a user's command; identify one or more specific values that are needed to carry out the user's command; or obtain other information from the user or provide other information to the user or both. Synthesis component **676** may generate speech synthesis which can be presented to the user through an audio operator interface mechanism, such as a speaker. Thus, the dialog managed by dialog management system **680** may be exclusively a spoken dialog or a combination of both a visual dialog and a spoken dialog.

Action signal generator **660** generates action signals to control operator interface mechanisms **218** based upon outputs from one or more of operator input command processing system **654**, other controller interaction system **656**, and speech processing system **658**. Visual control signal generator **684** generates control signals to control visual items in operator interface mechanisms **218**. The visual items may be lights, a display screen, warning indicators, or other visual items. Audio control signal generator **686** generates outputs that control audio elements of operator interface mechanisms **218**. The audio elements include a speaker, audible alert mechanisms, horns, or other audible elements. Haptic control signal generator **688** generates control signals that are output to control haptic elements of operator interface mechanisms **218**. The haptic elements include vibration elements that may be used to vibrate, for example, the operator's seat, the steering wheel, pedals, or joysticks used by the operator. The haptic elements may include tactile feedback or force feedback elements that provide tactile feedback or force feedback to the operator through operator interface mechanisms. The haptic elements may include a wide variety of other haptic elements as well.

FIG. **12** is a flow diagram illustrating one example of the operation of operator interface controller **231** in generating an operator interface display on an operator interface mechanism **218**, which can include a touch sensitive display screen. FIG. **12** also illustrates one example of how operator interface controller **231** can detect and process operator interactions with the touch sensitive display screen.

At block **692**, operator interface controller **231** receives a map. Block **694** indicates an example in which the map is a functional predictive map, and block **696** indicates an example in which the map is another type of map. At block **698**, operator interface controller **231** receives an input from geographic position sensor **204** identifying the geographic location of the agricultural harvester **100**. As indicated in block **700**, the input from geographic position sensor **204** can include the heading, along with the location, of agricultural harvester **100**. Block **702** indicates an example in which the input from geographic position sensor **204** includes the speed of agricultural harvester **100**, and block **704** indicates an example in which the input from geographic position sensor **204** includes other items.

At block **706**, visual control signal generator **684** in operator interface controller **231** controls the touch sensitive display screen in operator interface mechanisms **218** to generate a display showing all or a portion of a field represented by the received map. Block **708** indicates that the displayed field can include a current position marker showing a current position of the agricultural harvester **100** relative to the field. Block **710** indicates an example in which the displayed field includes a next work unit marker that identifies a next work unit (or area on the field) in which agricultural harvester **100** will be operating. Block **712** indicates an example in which the displayed field includes

an upcoming area display portion that displays areas that are yet to be processed by agricultural harvester **100**, and block **714** indicates an example in which the displayed field includes previously visited display portions that represent areas of the field that agricultural harvester **100** has already processed. Block **716** indicates an example in which the displayed field displays various characteristics of the field having georeferenced locations on the map. For instance, if the received map is a power map, such as functional predictive map **360** or **428**, the displayed field may show the different power characteristics existing in the field georeferenced within the displayed field. The mapped characteristics can be shown in the previously visited areas (as shown in block **714**), in the upcoming areas (as shown in block **712**), and in the next work unit (as shown in block **710**). Block **718** indicates an example in which the displayed field includes other items as well.

FIG. **13** is a pictorial illustration showing one example of a user interface display **720** that can be generated on a touch sensitive display screen. In other implementations, the user interface display **720** may be generated on other types of displays. The touch sensitive display screen may be mounted in the operator compartment of agricultural harvester **100** or on the mobile device or elsewhere. User interface display **720** will be described prior to continuing with the description of the flow diagram shown in FIG. **12**.

In the example shown in FIG. **13**, user interface display **720** illustrates that the touch sensitive display screen includes a display feature for operating a microphone **722** and a speaker **724**. Thus, the touch sensitive display may be communicably coupled to the microphone **722** and the speaker **724**. Block **726** indicates that the touch sensitive display screen can include a wide variety of user interface control actuators, such as buttons, keypads, soft keypads, links, icons, switches, etc. The operator **260** can actuate the user interface control actuators to perform various functions.

In the example shown in FIG. **13**, user interface display **720** includes a field display portion **728** that displays at least a portion of the field in which the agricultural harvester **100** is operating. The field display portion **728** is shown with a current position marker **708** that corresponds to a current position of agricultural harvester **100** in the portion of the field shown in field display portion **728**. In one example, the operator may control the touch sensitive display in order to zoom into portions of field display portion **728** or to pan or scroll the field display portion **728** to show different portions of the field. A next work unit **730** is shown as an area of the field directly in front of the current position marker **708** of agricultural harvester **100**. The current position marker **708** may also be configured to identify the direction of travel of agricultural harvester **100**, a speed of travel of agricultural harvester **100** or both. In FIG. **13**, the shape of the current position marker **708** provides an indication as to the orientation of the agricultural harvester **100** within the field which may be used as an indication of a direction of travel of the agricultural harvester **100**.

The size of the next work unit **730** marked on field display portion **728** may vary based upon a wide variety of different criteria. For instance, the size of next work unit **730** may vary based on the speed of travel of agricultural harvester **100**. Thus, when the agricultural harvester **100** is traveling faster, then the area of the next work unit **730** may be larger than the area of next work unit **730** if agricultural harvester **100** is traveling more slowly. In another example, the size of the next work unit **730** may vary based on the dimensions of the agricultural harvester **100**, including equipment on agricultural harvester **100** (such as header **102**). For example,

the width of the next work unit **730** may vary based on a width of header **102**. Field display portion **728** is also shown displaying previously visited area **714** and upcoming areas **712**. Previously visited areas **714** represent areas that are already harvested while upcoming areas **712** represent areas that still need to be harvested. The field display portion **728** is also shown displaying different characteristics of the field. In the example illustrated in FIG. **13**, the map that is being displayed is a predictive power map, such as functional predictive map **360** or functional predictive map **428**. Therefore, a plurality of power characteristic markers are displayed on field display portion **728**. There are a set of power characteristic display markers **732** shown in the already visited areas **714**. There are also a set of power characteristic display markers **732** shown in the upcoming areas **712**, and there are a set of power characteristic display markers **732** shown in the next work unit **730**. FIG. **13** shows that the ear size display markers **732** are made up of different symbols that indicate an area of similar power characteristics. In the example shown in FIG. **14**, the ! symbol represents areas of high power utilization; the * symbol represents areas of medium power utilization; and the # symbol represents an area of low power utilization. Power utilization is merely one example, other power characteristics, including power characteristics of particular subsystems, can also be displayed. The field display portion **728** shows different measured or predicted values (or characteristics indicated by the values) that are located at different areas within the field and represents those measured or predicted values (or characteristics indicated by or derived from the values) with a variety of display markers **732**. As shown, the field display portion **728** includes display markers, particularly power characteristic display markers **732** in the illustrated example of FIG. **13**, at particular locations associated with particular locations on the field being displayed. In some instances, each location of the field may have a display marker associated therewith. Thus, in some instances, a display marker may be provided at each location of the field display portion **728** to identify the nature of the characteristic being mapped for each particular location of the field. Consequently, the present disclosure encompasses providing a display marker, such as the power characteristic display marker **732** (as in the context of the present example of FIG. **13**), at one or more locations on the field display portion **728** to identify the nature, degree, etc., of the characteristic being displayed, thereby identifying the characteristic at the corresponding location in the field being displayed. As described earlier, the display markers **732** may be made up of different symbols, and, as described below, the symbols may be any display feature such as different colors, shapes, patterns, intensities, text, icons, or other display features. In some instances, each location of the field may have a display marker associated therewith. Thus, in some instances, a display marker may be provided at each location of the field display portion **728** to identify the nature of the characteristic being mapped for each particular location of the field. Consequently, the present disclosure encompasses providing a display marker, such as the loss level display marker **732** (as in the context of the present example of FIG. **11**), at one or more locations on the field display portion **728** to identify the nature, degree, etc., of the characteristic being displayed, thereby identifying the characteristic at the corresponding location in the field being displayed.

In other examples, the map being displayed may be one or more of the maps described herein, including information maps, prior information maps, the functional predictive maps, such as predictive maps or predictive control zone

maps, or a combination thereof. Thus, the markers and characteristics being displayed will correlate to the information, data, characteristics, and values provided by the one or more maps being displayed.

In the example of FIG. 13 user interface display 720 also has a control display portion 738. Control display portion 738 allows the operator to view information and to interact with user interface display 720 in various ways.

The actuators and display markers in portion 738 may be displayed as, for example, individual items, fixed lists, scrollable lists, drop down menus, or drop down lists. In the example shown in FIG. 13, display portion 738 shows information for the three different ear size categories that correspond to the three symbols mentioned above. Display portion 738 also includes a set of touch sensitive actuators with which the operator 260 can interact by touch. For example, the operator 260 may touch the touch sensitive actuators with a finger to activate the respective touch sensitive actuator. As shown in FIG. 13, display portion 738 also includes interactive tabs, such as machine wide tab 762, propulsion tab 764, threshing/separating tab 766, residue tab 768, and other tab 770. Activating one of the tabs can modify which values are displayed in portions 728 and 738. For instance, as shown, propulsion tab 764 is activated and thus the values mapped on portion 728 and shown in portion 738 correspond to power characteristics (shown as power utilization) across propulsion subsystem 250 of agricultural harvester 100. When the operator 260 touches the tab 762, touch gesture handling system 664 updates portion 728 and 738 to display characteristics relating to all subsystems of agricultural harvester 100. When the operator 260 touches the tab 766, touch gesture handling system 664 updates portion 728 and 738 to display characteristics relating to threshing and separating subsystems. When the operator 260 touches the tab 768, touch gesture handling system 664 updates portion 728 and 738 to display characteristics relating to residue subsystem 138. When the operator 260 touches the tab 770, touch gesture handling system 664 updates portion 728 and 738 to display characteristics relating to another set of components of agricultural harvester 100.

As shown in FIG. 13, display portion 738 includes an interactive flag display portion, indicated generally at 741. Interactive flag display portion 741 includes a flag column 739 that shows flags that have been automatically or manually set. Flag actuator 740 allows operator 260 to mark a location, such as the current location of the agricultural harvester, or another location on the field designated by the operator and add information indicating the characteristic, such as power utilization, found at the current location. For instance, when the operator 260 actuates the flag actuator 740 by touching the flag actuator 740, touch gesture handling system 664 in operator interface controller 231 identifies the current location as one where agricultural harvester 100 encountered high power utilization. When the operator 260 touches the button 742, touch gesture handling system 664 identifies the current location as a location where agricultural harvester 100 encountered medium power utilization. When the operator 260 touches the button 744, touch gesture handling system 664 identifies the current location as a location where agricultural harvester 100 encountered low power utilization. Upon actuation of one of the flag actuators 740, 742, or 744, touch gesture handling system 664 can control visual control signal generator 684 to add a symbol corresponding to the identified characteristic on field display portion 728 at a location the user identifies. In this way, areas of the field where the predicted

value did not accurately represent an actual value can be marked for later analysis, and can also be used in machine learning. In other examples, the operator may designate areas ahead of or around the agricultural harvester 100 by actuating one of the flag actuators 740, 742, or 744 such that control of the agricultural harvester 100 can be undertaken based on the value designated by the operator 260.

Display portion 738 also includes an interactive marker display portion, indicated generally at 743. Interactive marker display portion 743 includes a symbol column 746 that displays the symbols corresponding to each category of values or characteristics (in the case of FIG. 13, power utilization) that is being tracked on the field display portion 728. Display portion 738 also includes an interactive designator display portion, indicated generally at 745. Interactive designator display portion 745 includes a designator column 748 that shows the designator (which may be a textual designator or other designator) identifying the category of values or characteristics (in the case of FIG. 13, power utilization). Without limitation, the symbols in symbol column 746 and the designators in designator column 748 can include any display feature such as different colors, shapes, patterns, intensities, text, icons, or other display features, and can be customizable by interaction of an operator of agricultural harvester 100.

Display portion 738 also includes an interactive value display portion, indicated generally at 747. Interactive value display portion 747 includes a value display column 750 that displays selected values. The selected values correspond to the characteristics or values being tracked or displayed, or both, on field display portion 728. The selected values can be selected by an operator of the agricultural harvester 100. The selected values in value display column 750 define a range of values or a value by which other values, such as predicted values, are to be classified. Thus, in the example in FIG. 13, a predicted or measured propulsion subsystem power utilization meeting or greater than 45% is classified as "high power utilization" and a predicted or measured propulsion subsystem power utilization meeting or less than 35% is classified as "low power utilization." In some examples, the selected values may include a range, such that a predicted or measured value that is within the range of the selected value will be classified under the corresponding designator. As shown in FIG. 13, "medium power utilization" includes a range of 36%-44% such that a predicted or measured power utilization falling within the range of 36%-44% is classified as "medium power utilization". The selected values in value display column 750 are adjustable by an operator of agricultural harvester 100. In one example, the operator 260 can select the particular part of field display portion 728 for which the values in column 750 are to be displayed. Thus, the values in column 750 can correspond to values in display portions 712, 714 or 730.

Display portion 738 also includes an interactive threshold display portion, indicated generally at 749. Interactive threshold display portion 749 includes a threshold value display column 752 that displays action threshold values. Action threshold values in column 752 may be threshold values corresponding to the selected values in value display column 750. If the predicted or measured values of characteristics being tracked or displayed, or both, satisfy the corresponding action threshold values in threshold value display column 752, then control system 214 takes an action identified in column 754. In some instances, a measured or predicted value may satisfy a corresponding action threshold value by meeting or exceeding the corresponding action threshold value. In one example, operator 260 can select a

threshold value, for example, in order to change the threshold value by touching the threshold value in threshold value display column 752. Once selected, the operator 260 may change the threshold value. The threshold values in column 752 can be configured such that the designated action is performed when the measured or predicted value of the characteristic exceeds the threshold value, equals the threshold value, or is less than the threshold value. In some instances, the threshold value may represent a range of values, or range of deviation from the selected values in value display column 750, such that a predicted or measured characteristic value that meets or falls within the range satisfies the threshold value. For instance, in the example of propulsion subsystem power utilization, a predicted propulsion subsystem power utilization that falls within 10% of 45% power utilization will satisfy the corresponding action threshold value (of within 10% of 45% power utilization) and an action, such as reducing the vehicle speed, will be taken by control system 214. In other examples, the threshold values in column threshold value display column 752 are separate from the selected values in value display column 750, such that the values in value display column 750 define the classification and display of predicted or measured values, while the action threshold values define when an action is to be taken based on the measured or predicted values. For example, while a predicted or measured propulsion subsystem power utilization of 40% may be designated as a "medium power utilization" for purposes of classification and display, the action threshold value may be 41% power utilization such that no action will be taken until the propulsion subsystem power utilization satisfies the threshold value. In other examples, the threshold values in threshold value display column 752 may include distances or times. For instance, in the example of a distance, the threshold value may be a threshold distance from the area of the field where the measured or predicted value is georeferenced that the agricultural harvester 100 must be before an action is taken. For example, a threshold distance value of 5 feet would mean that an action will be taken when the agricultural harvester is at or within 5 feet of the area of the field where the measured or predicted value is georeferenced. In an example where the threshold value is time, the threshold value may be a threshold time for the agricultural harvester 100 to reach the area of the field where the measured or predictive value is georeferenced. For instance, a threshold value of 5 seconds would mean that an action will be taken when the agricultural harvester 100 is 5 seconds away from the area of the field where the measured or predicted value is georeferenced. In such an example, the current location and travel speed of the agricultural harvester can be accounted for.

Display portion 738 also includes an interactive action display portion, indicated generally at 751. Interactive action display portion 751 includes an action display column 754 that displays action identifiers that indicate actions to be taken when a predicted or measured value satisfies an action threshold value in threshold value display column 752. Operator 260 can touch the action identifiers in column 754 to change the action that is to be taken. When a threshold is satisfied, an action may be taken. For instance, at the bottom of column 754, an increase travel speed action, a decrease travel speed action, and an adjust header position action (such as adjusting a header height, a header pitch, or a header roll, relative to the surface or relative to the frame of agricultural harvester 100) are identified as actions that will be taken if the measured or predicted value meets the threshold value in column 752. In some examples, when a

threshold is met, multiple actions may be taken. For instance, a header position may be adjusted, a power output to one or more subsystems may be adjusted, and a speed of the agricultural machine may be adjusted. These are merely some examples.

The actions that can be set in column 754 can be any of a wide variety of different types of actions. For example, the actions can include a keep out action which, when executed, inhibits agricultural harvester 100 from further harvesting in an area. The actions can include a speed change action which, when executed, changes the travel speed of agricultural harvester 100 through the field. The actions can include a setting change action for changing a setting of an internal actuator or another WMA or set of WMAs or for implementing a settings change action that changes a setting, such as a header position setting along with various other settings. These are examples only, and a wide variety of other actions are contemplated herein.

The items shown on user interface display 720 can be visually controlled. Visually controlling the interface display 720 may be performed to capture the attention of operator 260. For instance, the items can be controlled to modify the intensity, color, or pattern with which the items are displayed. Additionally, the items may be controlled to flash. The described alterations to the visual appearance of the items are provided as examples. Consequently, other aspects of the visual appearance of the items may be altered. Therefore, the items can be modified under various circumstances in a desired manner in order, for example, to capture the attention of operator 260. Additionally, while a particular number of items are shown on user interface display 720, this need not be the case. In other examples, more or less items, including more or less of a particular item can be included on user interface display 720.

Returning now to the flow diagram of FIG. 12, the description of the operation of operator interface controller 231 continues. At block 760, operator interface controller 231 detects an input setting a flag and controls the touch sensitive user interface display 720 to display the flag on field display portion 728. The detected input may be an operator input, as indicated at 762, or an input from another controller, as indicated at 764. At block 766, operator interface controller 231 detects an in-situ sensor input indicative of a measured characteristic of the field from one of the in-situ sensors 208. At block 768, visual control signal generator 684 generates control signals to control user interface display 720 to display actuators for modifying user interface display 720 and for modifying machine control. For instance, block 770 represents that one or more of the actuators for setting or modifying the values in columns 739, 746, and 748 can be displayed. Thus, the user can set flags and modify characteristics of those flags. Block 772 represents that action threshold values in column 752 are displayed. Block 776 represents that the actions in column 754 are displayed, and block 778 represents that the selected value in column 750 is displayed. Block 780 indicates that a wide variety of other information and actuators can be displayed on user interface display 720 as well.

At block 782, operator input command processing system 654 detects and processes operator inputs corresponding to interactions with the user interface display 720 performed by the operator 260. Where the user interface mechanism on which user interface display 720 is displayed is a touch sensitive display screen, interaction inputs with the touch sensitive display screen by the operator 260 can be touch gestures 784. In some instances, the operator interaction

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inputs can be inputs using a point and click device **786** or other operator interaction inputs **788**.

At block **790**, operator interface controller **231** receives signals indicative of an alert condition. For instance, block **792** indicates that signals may be received by controller **5** input processing system **668** indicating that detected or predicted values satisfy threshold conditions present in column **752**. As explained earlier, the threshold conditions may include values being below a threshold, at a threshold, or above a threshold. Block **794** shows that action signal generator **660** can, in response to receiving an alert condition, alert the operator **260** by using visual control signal generator **684** to generate visual alerts, by using audio control signal generator **686** to generate audio alerts, by using haptic control signal generator **688** to generate haptic alerts, or by using any combination of these. Similarly, as indicated by block **796**, controller output generator **670** can generate outputs to other controllers in control system **214** **15**

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so that those controllers perform the corresponding action identified in column **754**. Block **798** shows that operator interface controller **231** can detect and process alert conditions in other ways as well.

Block **900** shows that speech handling system **662** may detect and process inputs invoking speech processing system **658**. Block **902** shows that performing speech processing may include the use of dialog management system **680** to conduct a dialog with the operator **260**. Block **904** shows that the speech processing may include providing signals to controller output generator **670** so that control operations are automatically performed based upon the speech inputs.

Table 1, below, shows an example of a dialog between operator interface controller **231** and operator **260**. In Table 1, operator **260** uses a trigger word or a wakeup word that is detected by trigger detector **672** to invoke speech processing system **658**. In the example shown in Table 1, the wakeup word is “Johnny”.

TABLE 1

Operator: “Johnny, tell me about current power utilization”
 Operator Interface Controller: “Propulsion Power Utilization is currently 40%”
 Operator: “Johnny, what should I do at the current power utilization?”
 Operator Interface Controller: “Power utilization can be increased to 45% if the machine speed is increased 1 MPH.”

Table 2 shows an example in which speech synthesis component **676** provides an output to audio control signal generator **686** to provide audible updates on an intermittent or periodic basis. The interval between updates may be time-based, such as every five minutes, or coverage or distance-based, such as every five acres, or exception-based, such as when a measured value is greater than a threshold value. **30**

TABLE 2

Operator Interface Controller: “Over last 10 minutes, propulsion power utilization has averaged 43%”
 Operator Interface Controller: “Next 1 acre predicted propulsion power utilization is 41%.”
 Operator Interface Controller: “Caution: power utilization falling below 35%. Machine speed increasing.”

The example shown in Table 3 illustrates that some actuators or user input mechanisms on the touch sensitive display **720** can be supplemented with speech dialog. The example in Table 3 illustrates that action signal generator **660** can generate action signals to automatically mark a power utilization area in the field being harvested. **50**

TABLE 3

Human: “Johnny, mark high power utilization area.”
 Operator Interface Controller: “High power utilization area marked.”

The example shown in Table 4 illustrates that action signal generator **660** can conduct a dialog with operator **260** to begin and end marking of a power utilization area.

TABLE 4

Human: “Johnny, start marking high power utilization area.”
 Operator Interface Controller: “Marking high power utilization area.”
 Human: “Johnny, stop marking high power utilization area.”
 Operator Interface Controller: “High power utilization area marking stopped.”

The example shown in Table 5 illustrates that action signal generator **160** can generate signals to mark a power utilization area in a different way than those shown in Tables 3 and 4.

TABLE 5

Human: "Johnny, mark next 100 feet as a low power utilization area."
Operator Interface Controller: "Next 100 feet marked as a low power utilization area."

Returning again to FIG. 12, block **906** illustrates that operator interface controller **231** can detect and process conditions for outputting a message or other information in other ways as well. For instance, other controller interaction system **656** can detect inputs from other controllers indicating that alerts or output messages should be presented to operator **260**. Block **908** shows that the outputs can be audio messages. Block **910** shows that the outputs can be visual messages, and block **912** shows that the outputs can be haptic messages. Until operator interface controller **231** determines that the current harvesting operation is completed, as indicated by block **914**, processing reverts to block **698** where the geographic location of harvester **100** is updated and processing proceeds as described above to update user interface display **720**.

Once the operation is complete, then any desired values that are displayed, or have been displayed on user interface display **720**, can be saved. Those values can also be used in machine learning to improve different portions of predictive model generator **210**, predictive map generator **212**, control zone generator **213**, control algorithms, or other items. Saving the desired values is indicated by block **916**. The values can be saved locally on agricultural harvester **100**, or the values can be saved at a remote server location or sent to another remote system.

It can thus be seen that one or more maps are obtained by an agricultural harvester that show agricultural characteristic values at different geographic locations of a field being harvested. An in-situ sensor on the harvester senses a characteristic that has values indicative of an agricultural characteristic as the agricultural harvester moves through the field. A predictive map generator generates a predictive map that predicts control values for different locations in the field based on the values of the agricultural characteristic in the map and the agricultural characteristic sensed by the in-situ sensor. A control system controls controllable subsystem based on the control values in the predictive map.

A control value is a value upon which an action can be based. A control value, as described herein, can include any value (or characteristics indicated by or derived from the value) that may be used in the control of agricultural harvester **100**. A control value can be any value indicative of an agricultural characteristic. A control value can be a predicted value, a measured value, or a detected value. A control value may include any of the values provided by a map, such as any of the maps described herein, for instance, a control value can be a value provided by an information map, a value provided by prior information map, or a value provided predictive map, such as a functional predictive map. A control value can also include any of the characteristics indicated by or derived from the values detected by any of the sensors described herein. In other examples, a control value can be provided by an operator of the agricultural machine, such as a command input by an operator of the agricultural machine.

The present discussion has mentioned processors and servers. In some examples, the processors and servers include computer processors with associated memory and timing circuitry, not separately shown. The processors and servers are functional parts of the systems or devices to which the processors and servers belong and are activated by and facilitate the functionality of the other components or items in those systems.

Also, a number of user interface displays have been discussed. The displays can take a wide variety of different forms and can have a wide variety of different user actuable operator interface mechanisms disposed thereon. For instance, user actuable operator interface mechanisms may include text boxes, check boxes, icons, links, drop-down menus, search boxes, etc. The user actuable operator interface mechanisms can also be actuated in a wide variety of different ways. For instance, the user actuable operator interface mechanisms can be actuated using operator interface mechanisms such as a point and click device, such as a track ball or mouse, hardware buttons, switches, a joystick or keyboard, thumb switches or thumb pads, etc., a virtual keyboard or other virtual actuators. In addition, where the screen on which the user actuable operator interface mechanisms are displayed is a touch sensitive screen, the user actuable operator interface mechanisms can be actuated using touch gestures. Also, user actuable operator interface mechanisms can be actuated using speech commands using speech recognition functionality. Speech recognition may be implemented using a speech detection device, such as a microphone, and software that functions to recognize detected speech and execute commands based on the received speech.

A number of data stores have also been discussed. It will be noted the data stores can each be broken into multiple data stores. In some examples, one or more of the data stores may be local to the systems accessing the data stores, one or more of the data stores may all be located remote from a system utilizing the data store, or one or more data stores may be local while others are remote. All of these configurations are contemplated by the present disclosure.

Also, the figures show a number of blocks with functionality ascribed to each block. It will be noted that fewer blocks can be used to illustrate that the functionality ascribed to multiple different blocks is performed by fewer components. Also, more blocks can be used illustrating that the functionality may be distributed among more components. In different examples, some functionality may be added, and some may be removed.

It will be noted that the above discussion has described a variety of different systems, components, logic, and interactions. It will be appreciated that any or all of such systems, components, logic and interactions may be implemented by hardware items, such as processors, memory, or other processing components, including but not limited to artificial intelligence components, such as neural networks, some of which are described below, that perform the functions asso-

ciated with those systems, components, logic, or interactions. In addition, any or all of the systems, components, logic and interactions may be implemented by software that is loaded into a memory and is subsequently executed by a processor or server or other computing component, as described below. Any or all of the systems, components, logic and interactions may also be implemented by different combinations of hardware, software, firmware, etc., some examples of which are described below. These are some examples of different structures that may be used to implement any or all of the systems, components, logic and interactions described above. Other structures may be used as well.

FIG. 14 is a block diagram of agricultural harvester 600, which may be similar to agricultural harvester 100 shown in FIG. 2. The agricultural harvester 600 communicates with elements in a remote server architecture 500. In some examples, remote server architecture 500 provides computation, software, data access, and storage services that do not require end-user knowledge of the physical location or configuration of the system that delivers the services. In various examples, remote servers may deliver the services over a wide area network, such as the internet, using appropriate protocols. For instance, remote servers may deliver applications over a wide area network and may be accessible through a web browser or any other computing component. Software or components shown in FIG. 2 as well as data associated therewith, may be stored on servers at a remote location. The computing resources in a remote server environment may be consolidated at a remote data center location, or the computing resources may be dispersed to a plurality of remote data centers. Remote server infrastructures may deliver services through shared data centers, even though the services appear as a single point of access for the user. Thus, the components and functions described herein may be provided from a remote server at a remote location using a remote server architecture. Alternatively, the components and functions may be provided from a server, or the components and functions can be installed on client devices directly, or in other ways.

In the example shown in FIG. 14, some items are similar to those shown in FIG. 2 and those items are similarly numbered. FIG. 14 specifically shows that predictive model generator 210 or predictive map generator 212, or both, may be located at a server location 502 that is remote from the agricultural harvester 600. Therefore, in the example shown in FIG. 14, agricultural harvester 600 accesses systems through remote server location 502.

FIG. 14 also depicts another example of a remote server architecture. FIG. 14 shows that some elements of FIG. 2 may be disposed at a remote server location 502 while others may be located elsewhere. By way of example, data store 202 may be disposed at a location separate from location 502 and accessed via the remote server at location 502. Regardless of where the elements are located, the elements can be accessed directly by agricultural harvester 600 through a network such as a wide area network or a local area network; the elements can be hosted at a remote site by a service; or the elements can be provided as a service or accessed by a connection service that resides in a remote location. Also, data may be stored in any location, and the stored data may be accessed by, or forwarded to, operators, users, or systems. For instance, physical carriers may be used instead of, or in addition to, electromagnetic wave carriers. In some examples, where wireless telecommunication service coverage is poor or nonexistent, another machine, such as a fuel truck or other mobile machine or vehicle, may have an

automated, semi-automated, or manual information collection system. As the combine harvester 600 comes close to the machine containing the information collection system, such as a fuel truck prior to fueling, the information collection system collects the information from the combine harvester 600 using any type of ad-hoc wireless connection. The collected information may then be forwarded to another network when the machine containing the received information reaches a location where wireless telecommunication service coverage or other wireless coverage—is available. For instance, a fuel truck may enter an area having wireless communication coverage when traveling to a location to fuel other machines or when at a main fuel storage location. All of these architectures are contemplated herein. Further, the information may be stored on the agricultural harvester 600 until the agricultural harvester 600 enters an area having wireless communication coverage. The agricultural harvester 600, itself, may send the information to another network.

It will also be noted that the elements of FIG. 2, or portions thereof, may be disposed on a wide variety of different devices. One or more of those devices may include an on-board computer, an electronic control unit, a display unit, a server, a desktop computer, a laptop computer, a tablet computer, or other mobile device, such as a palm top computer, a cell phone, a smart phone, a multimedia player, a personal digital assistant, etc.

In some examples, remote server architecture 500 may include cybersecurity measures. Without limitation, these measures may include encryption of data on storage devices, encryption of data sent between network nodes, authentication of people or processes accessing data, as well as the use of ledgers for recording metadata, data, data transfers, data accesses, and data transformations. In some examples, the ledgers may be distributed and immutable (e.g., implemented as blockchain).

FIG. 15 is a simplified block diagram of one illustrative example of a handheld or mobile computing device that can be used as a user's or client's hand held device 16, in which the present system (or parts of it) can be deployed. For instance, a mobile device can be deployed in the operator compartment of agricultural harvester 100 for use in generating, processing, or displaying the maps discussed above. FIGS. 16-17 are examples of handheld or mobile devices.

FIG. 15 provides a general block diagram of the components of a client device 16 that can run some components shown in FIG. 2, that interacts with them, or both. In the device 16, a communications link 13 is provided that allows the handheld device to communicate with other computing devices and under some examples provides a channel for receiving information automatically, such as by scanning. Examples of communications link 13 include allowing communication through one or more communication protocols, such as wireless services used to provide cellular access to a network, as well as protocols that provide local wireless connections to networks.

In other examples, applications can be received on a removable Secure Digital (SD) card that is connected to an interface 15. Interface 15 and communication links 13 communicate with a processor 17 (which can also embody processors or servers from other FIGS.) along a bus 19 that is also connected to memory 21 and input/output (I/O) components 23, as well as clock 25 and location system 27.

I/O components 23, in one example, are provided to facilitate input and output operations. I/O components 23 for various examples of the device 16 can include input components such as buttons, touch sensors, optical sensors,

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microphones, touch screens, proximity sensors, accelerometers, orientation sensors and output components such as a display device, a speaker, and or a printer port. Other I/O components 23 can be used as well.

Clock 25 illustratively comprises a real time clock component that outputs a time and date. It can also, illustratively, provide timing functions for processor 17.

Location system 27 illustratively includes a component that outputs a current geographical location of device 16. This can include, for instance, a global positioning system (GPS) receiver, a LORAN system, a dead reckoning system, a cellular triangulation system, or other positioning system. Location system 27 can also include, for example, mapping software or navigation software that generates desired maps, navigation routes and other geographic functions.

Memory 21 stores operating system 29, network settings 31, applications 33, application configuration settings 35, data store 37, communication drivers 39, and communication configuration settings 41. Memory 21 can include all types of tangible volatile and non-volatile computer-readable memory devices. Memory 21 may also include computer storage media (described below). Memory 21 stores computer readable instructions that, when executed by processor 17, cause the processor to perform computer-implemented steps or functions according to the instructions. Processor 17 may be activated by other components to facilitate their functionality as well.

FIG. 16 shows one example in which device 16 is a tablet computer 600. In FIG. 16, computer 601 is shown with user interface display screen 602. Screen 602 can be a touch screen or a pen-enabled interface that receives inputs from a pen or stylus. Tablet computer 600 may also use an on-screen virtual keyboard. Of course, computer 601 might also be attached to a keyboard or other user input device through a suitable attachment mechanism, such as a wireless link or USB port, for instance. Computer 601 may also illustratively receive voice inputs as well.

FIG. 17 is similar to FIG. 16 except that the device is a smart phone 71. Smart phone 71 has a touch sensitive display 73 that displays icons or tiles or other user input mechanisms 75. Mechanisms 75 can be used by a user to run applications, make calls, perform data transfer operations, etc. In general, smart phone 71 is built on a mobile operating system and offers more advanced computing capability and connectivity than a feature phone.

Note that other forms of the devices 16 are possible.

FIG. 18 is one example of a computing environment in which elements of FIG. 2 can be deployed. With reference to FIG. 18, an example system for implementing some embodiments includes a computing device in the form of a computer 810 programmed to operate as discussed above. Components of computer 810 may include, but are not limited to, a processing unit 820 (which can comprise processors or servers from previous FIGS.), a system memory 830, and a system bus 821 that couples various system components including the system memory to the processing unit 820. The system bus 821 may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. Memory and programs described with respect to FIG. 2 can be deployed in corresponding portions of FIG. 18.

Computer 810 typically includes a variety of computer readable media. Computer readable media may be any available media that can be accessed by computer 810 and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limita-

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tion, computer readable media may comprise computer storage media and communication media. Computer storage media is different from, and does not include, a modulated data signal or carrier wave. Computer readable media includes hardware storage media including both volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or other data. Computer storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by computer 810. Communication media may embody computer readable instructions, data structures, program modules or other data in a transport mechanism and includes any information delivery media. The term "modulated data signal" means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal.

The system memory 830 includes computer storage media in the form of volatile and/or nonvolatile memory or both such as read only memory (ROM) 831 and random access memory (RAM) 832. A basic input/output system 833 (BIOS), containing the basic routines that help to transfer information between elements within computer 810, such as during start-up, is typically stored in ROM 831. RAM 832 typically contains data or program modules or both that are immediately accessible to and/or presently being operated on by processing unit 820. By way of example, and not limitation, FIG. 18 illustrates operating system 834, application programs 835, other program modules 836, and program data 837.

The computer 810 may also include other removable/non-removable volatile/nonvolatile computer storage media. By way of example only, FIG. 18 illustrates a hard disk drive 841 that reads from or writes to non-removable, nonvolatile magnetic media, an optical disk drive 855, and nonvolatile optical disk 856. The hard disk drive 841 is typically connected to the system bus 821 through a non-removable memory interface such as interface 840, and optical disk drive 855 are typically connected to the system bus 821 by a removable memory interface, such as interface 850.

Alternatively, or in addition, the functionality described herein can be performed, at least in part, by one or more hardware logic components. For example, and without limitation, illustrative types of hardware logic components that can be used include Field-programmable Gate Arrays (FPGAs), Application-specific Integrated Circuits (e.g., ASICs), Application-specific Standard Products (e.g., ASSPs), System-on-a-chip systems (SOCs), Complex Programmable Logic Devices (CPLDs), etc.

The drives and their associated computer storage media discussed above and illustrated in FIG. 18, provide storage of computer readable instructions, data structures, program modules and other data for the computer 810. In FIG. 18, for example, hard disk drive 841 is illustrated as storing operating system 844, application programs 845, other program modules 846, and program data 847. Note that these components can either be the same as or different from operating system 834, application programs 835, other program modules 836, and program data 837.

A user may enter commands and information into the computer 810 through input devices such as a keyboard 862, a microphone 863, and a pointing device 861, such as a

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mouse, trackball or touch pad. Other input devices (not shown) may include a joystick, game pad, satellite dish, scanner, or the like. These and other input devices are often connected to the processing unit 820 through a user input interface 860 that is coupled to the system bus, but may be connected by other interface and bus structures. A visual display 891 or other type of display device is also connected to the system bus 821 via an interface, such as a video interface 890. In addition to the monitor, computers may also include other peripheral output devices such as speakers 897 and printer 896, which may be connected through an output peripheral interface 895.

The computer 810 is operated in a networked environment using logical connections (such as a controller area network—CAN, local area network—LAN, or wide area network WAN) to one or more remote computers, such as a remote computer 880.

When used in a LAN networking environment, the computer 810 is connected to the LAN 871 through a network interface or adapter 870. When used in a WAN networking environment, the computer 810 typically includes a modem 872 or other means for establishing communications over the WAN 873, such as the Internet. In a networked environment, program modules may be stored in a remote memory storage device. FIG. 18 illustrates, for example, that remote application programs 885 can reside on remote computer 880.

It should also be noted that the different examples described herein can be combined in different ways. That is, parts of one or more examples can be combined with parts of one or more other examples. All of this is contemplated herein.

Example 1 is the agricultural work machine of any or all previous examples, comprising:

- a communication system that receives a map that includes values of a power characteristic corresponding to different geographic locations in a field;

- a geographic position sensor that detects a geographic location of the agricultural work machine;

- an in-situ sensor that detects a value of an agricultural characteristic corresponding to the geographic location;

- a predictive map generator that generates a functional predictive agricultural map of the field that maps predictive control values to the different geographic locations in the field based on the values of the power characteristic in the map and based on the value of the agricultural characteristic;

- a controllable subsystem; and

- a control system that generates a control signal to control the controllable subsystem based on the geographic location of the agricultural work machine and based on the control values in the functional predictive agricultural map.

Example 2 is the agricultural work machine of any or all previous examples, wherein the map is a predictive power characteristic map generated based on values from a map and values of a power characteristic detected in-situ.

Example 3 is the agricultural work machine of any or all previous examples, wherein the predictive map generator comprises:

- a predictive agricultural characteristic map generator that generates, as the functional predictive agricultural map, a functional predictive agricultural characteristic map that maps, as the predictive control values, predictive values of the agricultural characteristic to the different geographic locations in the field.

Example 4 is the agricultural work machine of any or all previous examples, wherein the in-situ sensor detects, as the

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value of the agricultural characteristic, a value of an operator command indicative of a commanded action of the agricultural work machine.

Example 5 is the agricultural work machine of any or all previous examples, wherein the predictive map generator comprises:

- a predictive operator command map that generates, as the functional predictive agricultural map, a functional predictive operator command map that maps, as the predictive control values, predictive operator command values to the different geographic locations in the field.

Example 6 is the agricultural work machine of any or all previous examples, wherein the control system comprises:

- a settings controller that generates an operator command control signal indicative of an operator command based on the detected geographic location and the functional predictive operator command map and controls the controllable subsystem based on the operator command control signal to execute the operator command.

Example 7 is the agricultural work machine of any or all previous examples, wherein the control system generates the control signal to control the control subsystem to adjust a feed rate of material through the agricultural work machine.

Example 8 is the agricultural work machine of any or all previous examples, and further comprising:

- a predictive model generator that generates a predictive agricultural model that models a relationship between the power characteristic and the agricultural characteristic based on a value of the power characteristic in the map at the geographic location and the value of the agricultural characteristic detected by the in-situ sensor corresponding to the geographic location, wherein the predictive map generator generates the functional predictive agricultural map based on the values of the power characteristic in the map and based on the predictive agricultural model.

Example 9 is the agricultural work machine of any or all previous examples, wherein the control system further comprises:

- an operator interface controller that generates a user interface map representation of the functional predictive agricultural map, the user interface map representation comprising a field portion with one or more markers indicating the predictive control values at one or more geographic locations on the field portion.

Example 10 is the agricultural work machine of any or all previous examples, wherein the operator interface controller generates the user interface map representation to include an interactive display portion that displays a value display portion indicative of a selected value, an interactive threshold display portion indicative of an action threshold, and an interactive action display portion indicative of a control action to be taken when one of the predictive control values satisfies the action threshold in relation to the selected value, the control system generating the control signal to control the controllable subsystem based on the control action.

Example 11 is an example including any or all previous examples, comprising:

- obtaining a map that includes values of a power characteristic corresponding to different geographic locations in a field;

- detecting a geographic location of the agricultural work machine;

- detecting, with an in-situ sensor, a value of an agricultural characteristic corresponding to the geographic location;

- generating a functional predictive agricultural map of the field that maps predictive control values to the different geographic locations in the field based on the values of the

power characteristic in the map and based on the value of the agricultural characteristic; and

controlling a controllable subsystem based on the geographic location of the agricultural work machine and based on the control values in the functional predictive agricultural map.

Example 12 is an example including any or all previous examples, wherein obtaining the map comprises:

obtaining a predictive power characteristic map that includes, as values of the power characteristic, predictive values of the power characteristic corresponding to different geographic locations in the field.

Example 13 is an example including any or all previous examples, wherein generating the functional predictive agricultural map comprises:

generating a functional predictive agricultural characteristic map that maps, as the predictive control values, predictive values of the agricultural characteristic to the different geographic locations in the field.

Example 14 is an example including any or all previous examples, wherein detecting, with an in-situ sensor, the value of an agricultural characteristic comprises:

detecting, with the in-situ sensor, as the value of the agricultural characteristic, an operator command indicative of a commanded action of the agricultural work machine.

Example 15 is an example including any or all previous examples, wherein generating the functional predictive agricultural map comprises:

generating a functional predictive operator command map that maps, as the predictive control values, predictive operator command values to the different geographic locations in the field.

Example 16 is an example including any or all previous examples, wherein controlling the controllable subsystem comprises:

generating an operator command control signal indicative of an operator command based on the detected geographic location and the functional predictive operator command map; and

controlling the controllable subsystem based on the operator command control signal to execute the operator command.

Example 17 is an example including any or all previous examples, wherein controlling the controllable subsystem comprises:

controlling the controllable subsystem to adjust a feed rate of material through the agricultural work machine.

Example 18 is an example including any or all previous examples, and further comprising:

generating a predictive agricultural model that models a relationship between the power characteristic and the agricultural characteristic based on a value of the power characteristic in the map at the geographic location and the value of the agricultural characteristic detected by the in-situ sensor corresponding to the geographic location, wherein generating the functional predictive agricultural map comprises generating the functional predictive agricultural map based on the values of the power characteristic in the map and based on the predictive agricultural model.

Example 19 is the agricultural work machine of any or all previous examples comprising, a communication system that receives a map that includes values of a power characteristic corresponding to different geographic locations in a field;

a geographic position sensor that detects a geographic location of the agricultural work machine

an in-situ sensor that detects a value of an agricultural characteristic corresponding to the geographic location;

a predictive model generator that generates a predictive agricultural model that models a relationship between the power characteristic and the agricultural characteristic based on a value of the power characteristic in the map at the geographic location and the value of the agricultural characteristic detected by the in-situ sensor corresponding to the geographic location;

a predictive map generator that generates a functional predictive agricultural map of the field that maps predictive control values to the different geographic locations in the field based on the values of the power characteristic in the map and based on the predictive agricultural model;

a controllable subsystem; and

a control system that generates a control signal to control the controllable subsystem based on the geographic position of the agricultural work machine and based on the control values in the functional predictive agricultural map.

Example 20 is the agricultural work machine of any or all previous examples, wherein the control system comprises at least one of:

a feed rate controller that generates a feed rate control signal based on the detected geographic location and the functional predictive agricultural map and controls the controllable subsystem based on the feed rate control signal to control a feed rate of material through the agricultural work machine;

a settings controller that generates a speed control signal based on the detected geographic location and the functional predictive agricultural map and controls the controllable subsystem based on the speed control signal to control a speed of the agricultural work machine;

a header controller that generates a header control signal based on the detected geographic location and the functional predictive agricultural map and controls the controllable subsystem based on the header control signal to control a distance of at least a portion of a header on the agricultural work machine from a surface of the field; and

a settings controller that generates an operator command control signal indicative of an operator command based on the detected geographic location and the functional predictive agricultural map and controls the controllable subsystem based on the operator command control signal to execute the operator command.

Although the subject matter has been described in language specific to structural features or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of the claims.

What is claimed is:

1. An agricultural system, comprising:

a communication system that receives an information map that includes values of a power characteristic, as a first agricultural characteristic, corresponding to different geographic locations in a field;

a geographic position sensor that detects a geographic location of an agricultural work machine;

an in-situ sensor that detects a value of a second agricultural characteristic corresponding to a geographic location, wherein the second agricultural characteristic is different than the first agricultural characteristic;

a predictive map generator that generates a functional predictive agricultural map of the field that maps predictive values of the second agricultural characteristic

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to the different geographic locations in the field based on a value of the first agricultural characteristic in the information map corresponding to the geographic location and based on the value of the second agricultural characteristic detected by the in-situ sensor corresponding to the geographic location; and

a control system configured to control a controllable subsystem of the agricultural work machine based on the geographic location of the agricultural work machine and based on the predictive values of the second agricultural characteristic in the functional predictive agricultural map.

2. The agricultural system of claim 1, wherein the functional predictive agricultural map is a functional predictive power map that maps, as the values of the power characteristic, predictive values of the power characteristic corresponding to the different geographic locations in the field, the functional predictive power map generated based on values of a third agricultural characteristic in the information map and values of the power characteristic detected in-situ.

3. The agricultural system of claim 1, wherein the in-situ sensor detects, as the value of the second agricultural characteristic, a value of an operator command indicative of a commanded action of the agricultural work machine.

4. The agricultural system of claim 3, wherein the predictive map generator comprises:

a predictive operator command map generator that generates, as the functional predictive agricultural map, a functional predictive operator command map that maps, as the predictive values of the second agricultural characteristic, predictive operator command values to the different geographic locations in the field.

5. The agricultural system of claim 4, wherein the control system comprises:

a settings controller that generates an operator command control signal indicative of an operator command based on the detected geographic location of the agricultural work machine and the functional predictive operator command map and controls the controllable subsystem based on the operator command control signal to execute the operator command.

6. The agricultural system of claim 1, wherein the control system generates the control signal to control the control subsystem to adjust a feed rate of material through the agricultural work machine.

7. The agricultural system of claim 1 and further comprising:

a predictive model generator that generates a predictive agricultural model that models a relationship between the first agricultural characteristic and the second agricultural characteristic based on the value of the first agricultural characteristic in the information map corresponding to the geographic location and the value of the first agricultural characteristic detected by the in-situ sensor corresponding to the geographic location, wherein the predictive map generator generates the functional predictive agricultural map based on the value of the first agricultural characteristic in the information map corresponding to the geographic location and based on the predictive agricultural model.

8. The agricultural system of claim 1, wherein the control system further comprises;

an operator interface controller that generates a user interface map representation of the functional predictive agricultural map, the user interface map representation comprising a field portion with one or more markers indicating the predictive values of the second

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agricultural characteristic at one or more geographic locations on the field portion.

9. The agricultural system of claim 8, wherein the operator interface controller generates the user interface map representation to include an interactive display portion that displays a value display portion indicative of a selected value, an interactive threshold display portion indicative of an action threshold, and an interactive action display portion indicative of a control action to be taken when one of the predictive values of the second agricultural characteristic satisfies the action threshold in relation to the selected value, the control system generating the control signal to control the controllable subsystem based on the control action.

10. A computer implemented method of controlling an agricultural work machine comprising:

obtaining an information map that includes values of a power characteristic, as a first agricultural characteristic, corresponding to different geographic locations in a field;

detecting a geographic location of the agricultural work machine;

detecting, with an in-situ sensor, a value of a second agricultural characteristic corresponding to a geographic location, wherein the second agricultural characteristic is different than the first agricultural characteristic;

generating a functional predictive agricultural map of the field that maps predictive values of the second agricultural characteristic to the different geographic locations in the field based on a value of the first agricultural characteristic in the information map corresponding to the geographic location and based on the value of the second agricultural characteristic detected by the in-situ sensor corresponding to the geographic location; and

controlling a controllable subsystem of the agricultural work machine based on the geographic location of the agricultural work machine and based on the predictive values of the second agricultural characteristic in the functional predictive agricultural map.

11. The computer implemented method of claim 10, wherein obtaining the information map comprises:

obtaining a predictive power characteristic map that includes, as values of the power characteristic, predictive values of the power characteristic corresponding to different geographic locations in the field.

12. The computer implemented method of claim 10, wherein detecting, with an in-situ sensor, the value of the second agricultural characteristic comprises:

detecting, with the in-situ sensor, as the value of the second agricultural characteristic, an operator command indicative of a commanded action of the agricultural work machine.

13. The computer implemented method of claim 12, wherein generating the functional predictive agricultural map comprises:

generating a functional predictive operator command map that maps, as the predictive values of the second agricultural characteristic, predictive operator command values to the different geographic locations in the field.

14. The computer implemented method of claim 13, wherein controlling the controllable subsystem comprises: generating an operator command control signal indicative of an operator command based on the detected geographic location of the agricultural work machine and the functional predictive operator command map; and

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controlling the controllable subsystem based on the operator command control signal to execute the operator command.

15. The computer implemented method of claim 10, wherein controlling the controllable subsystem comprises: 5
controlling the controllable subsystem to adjust a feed rate of material through the agricultural work machine.

16. The computer implemented method of claim 10 and further comprising:

generating a predictive agricultural model that models a 10
relationship between the first agricultural characteristic and the second agricultural characteristic based on a value of the first agricultural characteristic in the map at the geographic location and the value of the second 15
agricultural characteristic detected by the in-situ sensor corresponding to the geographic location, wherein generating the functional predictive agricultural map comprises generating the functional predictive agricultural 20
map based on the values of the first agricultural characteristic in the map and based on the predictive agricultural model.

17. The computer implemented method of claim 10, wherein obtaining the information map comprises obtaining 25
the information map as the agricultural work machine operates at the field during a current operation, and wherein the controlling the controllable subsystem of the agricultural work machine comprises controlling the controllable sub- 30
system as the agricultural work machine operates at the field.

18. The computer implemented method of claim 17, wherein the information map, is a functional predictive 35
power characteristic map that includes, as the values of the power characteristic corresponding to the different geographic locations of the field, predictive values of the power characteristic corresponding to the different geographic 40
locations in the field, the method further comprising:

obtaining an additional information map that includes 45
values of a third agricultural characteristic corresponding to the different geographic locations in the field, the third agricultural characteristic different than both the first agricultural characteristic and the second agricul- 50
tural characteristic;

detecting, with an in-situ power sensor, a value of the 45
power characteristic corresponding to an additional geographic location; and

generating the functional predictive power characteristic 50
map that maps the predictive values of the power characteristic corresponding to the different geographic locations in the field based on a value of the third agricultural characteristic in the additional information map corresponding to the additional geographic loca- 55
tion and based on the value of the power characteristic detected by the in-situ power sensor corresponding to the additional geographic location.

19. An agricultural system comprising:

a communication system that receives an information map 55
that includes values of a power characteristic, as a first agricultural characteristic, corresponding to different geographic locations in a field;

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a geographic position sensor that detects a geographic location of an agricultural work machine

an in-situ sensor that detects a value of a second agricul-
tural characteristic corresponding to a geographic loca-
tion, wherein the second agricultural characteristic is
different than the first agricultural characteristic;

a predictive model generator that generates a predictive
agricultural model that models a relationship between
the first agricultural characteristic and the second agri-
cultural characteristic based on a value of the first
agricultural characteristic in the information map at the
geographic location to which the value of the second
agricultural characteristic detected by the in-situ sensor
corresponds and the value of the second agricultural
characteristic detected by the in-situ sensor correspond-
ing to the geographic location;

a predictive map generator that generates a functional
predictive agricultural map of the field that maps pre-
dictive values of the second agricultural characteristic
to the different geographic locations in the field based
on a value of the first agricultural characteristic in the
information map corresponding to the geographic loca-
tion and based on the predictive agricultural model;

a control system that controls a controllable subsystem of
the agricultural work machine based on the geographic
location of the agricultural work machine and based on
the predictive values of the second agricultural char-
acteristic in the functional predictive agricultural map.

20. The agricultural system of claim 19, wherein the
control system comprises at least one of:

a feed rate controller that generates a feed rate control
signal based on the detected geographic location of the
agricultural work machine and the functional predictive
agricultural map and controls the controllable subsys-
tem based on the feed rate control signal to control a
feed rate of material through the agricultural work
machine;

a settings controller that generates a speed control signal
based on the detected geographic location of the agri-
cultural work machine and the functional predictive
agricultural map and controls the controllable subsys-
tem based on the speed control signal to control a speed
of the agricultural work machine;

a header controller that generates a header control signal
based on the detected geographic location of the agri-
cultural work machine and the functional predictive
agricultural map and controls the controllable subsys-
tem based on the header control signal to control a
distance of at least a portion of a header on the
agricultural work machine from a surface of the field;
and

a settings controller that generates an operator command
control signal indicative of an operator command based
on the detected geographic location of the agricultural
work machine and the functional predictive agricultural
map and controls the controllable subsystem based on
the operator command control signal to execute the
operator command.

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